

MCS4206: Control Engineering II

Chapter 1

Review of Basic Control Theory

Introduction

Control Engineering is the professional art of making control systems

Control means to regulate, direct, command, or govern. A system is a collection, set, or arrangement of elements (subsystems). A control system is an interconnection of components forming a system configuration that will provide a desired system response. Hence, a control system is an arrangement of physical components connected or related in such a manner as to command, regulate, direct, or govern itself or another system.

Control systems can be broadly classified into open-loop and closed-loop systems.

An **open-loop** control system utilizes a controller or control actuator to obtain the desired response (output) as shown in Figure 1. The actuating device controls the process (plant) directly without monitoring the actual output.

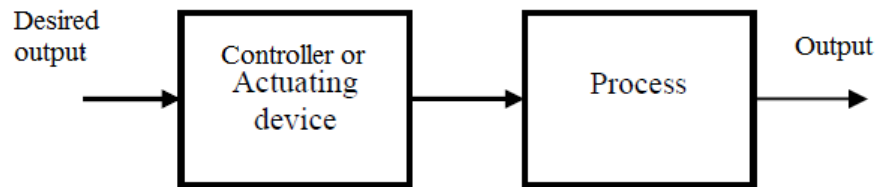


Figure 1: Open loop control system

Examples of open-loop control system include electric toaster, traffic light and manual washing machine among others.

A closed-loop control system utilizes an actual output monitoring devices in addition to the plant and the controller as illustrated in figure 2.

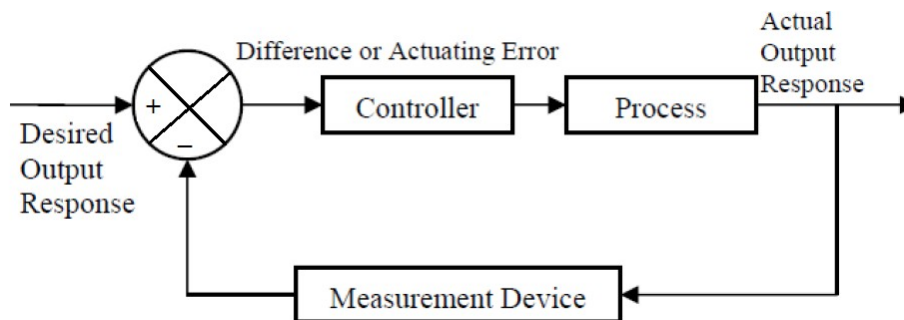


Figure 2: Closed loop control system

The actual output is compared to the desired output via measurement device (sensor). If they are the same the controller should not act otherwise the controller adjusts the actual output. A closed-loop control system is also called **feedback control system**, because the output of the process of the system is fed back to the input of the controller.

Mathematical Modeling

One of the most important tasks in the analysis and design of control systems is mathematical modeling of the system in which physical systems are represented by Dynamic equations in form of differential equations. The differential equations describing the dynamic performance of a physical system are obtained by utilizing the physical laws of the process (such as Newton's Laws, Kirchhoff's laws, Ohm's law etc.).

For example consider the following differential equation:

$$\ddot{x}(t) + 3\dot{x}(t) + 2x(t) = y(t) \quad (1)$$

The equation (1) can represent a 2nd order control system with an input $y(t)$ and output $x(t)$.

Transfer Function (TF)

The transfer function is the ratio of the Laplace transform of the output (response function) to the Laplace transform of the input (driving function) under the assumption that all initial conditions are zero.

The transfer function of a control system can be obtained from its dynamic equations (differential equations).

For example the transfer function of the control system represented by equation (1) can be obtained as follows:

Taking the Laplace transforms yields:

$$s^2 X(s) - s\dot{x}(0) - x(0) + 3sX(s) - 3x(0) + 2X(s) = Y(s)$$

Assuming zero initial condition:

$$s^2 X(s) + 3sX(s) + 2X(s) = Y(s)$$

$$(s^2 + 3s + 2)X(s) = Y(s)$$

Therefore,

$$TF = \frac{X(s)}{Y(s)} = \frac{1}{s^2 + 3s + 2}$$

Consider the closed loop control system in figure 3.

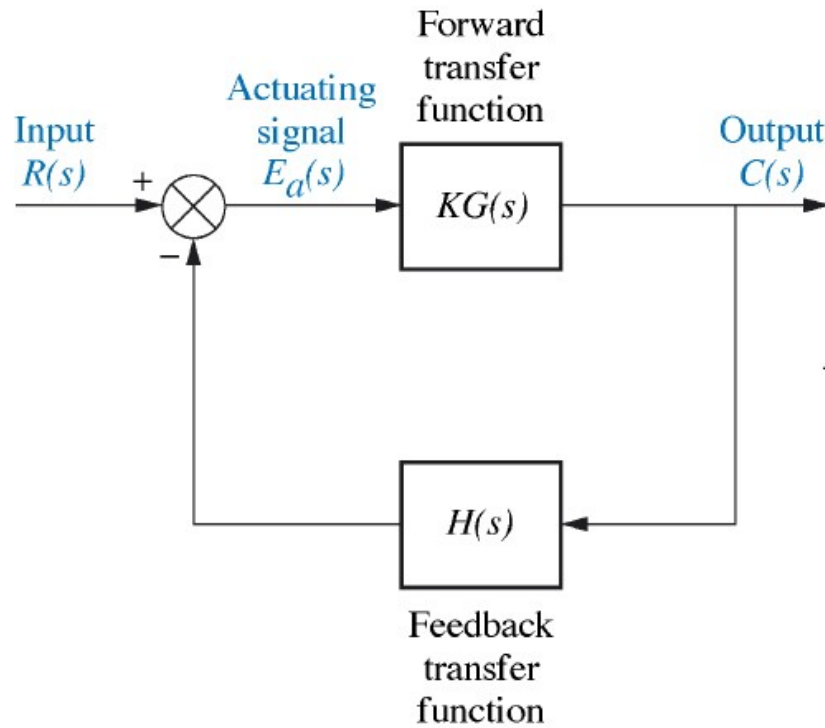


Figure 3: Closed loop control system

The open loop transfer function is given by:

$$\frac{C(s)}{R(s)} = KG(s)H(s)$$

The closed loop transfer function is given by:

$$\frac{C(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)H(s)}$$

System poles, zeros, order and type

Consider the general transfer function below:

$$G(s) = \frac{C(s)}{R(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}$$

Zeros: The values of s that makes $G(s)=0$ are called the zeros of the system. In other words they are the roots of the numerator of the transfer function.

Poles: The values of s that makes $G(s)=\infty$ are called the poles of the system. In other words they are the roots of the denominator of the transfer function.

Order of system: The highest power of s in the denominator is called the order of the system. Hence m is the order of $G(s)$. In other words it is the number of poles of the system.

Type of system: The system type is highest power of s in the denominator which is equal to zero. In other words it is the number of poles that are equal to zero.

Given the following T.F find the zeros, poles, system order and type.

$$\text{a. } G(s) = \frac{s^2 + 4s + 3}{s^5 + 8s^4 + 20s^3 + 16s^2}$$

$$\text{b. } G(s) = \frac{s^2 + 4s}{s^3 + 4s^2 + 20s + 16}$$

Solutions:

a. If we factorize $G(s)$ yields

$$G(s) = \frac{(s + 3)(s + 1)}{s^2(s + 2)^2(s + 4)}$$

The zeros are $z_1 = -1, z_2 = -3$, the poles are $p_{1,2} = 0, p_{3,4} = -2, p_5 = -4$

Now, 2 poles are equal to zero, hence the type is 2. And the order is 5, since 5 is the highest power of the system.

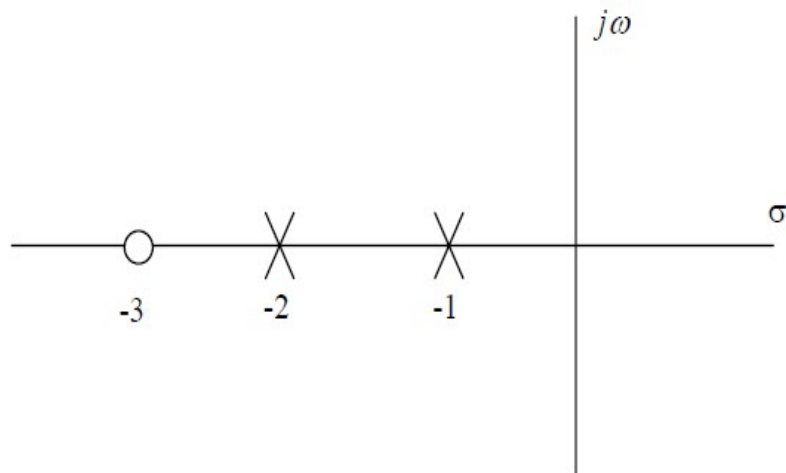
b. Zeros are $z_1 = -4, z_2 = 0$. Poles are $p_{1,2} = -1.5332 \pm 3.8453i, p_3 = 0.9336$. Order is 3, and type is 0.

The poles and zeros can be mapped on an s-plane (pole: **x**, zero: **o**) as illustrated below:

Consider the closed loop transfer function below:

$$\frac{C(s)}{R(s)} = \frac{K(s+3)}{(s+1)(s+2)}$$

The corresponding s-plane plot is shown below:

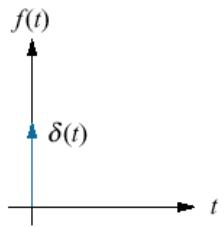
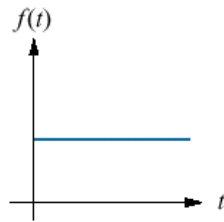
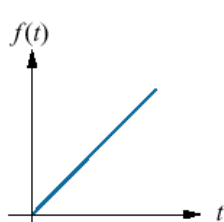
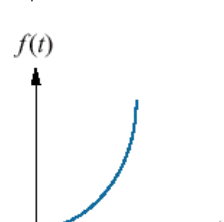
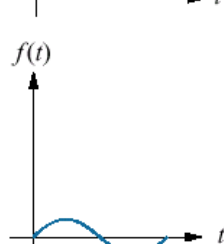


The response of control systems is governed by the location of closed loop poles of the system in the s-plane.

For a stable system all the closed loop poles must be located in the left hand side of the s-plane.

Input Signals

To obtain an output of a control system the system must be excited with input signal or excitation signal or driving force function. The commonly used input signals are given in the table below:

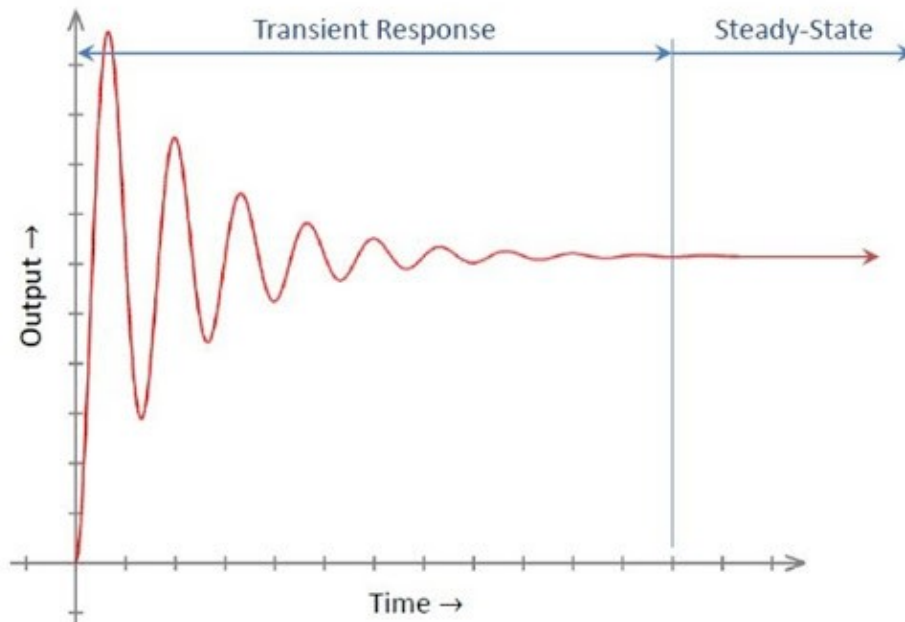
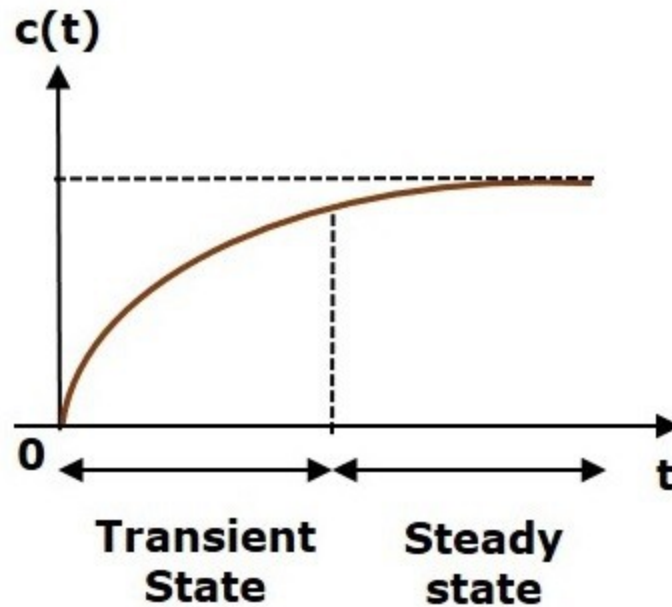
Input	Function	Description	Sketch	Use
Impulse	$\delta(t)$	$\delta(t) = \infty$ for $0- < t < 0+$ $= 0$ elsewhere $\int_{0-}^{0+} \delta(t) dt = 1$		Transient response Modeling
Step	$u(t)$	$u(t) = 1$ for $t > 0$ $= 0$ for $t < 0$		Transient response Steady-state error
Ramp	$tu(t)$	$tu(t) = t$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Parabola	$\frac{1}{2}t^2u(t)$	$\frac{1}{2}t^2u(t) = \frac{1}{2}t^2$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Sinusoid	$\sin \omega t$			Transient response Modeling Steady-state error

Transient and Steady State Response

When a system is subjected to an input, the response in time may be divided in two parts. The first part of the response immediately after the application of the input is called **transient response** whereas the response long after the application of the input, as $t \rightarrow \infty$ is known as **steady state response**. In stable control systems, the transient response dies as time becomes large compared to the smallest time constant of the system and the steady state response begins. The time response of an analog control system, denoted by $y(t)$, can, in general, be written in terms of its two components as:

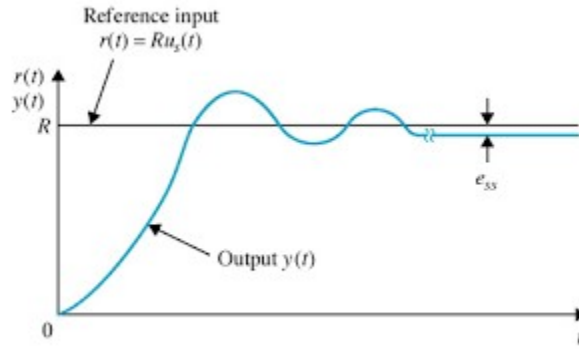
$$y(t) = y_t(t) + y_{ss}(t)$$

where $y_t(t)$ denotes the *transient response*, and $y_{ss}(t)$ denotes the *steady-state response*.



If the steady-state response of the output does not agree exactly with the input, the system is said to have a finite **steady-state error**. Therefore, the steady state error is the deviation of the output of control system from desired response during steady state.

The figure below illustrated the steady state error:



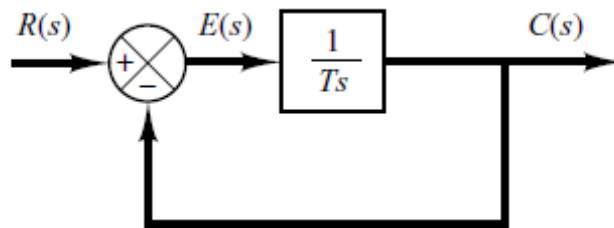
Time domain performance measures

The study of a control system in the time domain essentially involves the evaluation of the transient and the steady-state responses of the system. In the design stage of a controller, specifications are usually given in terms of the transient and the steady-state performance, which are to be satisfied by the closed loop system.

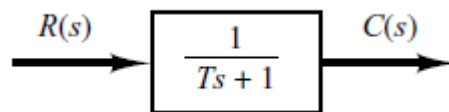
The response of control systems is governed by the location of closed loop poles of the system in the s-plane

First order system Response

Consider the first-order system shown in the figure below.



Physically, this system may represent an RC circuit, thermal system, fluid level system, or the like. A simplified block diagram is shown in the figure below:



The input-output relationship is given by:

$$\frac{C(s)}{R(s)} = \frac{1}{Ts + 1} \quad (1)$$

The initial conditions are assumed to be zero.

Unit-Step Response of First-Order Systems

The system response will be analyzed using unit-step function as an input.

Since the Laplace transform of the unit-step function is $1/s$, substituting $R(s) = 1/s$ into Equation (1), we obtain:

$$C(s) = \frac{1}{Ts+1} R(s) = \frac{1}{Ts+1} \frac{1}{s} \quad (2)$$

Expanding in to partial fraction gives:

$$C(s) = \frac{A}{s} - \frac{B}{Ts+1} = \frac{1}{s(Ts+1)}$$

$$(AT+B)s + A = 1$$

Comparing coefficients gives:

$$A = 1$$

$$AT+B=0$$

Hence

$$B = -T$$

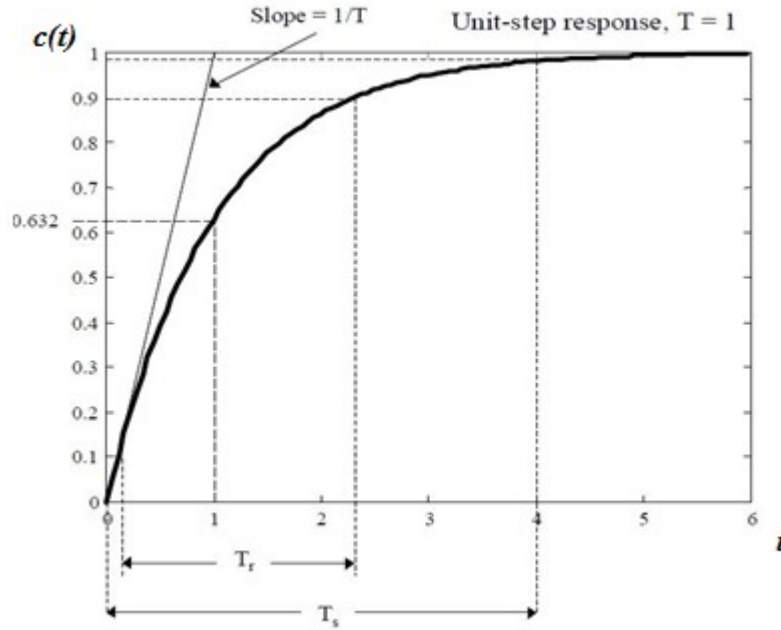
Therefore

$$C(s) = \frac{1}{s} - \frac{T}{Ts+1} = \frac{1}{s} - \frac{1}{s+\cancel{\cancel{\cancel{T}}}} \quad (3)$$

Taking the inverse Laplace transform of Equation (3), gives:

$$c(t) = 1 - e^{\frac{-t}{T}}, \quad \text{for } t \geq 0 \quad (4)$$

Equation (4) can be plotted as shown below:



Equation (4) can also be used to define 3 transient performance measures:

Time constant (τ): it is the time it takes a step response to rise to 63% of its final value.

From equation (4):

$$1 - e^{\frac{-t}{T}} = 0.63$$

$$1 - 0.63 = e^{\frac{-t}{T}}$$

$$\frac{-t}{T} = \ln(0.37) = -1$$

$$\tau = T$$

A sluggish system has large τ

A fast system has small τ

Rise time (T_r): is defined as the time taken for the response to rise from 10% to 90% of its final value.

$$T_r = t_1 - t_2$$

For t_1

$$1 - e^{\frac{-t_1}{T}} = 0.9$$

$$\frac{-t_1}{T} = \ln(0.1) = -2.3$$

$$t_1 = 2.3 T$$

Similarly for t_2

$$1 - e^{\frac{-t_2}{T}} = 0.1$$

$$\frac{-t_2}{T} = \ln(0.9) = -1.1$$

$$t_2 = 1.1 T$$

Therefore

$$T_r = 2.3 T - 1.1 T = 1.2 T$$

Settling time (T_s): is defined as the time taken for the response to reach 98% and stay within the band of 98% to 102% of its final value.

$$1 - e^{\frac{-T_s}{T}} = 0.98$$

$$\frac{-T_s}{T} = \ln(0.02) = -4$$

$$T_s = 4 T$$

Second order systems response

Compared to the simplicity of first order systems, a second order system exhibits a wide range of responses that must be analyzed.

Whereas varying a first order system's parameters simply changes the speed of the response, changes in the parameters of a second order system can change the form of the response.

The general second order transfer function can be given by

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Where ω_n = natural frequency

ζ = damping ratio

The poles of the system can be determined from the characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$a=1, b=2\zeta\omega_n, \omega_n^2$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$s = \frac{-2\zeta\omega_n \pm 2\omega_n\sqrt{\zeta^2 - 1}}{2}$$

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \quad (1)$$

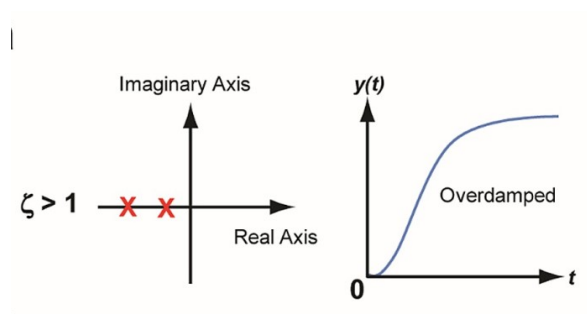
Based on equation (1) there are four possibilities as illustrated below:

a. When $\zeta > 1$

The roots (poles) of the system are real and distinct,

$$s_{1,2} = -\zeta\omega_n \pm k$$

Then the system is said to be over damped as illustrated below:

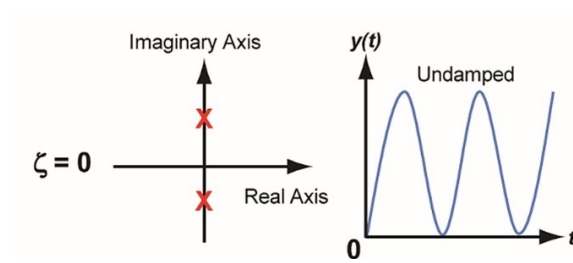


b. When $\zeta = 0$

The roots are purely imaginary,

$$s_{1,2} = \pm j\omega_n$$

Then the system is said to be undamped as illustrated below:

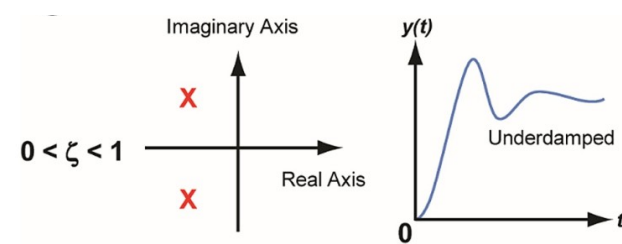


c. When $0 < \zeta < 1$

The roots are complex

$$s_{1,2} = -\zeta \omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$$

Then the system is said to be under damped as illustrated below:

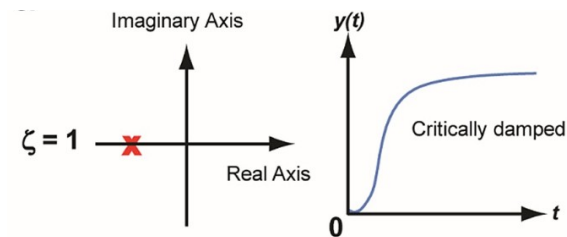


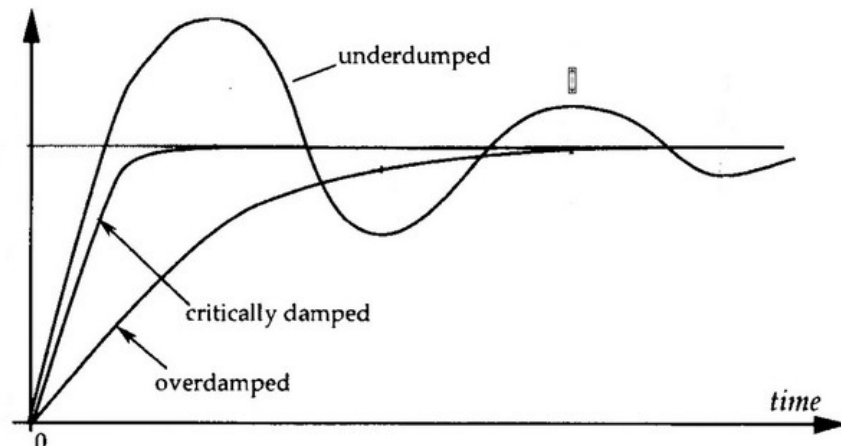
d. When $\zeta = 1$

The roots are real and the same

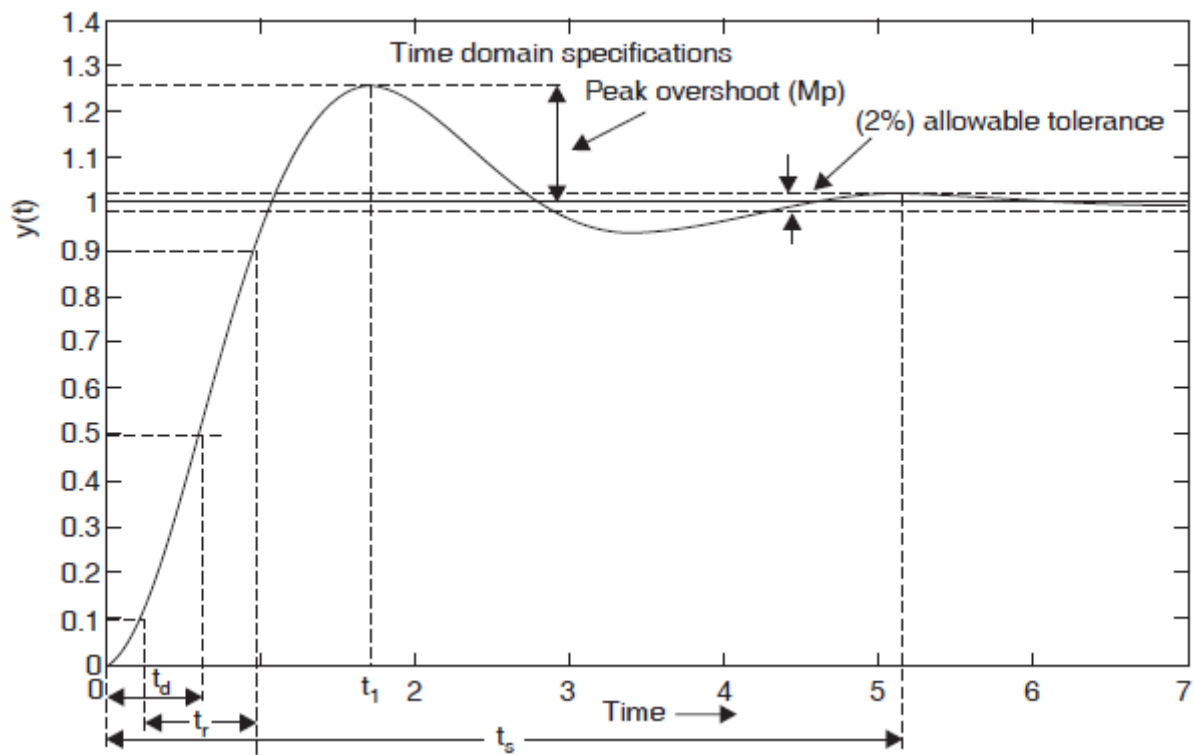
$$s_{1,2} = -\zeta \omega_n$$

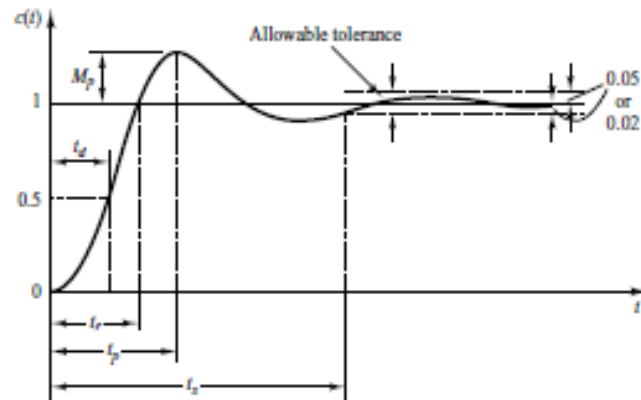
Then the system is said to be critically damped as illustrated below:





The under damped second order system is the common model for physical problem.

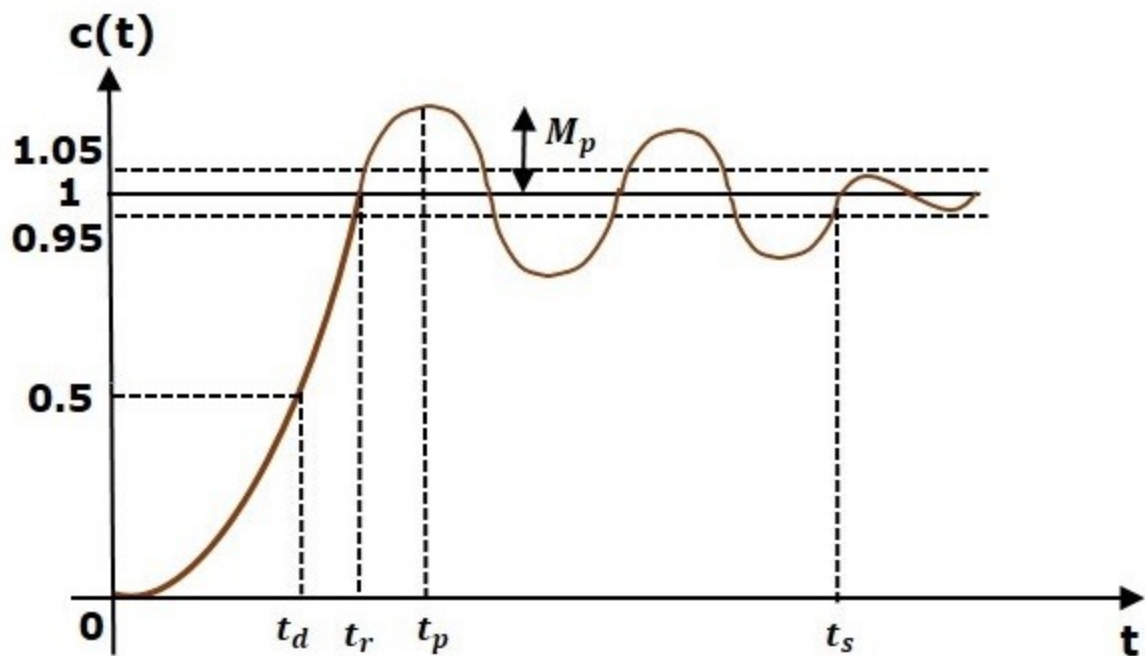




Other parameters associated with the under damped response are:

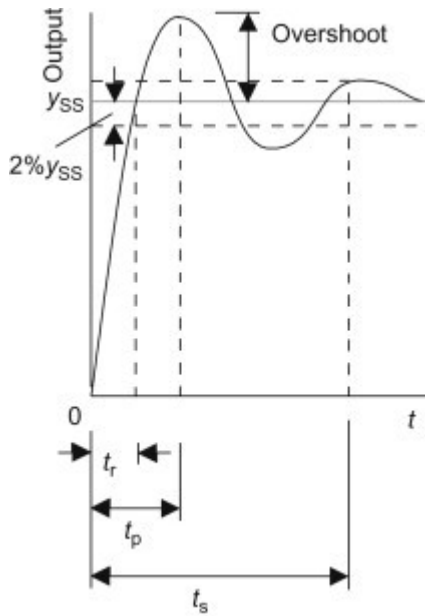
Peak time (t_p): is the time required for the response to reach the maximum or peak:

it is given as:
$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$



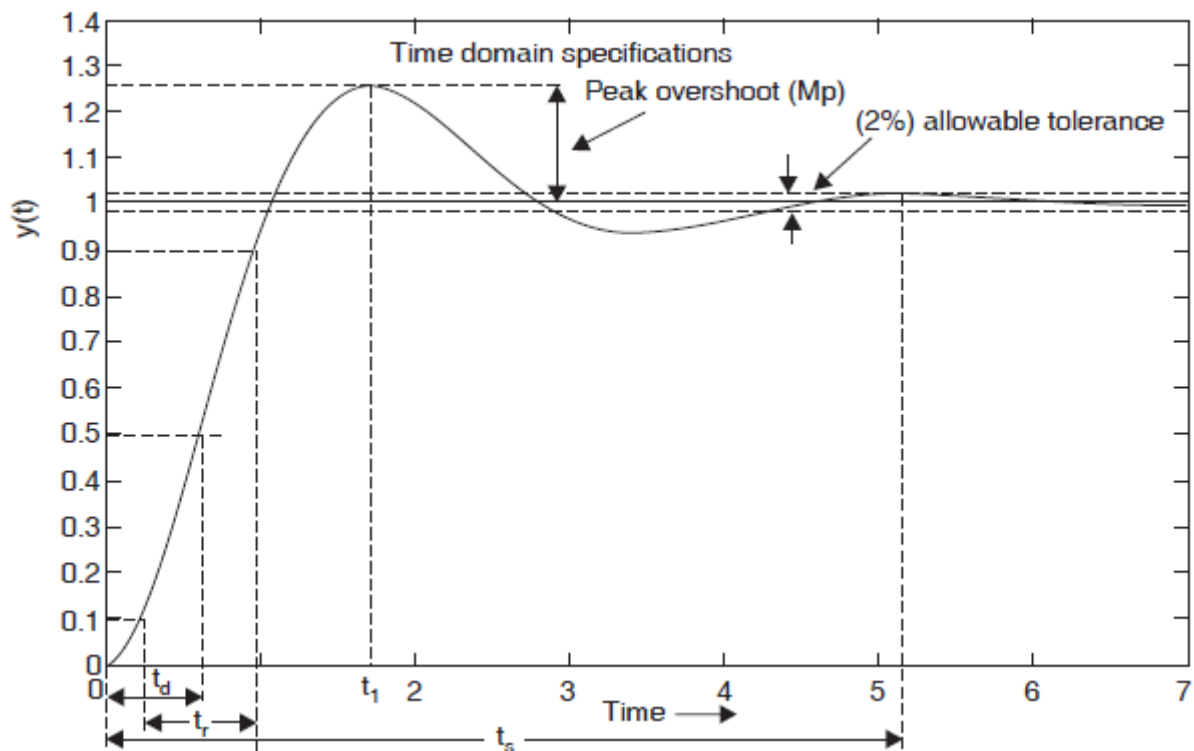
Percent Overshoot. Let $y_{\max}$ denote the maximum value of the unit-step response $y(t)$, and y_{ss} be the steady-state value of $y(t)$, and $y_{\max} \geq y_{ss}$. The maximum percent overshoot of $y(t)$ is defined as

$$\text{Percent overshoot} = \frac{y_{\max} - y_{ss}}{y_{ss}} \times 100 = e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100$$



The maximum overshoot is often used as a measure of the relative stability of a control system. A system with a large overshoot is usually not desirable in many applications.

Settling time(t_s): the time required for the transient damped oscillations to reach and stay within of the steady state value.



$$t_s = \frac{4}{\zeta \omega_n}$$

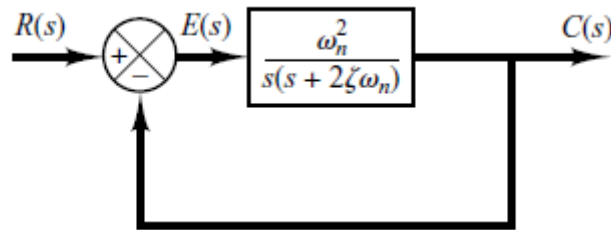
Delay Time: The delay time t_d is normally defined as the time required for the step response to reach 50 percent of its final value

Rise Time. The rise time t_r is defined as the time required for the step response to rise from 10 to 90 percent of its final value.

$$T_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)}{\omega_n \sqrt{1-\zeta^2}}$$

Example1

Consider the system shown in the figure below, where $\zeta = 0.6$ and $\omega_n = 5 \text{ rad/s}$. Obtain the rise time, peak time, percent overshoot, and settling time when the system is subjected to a unit-step input.



Solution:

Given:

$$\zeta = 0.6$$

$$\omega_n = 5 \text{ rad/s}$$

Rise time

$$T_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)}{\omega_n \sqrt{1-\zeta^2}} = 0.55 \text{ seconds}$$

Peak time

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = 0.785 \text{ seconds}$$

Percent overshoot

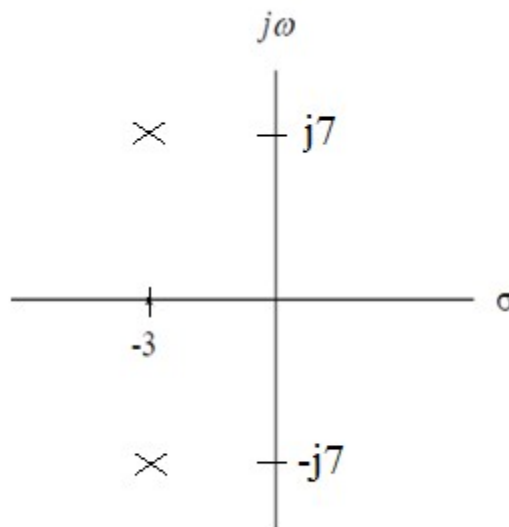
$$\text{Percent overshoot} = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 9.5\%$$

Settling time

$$t_s = \frac{4}{\zeta \omega_n} = 1.33 \text{ seconds}$$

Example 2

Given the pole plot below, find ζ , ω_n , T_p , t_s and %OS



Solution

The characteristic equation can be obtained as:

$$\begin{aligned} \phi(s) &= (s+3-j7)(s+3+j7) = 0 \\ s^2 + 6s + 58 &= 0 \end{aligned} \quad 1$$

The general second order characteristic equation can be written as:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad 2$$

Comparing (1) and (2) gives:

$$\omega_n = \sqrt{58} = 7.616$$

$$2\zeta\omega_n = 6$$

$$\zeta = 0.394$$

Peak time

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = 0.449 \text{ seconds}$$

Percent overshoot

$$\text{Percent overshoot} = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 26\%$$

Settling time

$$t_s = \frac{4}{\zeta \omega_n} = 1.333 \text{ seconds}$$

Effect of Additional Poles and Zeros

The addition of an extra pole in the left-half s-plane (LHP) tends to slow the system down—the rise time of the system, for example, will become larger. When the pole is far to the left of the imaginary axis, its effect tends to be small. The effect becomes more pronounced as the pole moves toward the imaginary axis.

The addition of a zero in the LHP has the opposite effect, as it tends to speed the system up. Again the effect of a zero far away to the left of the imaginary axis tends to be small. It becomes more pronounced as the zero moves closer to the imaginary axis.

The addition of a zero in the right-half s-plane (RHP) has a delaying effect much more severe than the addition of a LHP pole. In fact a RHP zero causes the response (say, to a step input) to start toward the wrong direction. It will move down first and become negative, for example, before it becomes positive again and starts toward its steady-state value. Systems with RHP zeros are called **non-minimum phase systems** (for reasons that will become clearer after the discussion of the frequency design methods) and are typically difficult to control.

Systems with only LHP poles (stable) and LHP zeros are called **minimum phase systems**.

MCS4206: Control Engineering II

Chapter 2

Analysis and design of control systems using Root locus

Introduction

Root locus (RL) is a powerful method of analysis and design for stability and transient response of a control system.

Consider the closed loop control system in figure 1.

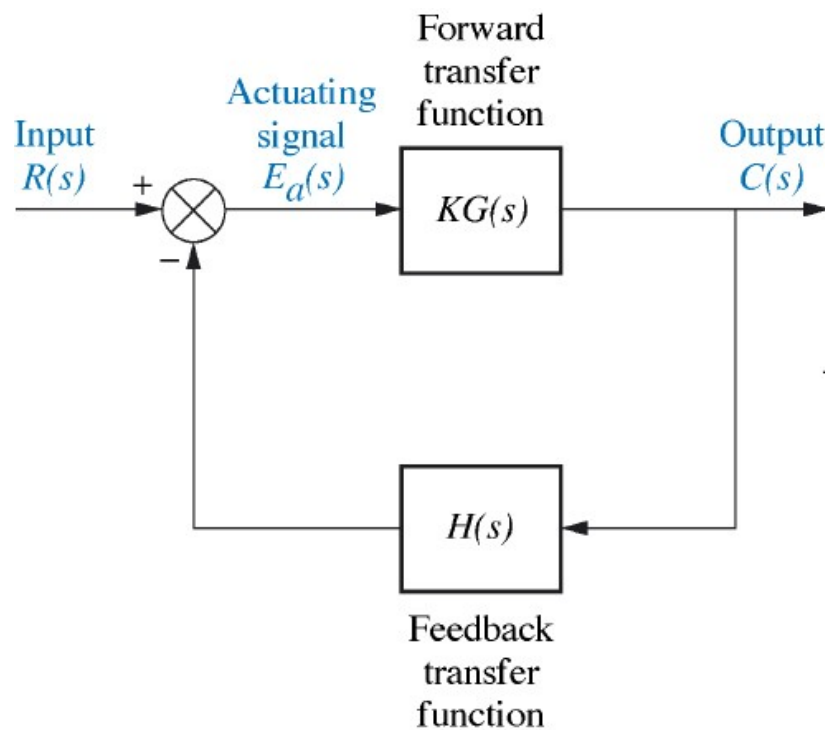


Figure 1: Closed loop control system

The open loop transfer function is given by:

$$\frac{C(s)}{R(s)} = KG(s)H(s)$$

The closed loop transfer function is given by:

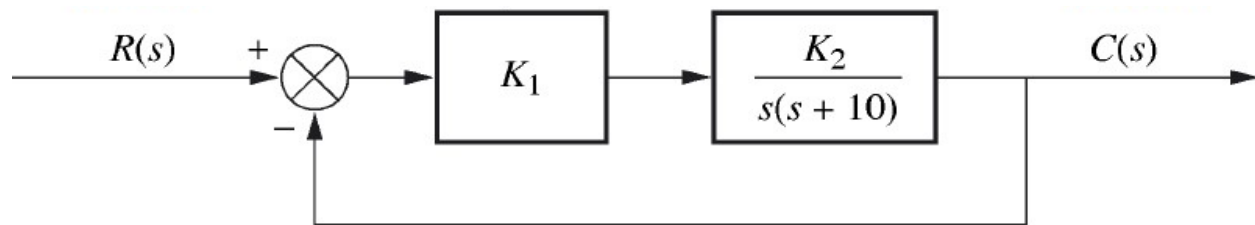
$$\frac{C(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)H(s)}$$

RL is a graphical representation of the closed-loop poles as a system parameter is varied i.e. the gain K of the system.

Thus, RL gives the transient response specifications (OS , T_p , T_s , e_{ss} , stability) as a system parameter is varied.

Example 1

Consider the closed-loop system in figure 2. Investigate how the gain, $K=K_1K_2$ affected the closed-loop poles and system specifications?



Solution

The closed loop transfer function can be obtained as:

$$\frac{C(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)H(s)} = \frac{\frac{K_1 K_2}{s(s+10)}}{1 + \frac{K_1 K_2}{s(s+10)}} = \frac{K_1 K_2}{s(s+10) + K_1 K_2} = \frac{K}{s^2 + 10s + K}$$

The locations of the poles as a function of the gain, K are given in table 1.

Table 1: locations of poles as a function of the gain, K

K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

Table 1 can be represented on the s-plane as shown in figure 3.

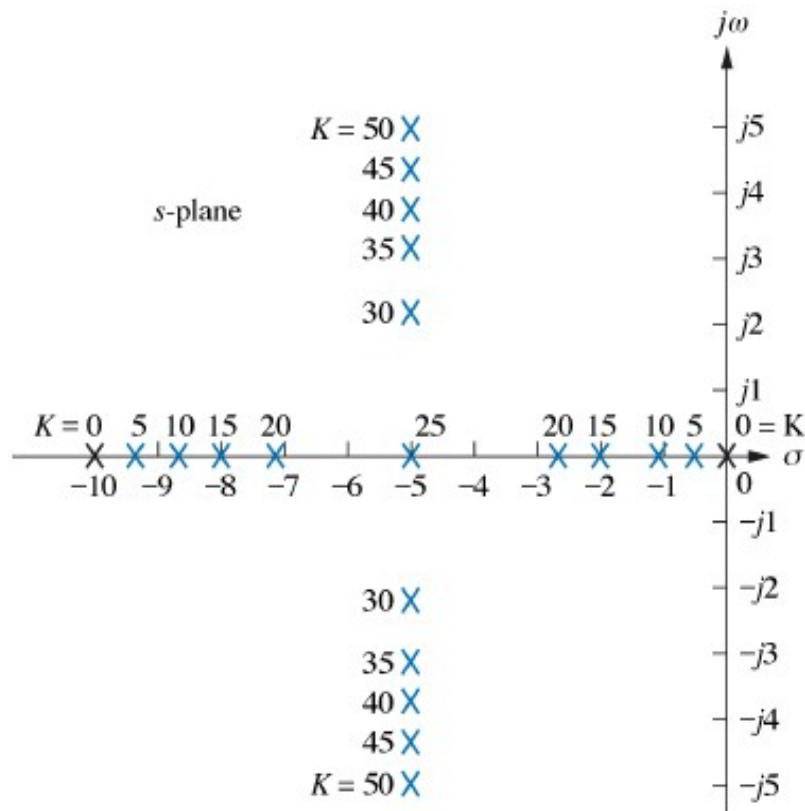


Figure 3: S-plane representation of table 1

From figure 3, it can be seen that:

- When $K=0$, the closed loop poles are located at **-10** and **0**
- The closed loop pole at **-10** moves towards the **right** and closed loop pole at **0** moves towards the **left** as **K increases**.
- The closed loop poles meet at **-5**, *breakaway* from the real axis, move into complex plane. One move **upwards** and the other moves **downwards**.

Individual closed loop pole locations are removed and their paths are represented with lines as shown in figure 4.

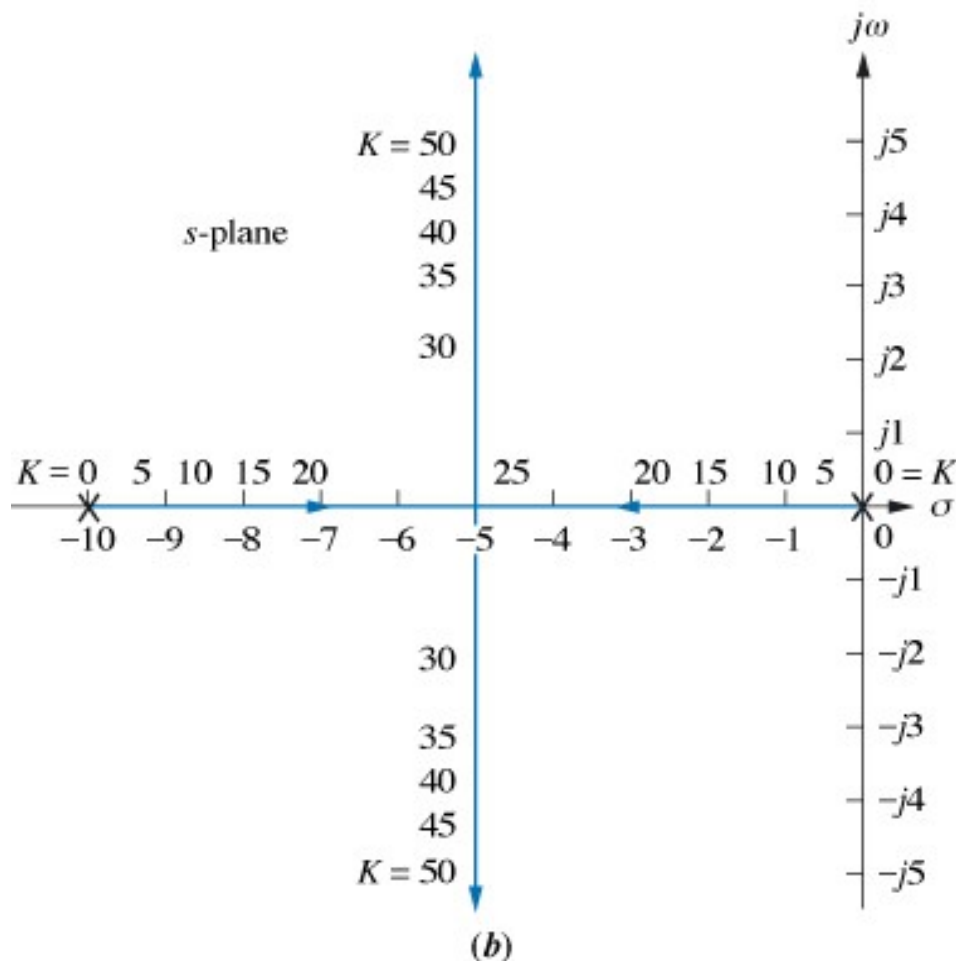


Figure 4: S-plane representation with pole locations paths represented with lines

The representation of the path of the closed loop poles as the gain varied illustrated in figure 4 is called the **root locus**.

From the root locus of figure 4, it can be seen that:

- $K < 25$, the poles are real, over-damped i.e $\zeta > 1$

- **K = 25**, the poles are real and multiple (overlapped), critically damped i.e. $\zeta = 1$
- **K > 25**, complex poles, under-damped i.e. $0 < \zeta < 1$
- For the under-damped system, regardless of the gain value, the *real parts are always the same*. Thus the *settling time remains the same*.
- Since the root locus never crosses into the right half of the s-plane, the system is always stable for all value of K.

Rules for sketching root locus

From the example 1, it can be seen that, sketching the root locus requires calculating the locations of all the closed loop poles of the system at a particular gain. This is tedious especially for higher order systems with much number of poles.

The following rules allow sketching the root locus using minimal calculations:

- **Rule 1: Number of Branches:** *The number of branches is equal to the number of opened-loop poles.*
- **Rule 2: Symmetry:** *The root locus is symmetrical about the real axis.*
- **Rule 3: Real-axis Segment:** *On the real axis, for $K > 0$, the root locus exists to the left of an odd number of real-axis, finite OL poles and/or zeros.*
- **Rule 4: Starting and Ending Point:** *The root locus begins at the finite and infinite poles of $G(s)H(s)$ at $K=0$ and ends at the finite and infinite zeros of $G(s)H(s)$ at $K \rightarrow \infty$.*
- **Rule 5: Behaviour at Infinity:** *The root locus approaches straight lines known as asymptotes as the locus approaches infinity.*

Number of asymptotes = # finite poles - # finite zeros.

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\cancel{\text{finite poles}} - \cancel{\text{finite zeros}}}$$

where $k = 0, \pm 1, \pm 2, \dots$

- **Rule 6: Real-axis Breakaway and Break-in Points:** The root locus breaks away from the real axis at a point of **maximum gain** and breaks into the real axis at a point where the **gain is minimum**.

The breakaway point occurs at a point of maximum gain on the real axis between two OL poles.

The break-in point occurs at a point of minimum gain on the real axis between two OL zeros.

To find the break point $s = \sigma$,

- Maximize and minimize the gain, K through differentiation.

$$\frac{d}{ds}(G(s)H(s))=0$$

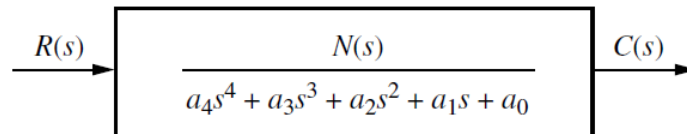
- Solve for s.
- The s obtained from above equation is the breakpoint, σ

- **Rule 7: The $j\omega$ - crossing:** The $j\omega$ -axis crossing is a point on the root locus that separates the stable and unstable operations. *Routh-Hurwitz can be used to find the $j\omega$ -crossing based on the closed loop characteristic equation which is given as:*

$$1 + KG(s)H(s) = 0$$

The following example illustrate the procedure for finding the $j\omega$ -crossing:

- Consider the closed loop system below:



- The Ruth table can be obtained from the characteristic equation as follows:

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2	$-\frac{\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$	$-\frac{\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$	$-\frac{\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$
s^1	$-\frac{\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$	$-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$	$-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$
s^0	$-\frac{\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$	$-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$	$-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$

- The $j\omega$ -crossing occur at the limit of stability (when all the elements of an entire row in the Routh table are zeros) that is $c_1=0$
- Then an auxiliary equation is formed using the row above the zero row as:

$$b_1 s^2 + b_2 = 0$$

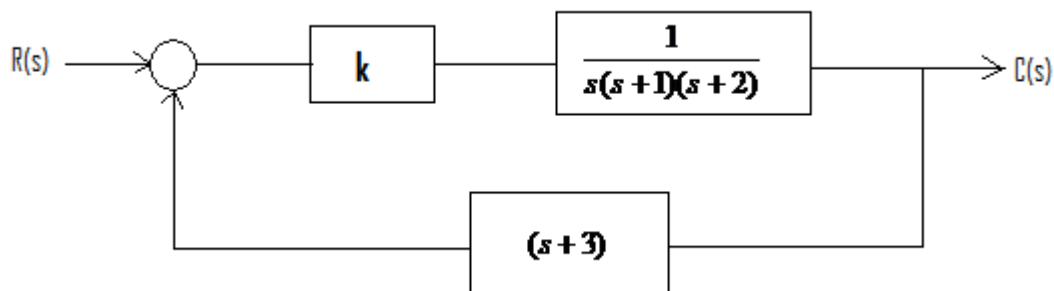
- Solving for s in the auxiliary equation gives the $j\omega$ -crossing values.
- **Rule 8: Angles of Departure and Arrival:** Angles of departure (pole) and angles of arrival (zero) are used for complex OL poles and zeros. The root locus departs from complex OL poles and arrives at complex OL zeros. The angles can be found using the angle property:

$$\theta_a = \sum (\text{angles of all poles to } B) - \sum (\text{angles of all zeros to } B) + 180^\circ$$

$$\theta_d = \sum (\text{angles of all zeros to } A) - \sum (\text{angles of all poles to } A) + 180^\circ$$

Example 1

Consider the system in the figure below:



Sketch the root locus for the system.

Solution

The open loop transfer function can be obtained as:

$$OLTF = kG(s)H(s) = \frac{k(s+3)}{s(s+1)(s+2)}$$

There are three finite open loop poles located at 0, -1 and -2.

There is one finite open loop zero located at -3.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 3-1=2

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$
$$\sigma_a = \frac{(0 + (-1) + (-2)) - (-3)}{3 - 1} = \frac{0}{2} = 0$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\text{\textcolor{red}{i}} \text{finite poles} - \text{\textcolor{red}{i}} \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{3-1} = \frac{(2k+1)180^\circ}{2} = (2k+1)90^\circ$$

There are two asymptotes, therefore $k = 0, 1$

For $k = 0$

$$\theta_a = (0+1)90^\circ = 90^\circ$$

For $k = 1$

$$\theta_a = (2+1)90^\circ = 270^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s))=0$$

$$G(s)H(s)=\frac{(s+3)}{s(s+1)(s+2)}$$

Let

$$\frac{(s^3+3s^2+2s)}{(s+3)}=\frac{u}{v}$$

$$\frac{vdu-udv}{v^2}=\frac{(s+3)(3s^2+6s+2)-(s^3+3s^2+2s)}{(s+3)^2}=0$$

$$(s+3)(3s^2+6s+2)-(s^3+3s^2+2s)=0$$

$$(3s^3+6s^2+2s+9s^2+18s+6)-(s^3+3s^2+2s)=0$$

$$(2s^3+12s^2+18s+6)=0$$

$$(s^3+6s^2+9s+3)=0$$

Therefore $s=-3.8794$ or -1.6527 or -0.4679 , but $s=-3.8794$ and -1.6527 are not part of the root locus, therefore the breakaway point is at $s=-0.4679$

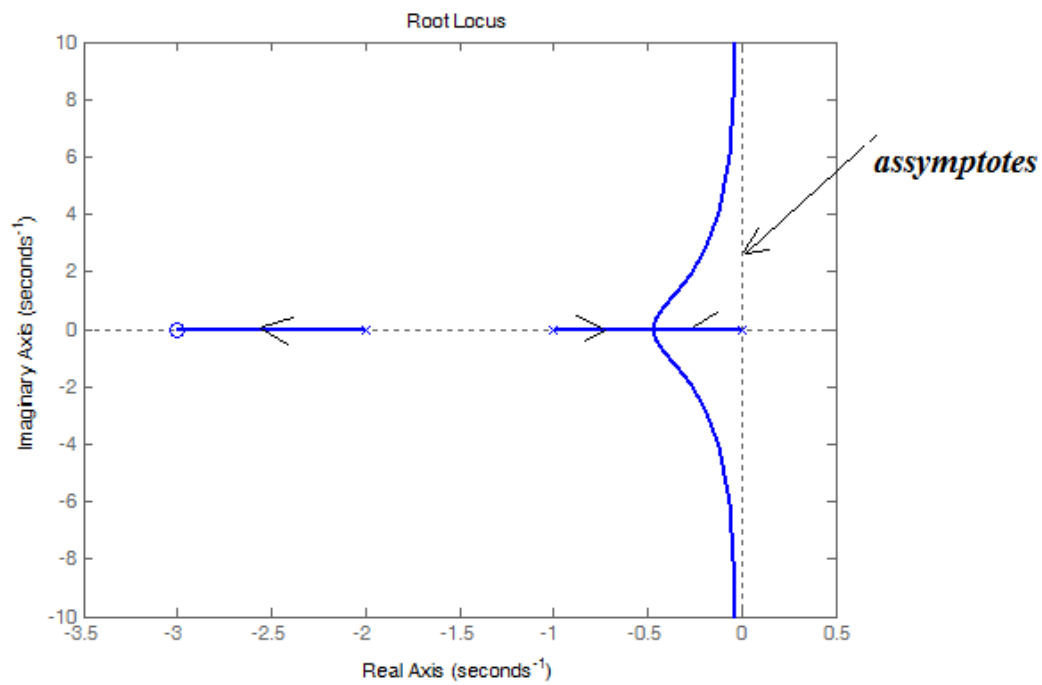
j ω - crossing

There is no j ω - crossing because the asymptotes are super imposed on the imaginary axis

Angles of Departure and Arrival

There are no angles of departure or arrival because there are no complex open loop poles or zeros.

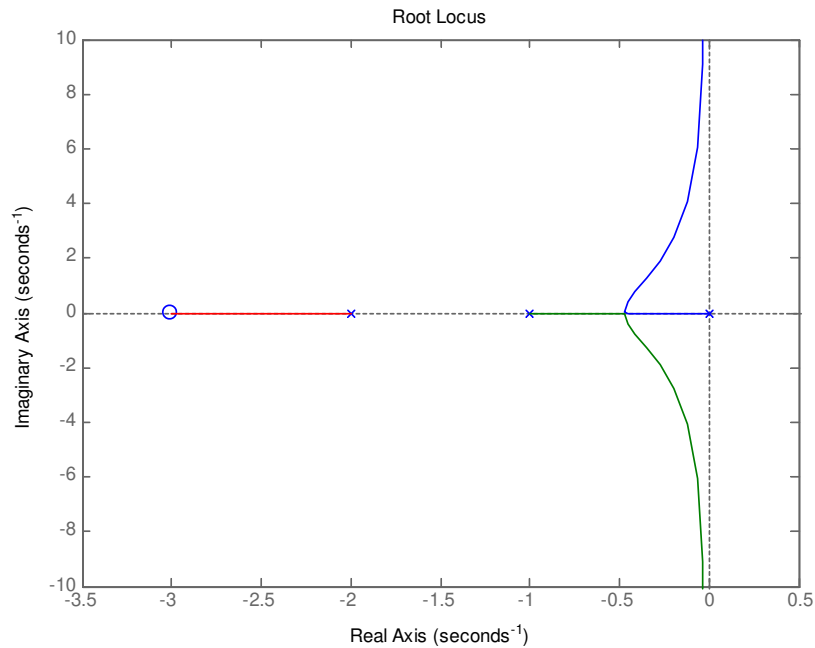
Using the above information the root locus is plotted as:



Matlab can be used to generate the root locus using the following code:

```
num=[1 3]
den=[1 3 2 0]
rlocus(num,den)
grid
```

Running any of the above codes gives the following:



Example 2

Obtain the root-locus for $K > 0$ for a system whose open-loop transfer function is given below.

$$KG(s)H(s) = \frac{K}{(s+1)(s+3)(s+4)} = \frac{K}{s^3 + 8s^2 + 19s + 12}$$

Solution

There are three finite open loop poles located at -1, -3 and -4.

There is no finite open loop zero.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 3 - 0 = 3

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((-1) + (-3) + (-4)) - (0)}{3 - 0} = \frac{-8}{3} = -2.67$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\text{finite poles} - \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{3-0} = \frac{(2k+1)180^\circ}{3} = (2k+1)60^\circ$$

There are three asymptotes, therefore $k = 0, 1, 2$

For $k = 0$

$$\theta_a = (0+1)60^\circ = 60^\circ$$

For $k = 1$

$$\theta_a = (2+1)60^\circ = 180^\circ$$

For $k = 2$

$$\theta_a = (4+1)60^\circ = 300^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{1}{s^3 + 8s^2 + 19s + 12}$$

Let

$$\frac{1}{s^3 + 8s^2 + 19s + 12} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^3 + 8s^2 + 19s + 12)(0) - (3s^2 + 16s + 19)}{(s^3 + 8s^2 + 19s + 12)^2} = 0$$

$$0 - (3s^2 + 16s + 19) = 0$$

$$3s^2 + 16s + 19 = 0$$

The roots are $s = -3.55$ or -1.78 , but $s = -3.55$ is not part of the root locus, therefore the breakaway point is at $s = -1.78$

j ω - crossing

$$1 + KG(s)H(s) = 0$$

$$1 + \frac{K}{s^3 + 8s^2 + 19s + 12} = 0$$

$$\frac{s^3 + 8s^2 + 19s + 12 + K}{s^3 + 8s^2 + 19s + 12} = 0$$

$$s^3 + 8s^2 + 19s + 12 + K = 0$$

Routh table

s^3	1	19
s^2	8	$12 + K$
s^1	$\frac{140 - K}{8}$	0
s^0	$12 + K$	

$$\text{At the limit of stability } \frac{140 - K}{8} = 0$$

Hence $K = 140$

The auxiliary equation is formed using the row above the zero row as:

$$8s^2 + 12 + K = 0$$

$$8s^2 + 12 + 140 = 0$$

$$8s^2 + 152 = 0$$

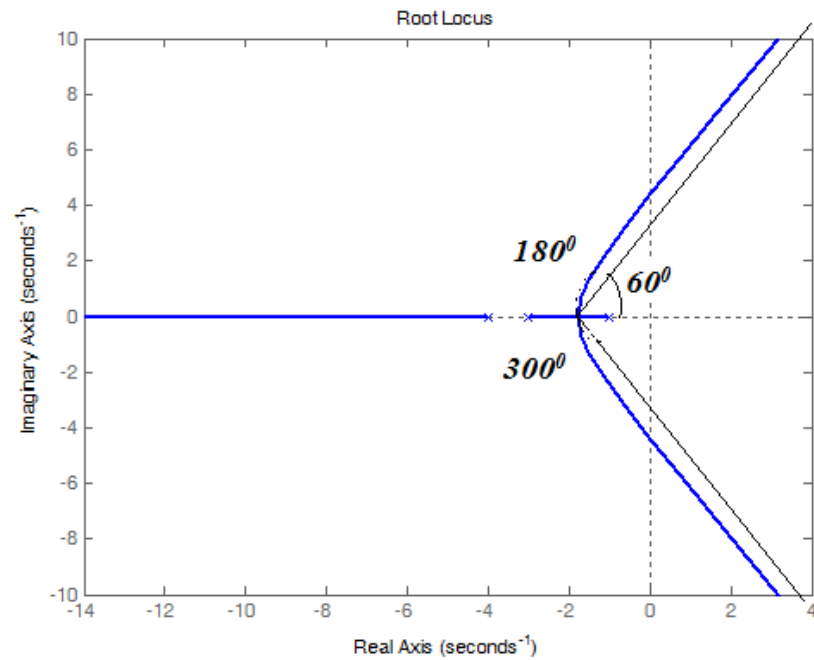
Solving for s in the auxiliary equation gives the *j ω -crossing* values as:

$$s = \pm \sqrt{\frac{-152}{8}} = \pm j 4.36$$

Angles of Departure and Arrival

There are no angles of departure or arrival because there are no complex open loop poles or zeros.

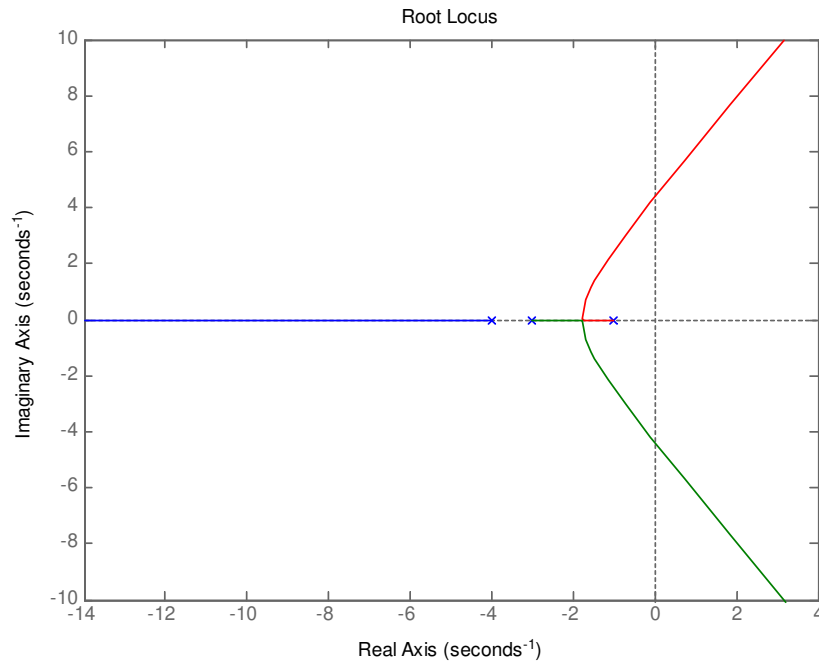
Using the above information the root locus is plotted as:



Matlab can be used to generate the root locus using the following code:

```
num=[1]
den=[1 8 19 12]
rlocus(num,den)
grid
```

Running any of the above codes gives the following:



Example 3

Obtain the root-locus for $K > 0$ for a system whose open-loop transfer function is given below.

$$KG(s)H(s) = \frac{K}{s^3 + 7s^2 + 19s + 13}$$

Solution

$$KG(s)H(s) = \frac{K}{(s+1)(s+3+j2)(s+3-j2)}$$

There are three finite open loop poles located at -1, -3+j2 and -3-j2.

There is no finite open loop zero.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 3 - 0 = 3

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((-1) + (-3 + j2) + (-3 - j2)) - (0)}{3 - 0} = \frac{-7}{3} = -2.33$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\textcolor{red}{j} \text{ finite poles} - \textcolor{red}{j} \text{ finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{3-0} = \frac{(2k+1)180^\circ}{3} = (2k+1)60^\circ$$

There are three asymptotes, therefore $k = 0, 1, 2$

For $k = 0$

$$\theta_a = (0+1)60^\circ = 60^\circ$$

For $k = 1$

$$\theta_a = (2+1)60^\circ = 180^\circ$$

For $k = 2$

$$\theta_a = (4+1)60^\circ = 300^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{1}{s^3 + 7s^2 + 19s + 13}$$

Let

$$\frac{1}{s^3 + 7s^2 + 19s + 13} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^3 + 7s^2 + 19s + 13)(0) - (3s^2 + 14s + 19)}{(s^3 + 7s^2 + 19s + 13)^2} = 0$$

$$0 - (3s^2 + 14s + 19) = 0$$

$$3s^2 + 14s + 19 = 0$$

The roots are $s = -2.3333 + j0.9428$ or $-2.3333 - j0.9428$, which shows no intersection with the real axis.

j ω - crossing

$$1 + KG(s)H(s) = 0$$

$$1 + \frac{K}{s^3 + 7s^2 + 19s + 13} = 0$$

$$\frac{s^3 + 7s^2 + 19s + 13 + K}{s^3 + 7s^2 + 19s + 13} = 0$$

$$s^3 + 7s^2 + 19s + 13 + K = 0$$

Routh table

s^3	1	19
s^2	7	$13 + K$
s^1	$\frac{120 - K}{7}$	0
s^0	$13 + K$	

$$\text{At the limit of stability } \frac{120 - K}{7} = 0$$

Hence $K = 120$

The auxiliary equation is formed using the row above the zero row as:

$$7s^2 + 13 + K = 0$$

$$7s^2 + 13 + 120 = 0$$

$$7s^2 + 133 = 0$$

Solving for s in the auxiliary equation gives the $j\omega$ -crossing values as:

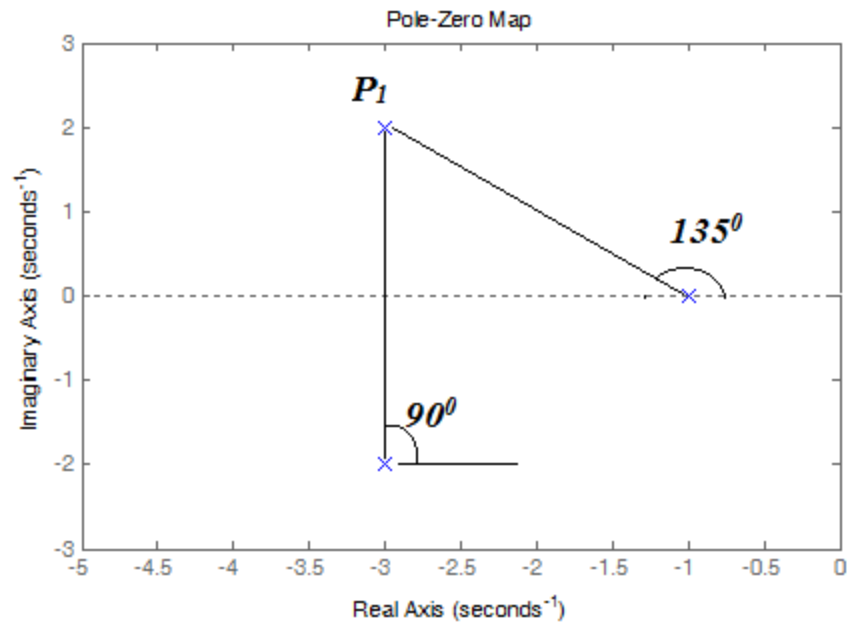
$$s = \pm \sqrt{\frac{-133}{7}} = \pm j4.36$$

Angles of Departure and Arrival

$$\theta_d = \sum (\text{angles of all zeros to } A) - \sum (\text{angles of all poles to } A) + 180^\circ$$

There are two open loop complex poles $P_1 = -3 + j2$ and $P_2 = -3 - j2$

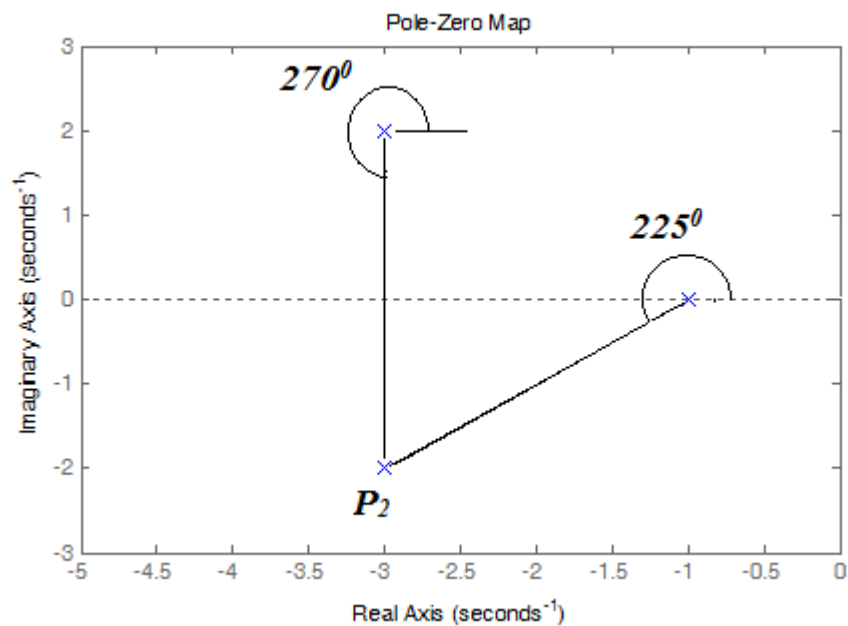
For P_1 , the angle of departure can be obtained as follows:



$$\theta_{d1} = \sum (\text{angles of all zeros to } P_1) - \sum (\text{angles of all poles to } P_1) + 180^\circ$$

$$\theta_{d1} = 0 - (90^\circ + 135^\circ) + 180^\circ = -45^\circ$$

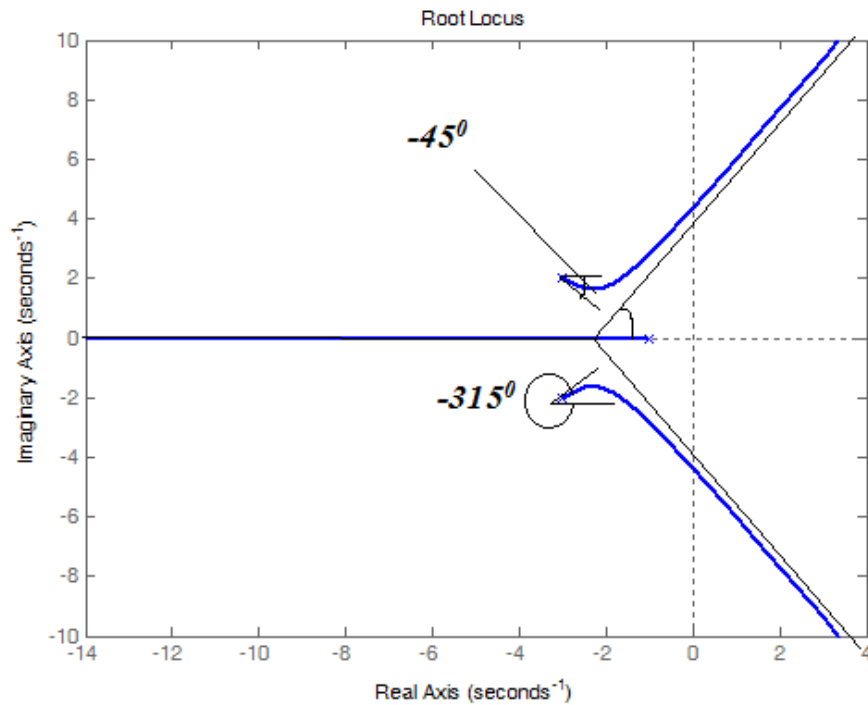
For P_2 , the angle of departure can be obtained as follows:



$$\theta_{d2} = \Sigma (\text{angles of all zeros to } P_2) - \Sigma (\text{angles of all poles to } P_2) + 180^\circ$$

$$\theta_{d2} = 0 - (270^\circ + 225^\circ) + 180^\circ = -315^\circ$$

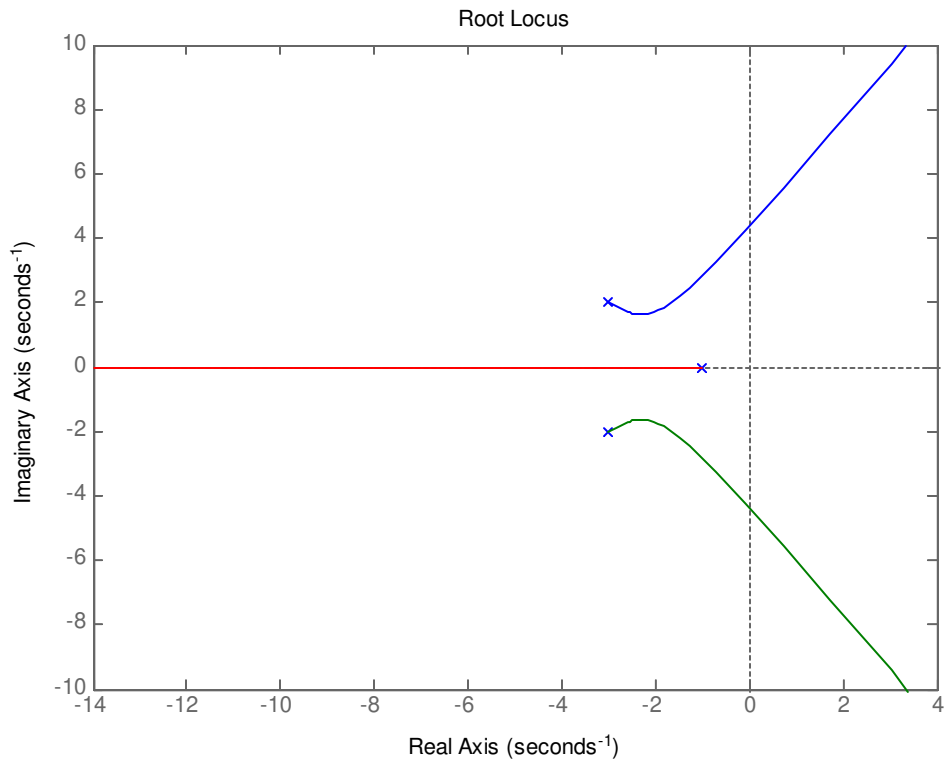
Using the above information the root locus is plotted as:



Matlab can be used to generate the root locus using the following code:

```
num=[1]
den=[1 7 19 13]
rlocus(num,den)
grid
```

Running any of the above codes gives the following:



Example 4

Obtain the root-locus for $K > 0$ for a system whose open-loop transfer function is given below.

$$KG(s)H(s) = \frac{K}{s^3 + 7s^2 + 16s + 10}$$

Solution

$$KG(s)H(s) = \frac{K}{(s+1)(s+3+j1)(s+3-j1)}$$

There are three finite open loop poles located at -1, -3+j1 and -3-j1.

There is no finite open loop zero.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 3-0=3

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{ finite poles} - \# \text{ finite zeros}}$$

$$\sigma_a = \frac{((-1) + (-3 + j1) + (-3 - j1)) - (0)}{3 - 0} = \frac{-7}{3} = -2.33$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\text{\textcolor{red}{j}} \text{ finite poles} - \text{\textcolor{red}{j}} \text{ finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{3-0} = \frac{(2k+1)180^\circ}{3} = (2k+1)60^\circ$$

There are three asymptotes, therefore $k = 0, 1, 2$

For $k=0$

$$\theta_a = (0+1)60^\circ = 60^\circ$$

For $k=1$

$$\theta_a = (2+1)60^\circ = 180^\circ$$

For $k=2$

$$\theta_a = (4+1)60^\circ = 300^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{1}{s^3 + 7s^2 + 16s + 10}$$

Let

$$\frac{1}{s^3 + 7s^2 + 16s + 10} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^3 + 7s^2 + 16s + 10)(0) - (3s^2 + 14s + 16)}{(s^3 + 7s^2 + 16s + 10)^2} = 0$$

$$0 - (3s^2 + 14s + 16) = 0$$

$$3s^2 + 14s + 16 = 0$$

The roots are $s = -2.67$ or -2 , Both points lie on the root-locus for $K > 0$; they are the breakaway point and break-in point.

j ω - crossing

$$1 + KG(s)H(s) = 0$$

$$1 + \frac{K}{s^3 + 7s^2 + 16s + 10} = 0$$

$$\frac{s^3 + 7s^2 + 16s + 10 + K}{s^3 + 7s^2 + 16s + 10} = 0$$

$$s^3 + 7s^2 + 16s + 10 + K = 0$$

Routh table

s^3	1	16
s^2	7	$10 + K$
s^1	$\frac{102 - K}{7}$	0
s^0	$10 + K$	

$$\text{At the limit of stability } \frac{102 - K}{7} = 0$$

Hence $K = 102$

The auxiliary equation is formed using the row above the zero row as:

$$7s^2 + 10 + K = 0$$

$$7s^2 + 10 + 102 = 0$$

$$7s^2 + 112 = 0$$

Solving for s in the auxiliary equation gives the *j ω -crossing* values as:

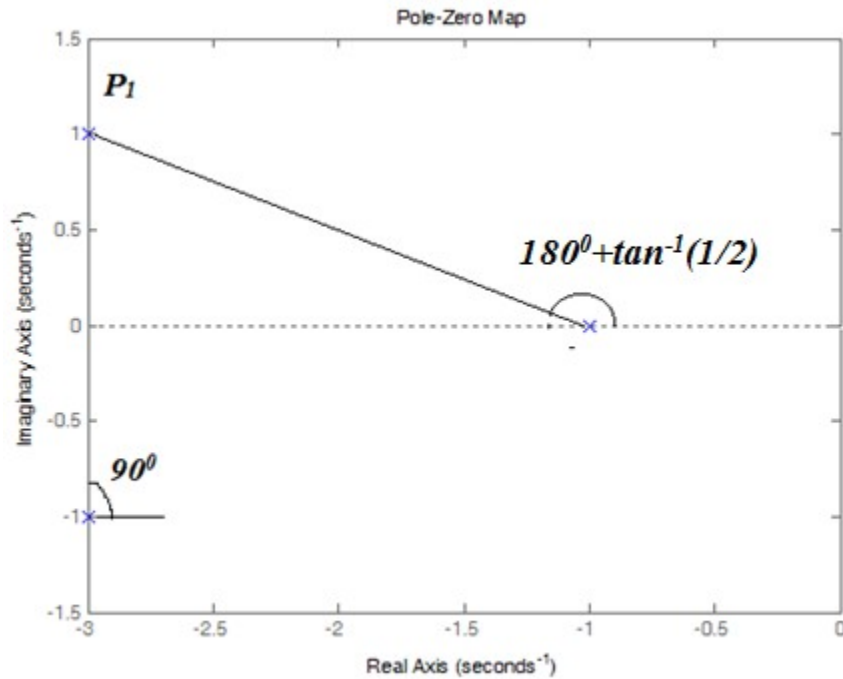
$$s = \pm \sqrt{\frac{-112}{7}} = \pm j4$$

Angles of Departure and Arrival

$$\theta_d = \sum (\text{angles of all zeros to } A) - \sum (\text{angles of all poles to } A) + 180^\circ$$

There are two open loop complex poles $P_1 = -3 + j1$ and $P_2 = -3 - j1$

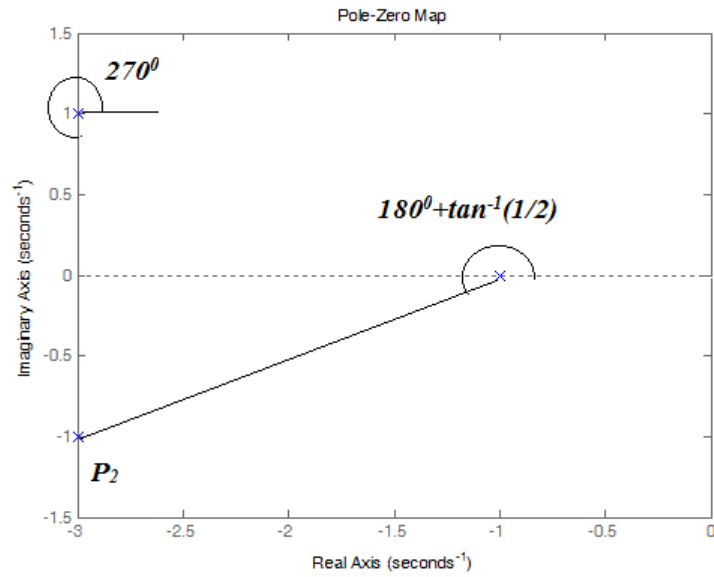
For P_1 , the angle of departure can be obtained as follows:



$$\theta_{d1} = \sum (\text{angles of all zeros to } P_1) - \sum (\text{angles of all poles to } P_1) + 180^\circ$$

$$\theta_{d1} = 0 - \left(90^\circ + 180^\circ - \tan^{-1}\left(\frac{1}{2}\right) \right) + 180^\circ = -63.43^\circ$$

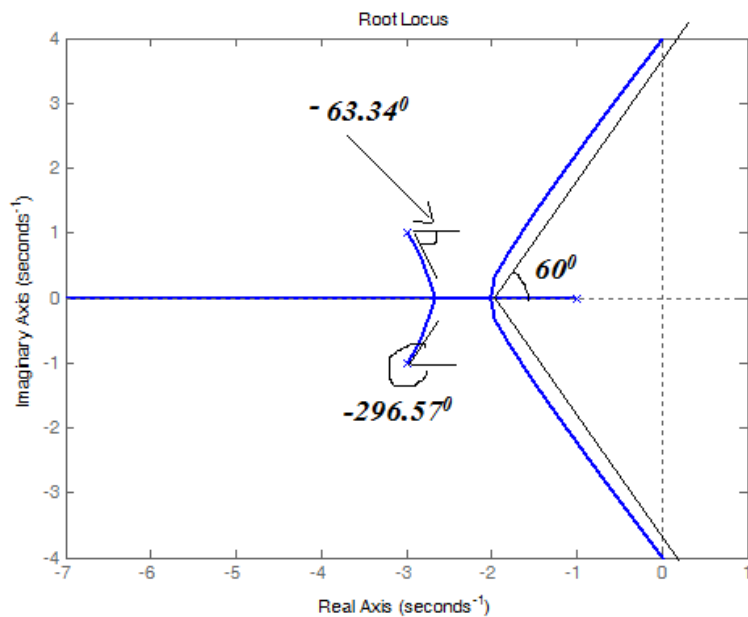
For P_2 , the angle of departure can be obtained as follows:



$$\theta_{d2} = \sum (\text{angles of all zeros to } P_2) - \sum (\text{angles of all poles to } P_2) + 180^\circ$$

$$\theta_{d2} = 0 - \left(270^\circ + 180^\circ + \tan^{-1}\left(\frac{1}{2}\right) \right) + 180^\circ = -296.57^\circ$$

Using the above information the root locus is plotted as:

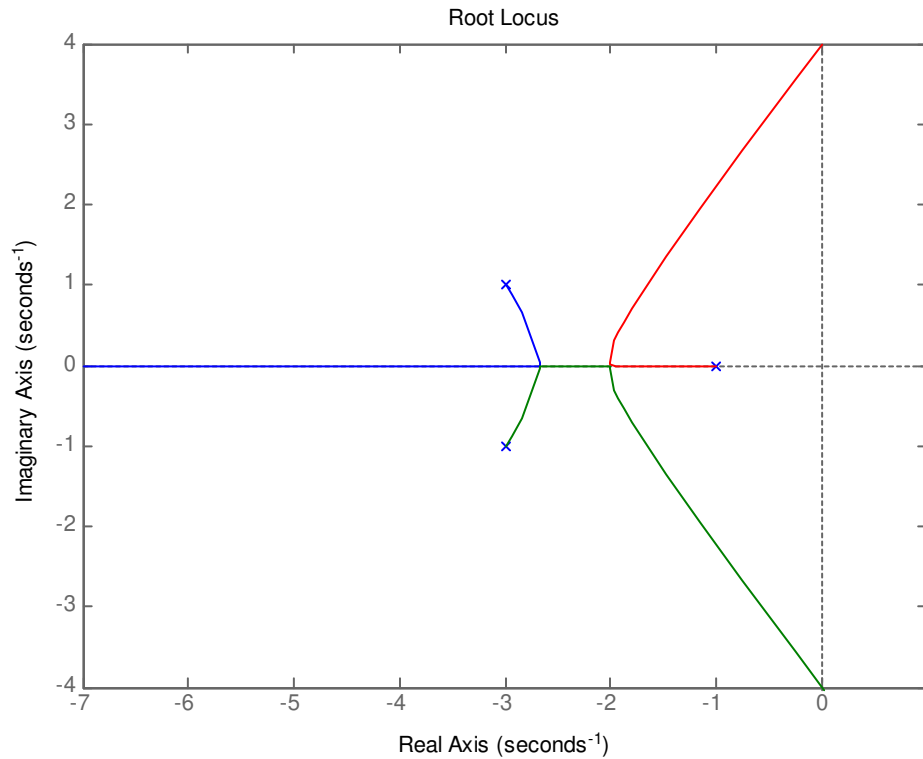


Matlab can be used to generate the root locus using the following code:

```
num=[1]
den=[1 7 19 13]
```

rlocus(num,den)
grid

Running any of the above codes gives the following:



Example 5

Obtain the root-locus for $K > 0$ for a system whose open-loop transfer function is given below.

$$KG(s)H(s) = \frac{K(s+5)}{s^2+4s+3}$$

Solution

$$KG(s)H(s) = \frac{K(s+5)}{s^2+4s+3} = \frac{K(s+5)}{(s+1)(s+3)}$$

There are two finite open loop poles located at -1 and -3.

There is one finite open loop zero located at -5.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 2-1=1

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((-1) + (-3)) - (-5)}{2 - 1} = \frac{1}{1} = 1$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\text{\textcolor{red}{i}} \text{finite poles} - \text{\textcolor{red}{i}} \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{2-1} = \frac{(2k+1)180^\circ}{1} = (2k+1)180^\circ$$

There is one asymptote, therefore $k=0$

For $k=0$

$$\theta_a = (0+1)180^\circ = 180^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{(s+5)}{s^2+4s+3}$$

Let

$$\frac{(s+5)}{s^2+4s+3} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^2+4s+3)(1) - (s+5)(2s+4)}{(s^2+4s+3)^2} = 0$$

$$(s^2+4s+3) - (2s^2+14s+20) = 0$$

$$s^2 + 10s + 17 = 0$$

The roots are $s = -2.17$ or -7.83 , Both points lie on the root-locus for $K > 0$; they are the breakaway point and break-in point respectively.

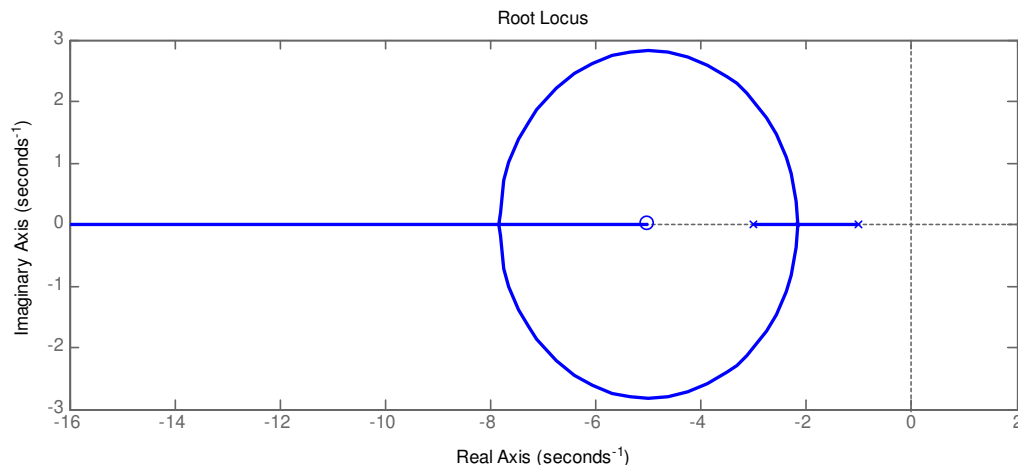
j ω - crossing

There is no j ω - crossing

Angles of Departure and Arrival

There are no angles of departure or arrival because there are no complex open loop poles or zeros.

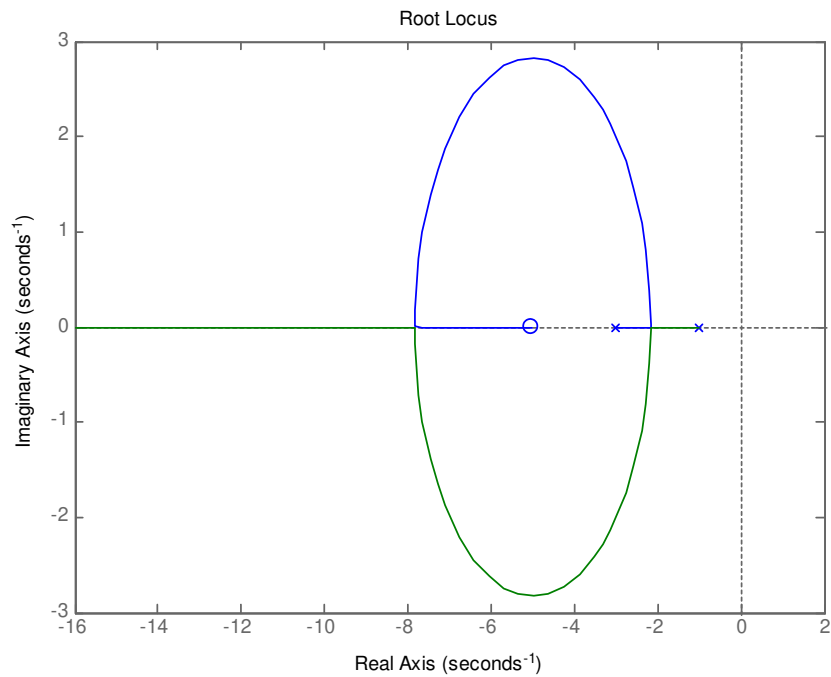
Using the above information the root locus is plotted as:



Matlab can be used to generate the root locus using the following code:

```
num=[1 5]
den=[1 4 3]
rlocus(num,den)
grid
```

Running any of the above codes gives the following:



Example 6

Obtain the root-locus for $K > 0$ for a system whose open-loop transfer function is given below.

$$KG(s)H(s) = \frac{K(s+2)}{s^3 + 10s^2 + 27s + 18}$$

Solution

$$KG(s)H(s) = \frac{K(s+2)}{s^3 + 10s^2 + 27s + 18} = \frac{K(s+2)}{(s+1)(s+3)(s+6)}$$

There are three finite open loop poles located at -1, -3 and -6.

There is one finite open loop zero located at -2.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 3 - 1 = 2

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((-1)+(-3)+(-6))-(-2)}{3-1} = \frac{-8}{2} = -4$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\text{finite poles} - \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{3-1} = \frac{(2k+1)180^\circ}{2} = (2k+1)90^\circ$$

There are two asymptotes, therefore $k = 0, 1$

For $k = 0$

$$\theta_a = (0+1)90^\circ = 90^\circ$$

For $k = 1$

$$\theta_a = (2+1)90^\circ = 270^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{(s+2)}{s^3+10s^2+27s+18}$$

Let

$$\frac{(s+2)}{s^3+10s^2+27s+18} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^3+10s^2+27s+18)(1) - (s+2)(3s^2+20s+27)}{(s^3+10s^2+27s+18)^2} = 0$$

$$(s^3+10s^2+27s+18) - (3s^3+26s^2+67s+54) = 0$$

$$2s^3+16s^2+40s+36=0$$

The roots are $s = -4.36$ or $-1.82 - j0.9$ or $-1.82 + j0.9$. $s = -4.36$ lies on the root-locus for $K > 0$, therefore this is the breakaway point.

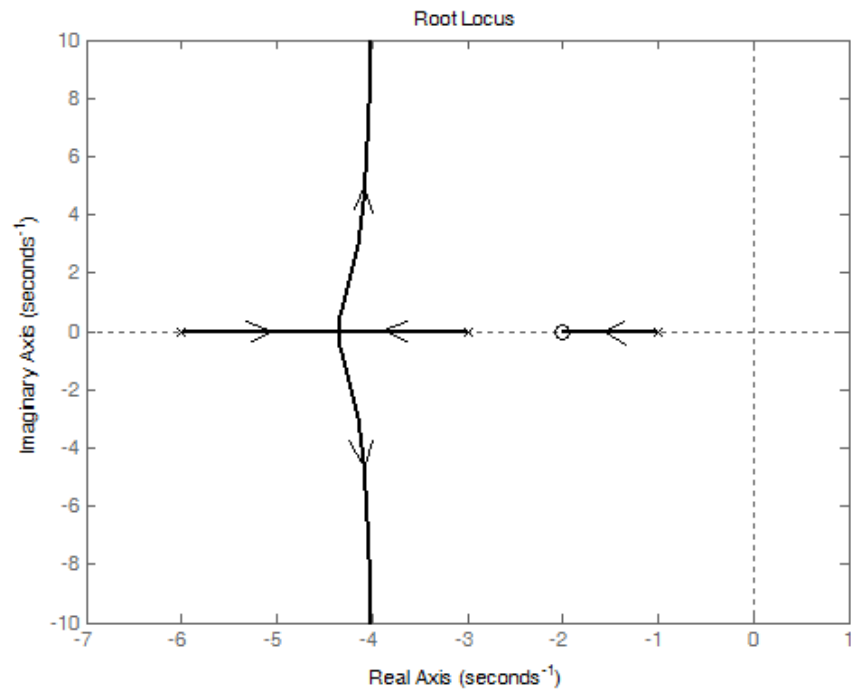
$j\omega$ - crossing

There is no $j\omega$ - crossing

Angles of Departure and Arrival

There are no angles of departure or arrival because there are no complex open loop poles or zeros.

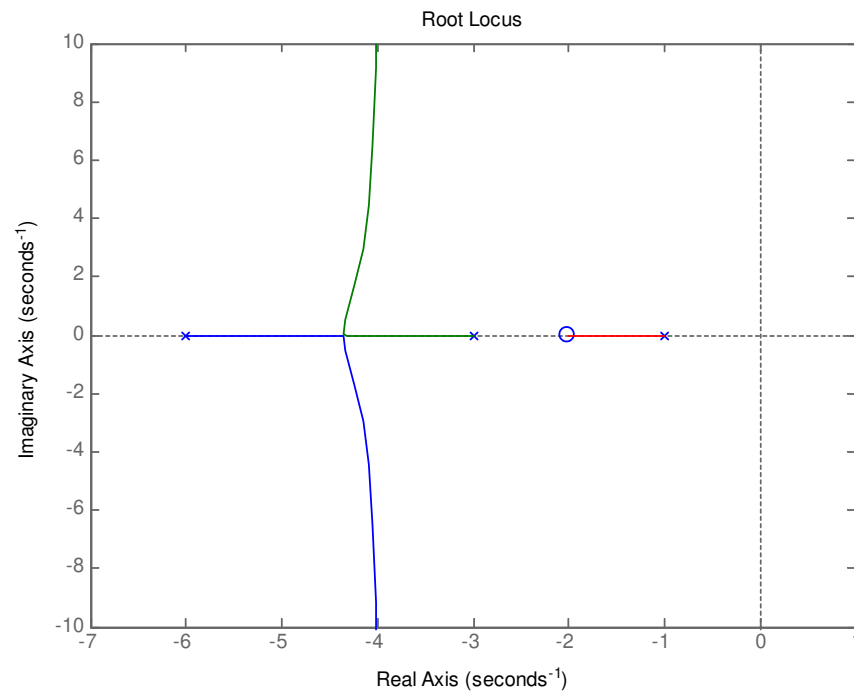
Using the above information the root locus is plotted as:



Matlab can be used to generate the root locus using the following code:

```
num=[1 2]  
den=[1 10 27 18]  
rlocus(num,den)  
grid
```

Running any of the above codes gives the following:



Example 7

Obtain the root-locus for $K > 0$ for a system whose open-loop transfer function is given below.

$$KG(s)H(s) = \frac{K(s^2 + 2s + 2)}{s^3 - 2s + 4} = \frac{K(s+1-j1)(s+1+j1)}{(s+2)(s-1-j1)(s-1+j1)}$$

Solution

There are three finite open loop poles located at -2, $1+j1$ and $1-j1$.

There are two finite open loop zero located at $-1+j1$ and $-1-j1$.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = $3-2=1$

only one branch going to infinity

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\text{finite poles} - \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{3-2} = \frac{(2k+1)180^\circ}{1} = (2k+1)180^\circ$$

There is one asymptotes, therefore $k=0$

For $k=0$

$$\theta_a = (0+1)180^\circ = 180^\circ$$

Real-axis Breakaway and Break-in Points

There are no Real-axis Breakaway and Break-in Points

$j\omega$ - crossing

$$1 + KG(s)H(s) = 0$$

$$1 + \frac{K(s^2 + 2s + 2)}{s^3 - 2s + 4} = 0$$

$$\frac{s^3 - 2s + 4 + K(s^2 + 2s + 2)}{s^3 - 2s + 4} = 0$$

$$s^3 - 2s + 4 + K(s^2 + 2s + 2) = 0$$

$$s^3 + Ks^2 + (2K - 2)s + 2K + 4 = 0$$

Routh table

s^3	1	$2K - 2$
s^2	K	$2K + 4$
s^1	$\frac{2K^2 - 4K - 4}{K}$	0
s^0	$2K + 4$	

$$\text{At the limit of stability } \frac{2K^2 - 4K - 4}{K} = 0$$

$$K^2 - 2K - 2 = 0$$

Hence $K = 2.73$

The auxiliary equation is formed using the row above the zero row as:

$$Ks^2 + 2K + 4 = 0$$

$$2.73 s^2 + 5.46s + 4 = 0$$

$$2.73 s^2 + 9.46 = 0$$

Solving for s in the auxiliary equation gives the $j\omega$ -crossing values as:

$$s = \pm \sqrt{\frac{-9.46}{2.73}} = \pm j 1.86$$

Angles of Departure and Arrival

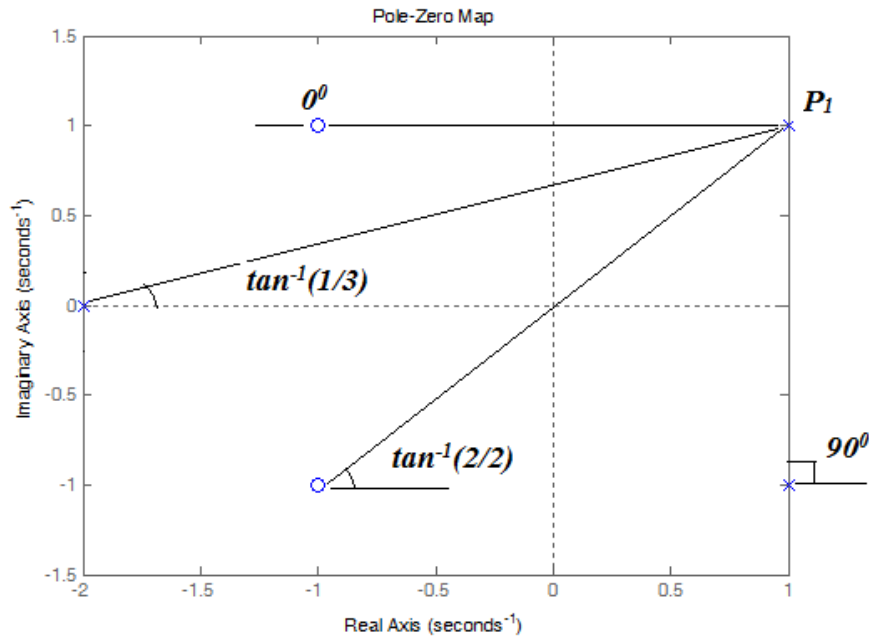
There are two open loop complex poles $P_1 = 1 + j1$ and $P_2 = 1 - j1$

There are also two open loop complex zeros $Z_1 = -1 + j1$ and $Z_2 = -1 - j1$

Angles of Departure

$$\theta_d = \sum (\text{angles of all zeros to } A) - \sum (\text{angles of all poles to } A) + 180^\circ$$

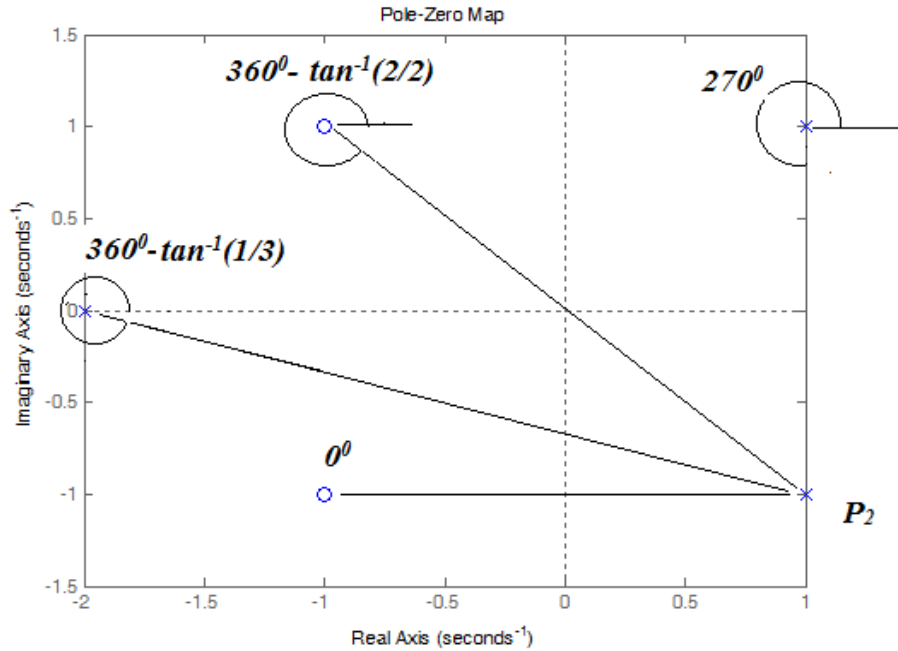
For P_1 , the angle of departure can be obtained as follows:



$$\theta_{d1} = \sum (\text{angles of all zeros to } P_1) - \sum (\text{angles of all poles to } P_1) + 180^\circ$$

$$\theta_{d1} = \left(0^\circ + \tan^{-1}\left(\frac{2}{2}\right) \right) - \left(90^\circ + \tan^{-1}\left(\frac{1}{3}\right) \right) + 180^\circ = 116.57^\circ$$

For P_2 , the angle of departure can be obtained as follows:



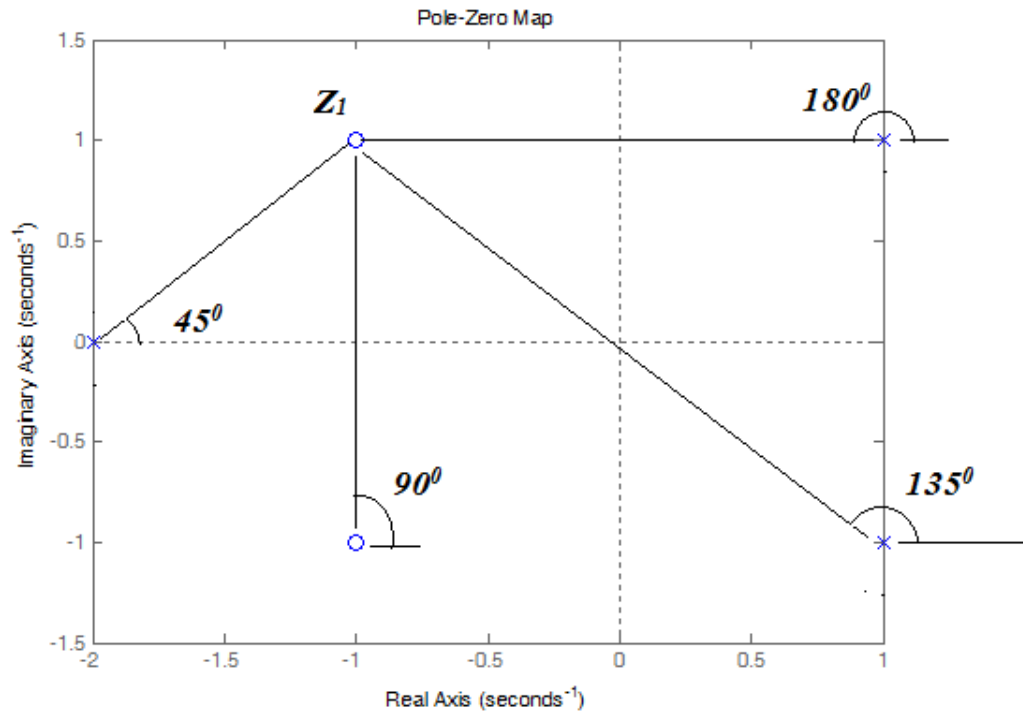
$$\theta_{d2} = \sum (\text{angles of all zeros to } P_2) - \sum (\text{angles of all poles to } P_2) + 180^\circ$$

$$\theta_{d2} = \left(0^\circ + 360^\circ - \tan^{-1}\left(\frac{2}{2}\right) \right) - \left(270^\circ + 360^\circ - \tan^{-1}\left(\frac{1}{3}\right) \right) + 180^\circ = -116.57^\circ$$

Angles of Arrival

$$\theta_a = \sum (\text{angles of all poles to } B) - \sum (\text{angles of all zeros to } B) + 180^\circ$$

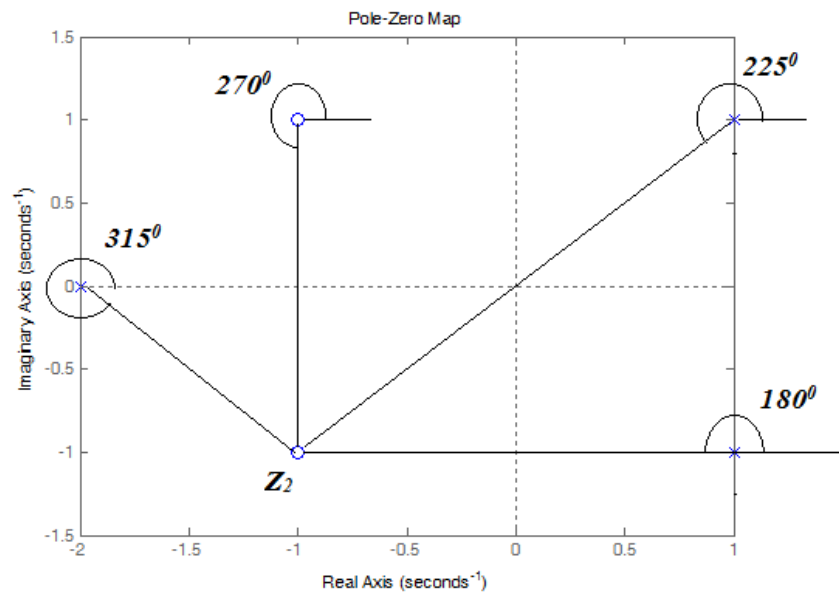
For Z_1 , the angle of departure can be obtained as follows:



$$\theta_{a1} = \sum (\text{angles of all poles to } Z_1) - \sum (\text{angles of all zeros to } Z_1) + 180^\circ$$

$$\theta_{a1} = (180^\circ + 45^\circ + 135^\circ) - (90^\circ) + 180^\circ = 450^\circ = 90^\circ$$

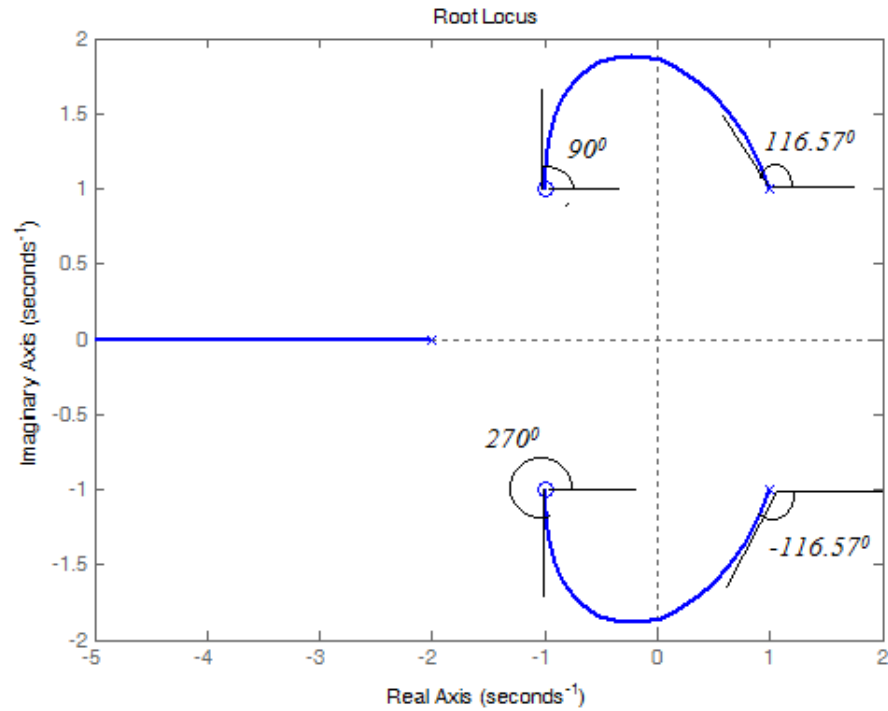
For Z_2 , the angle of departure can be obtained as follows:



$$\theta_{a2} = \sum (\text{angles of all poles to } Z_2) - \sum (\text{angles of all zeros to } Z_2) + 180^\circ$$

$$\theta_{a2} = (180^\circ + 225^\circ + 315^\circ) - (270^\circ) + 180^\circ = 630^\circ = 270^\circ$$

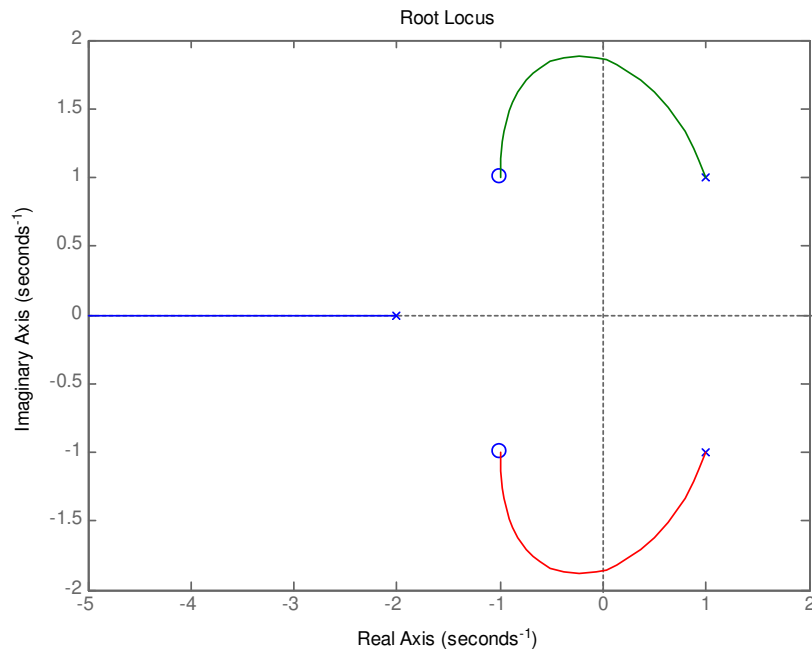
Using the above information the root locus is plotted as:



Matlab can be used to generate the root locus using the following code:

```
num=[1 2 2]
den=[1 0 -2 4]
rlocus(num,den)
grid
```

Running any of the above codes gives the following:



Example 8

Obtain the root-locus for $K > 0$ for a system whose open-loop transfer function is given below.

$$KG(s)H(s) = \frac{K(s^2 + 5s + 6)}{s^4 + 12s^3 + 41s^2 + 30s} = \frac{K(s+2)(s+3)}{s(s+1)(s+5)(s+6)}$$

Solution

There are four finite open loop poles located at 0, -1, -5 and -6.

There are two finite open loop zeros located at -2 and -3.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 4 - 2 = 2

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((0) + (-1) + (-5) + (-6)) - ((-2) + (-3))}{4 - 2} = \frac{-7}{2} = -3.5$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\text{finite poles} - \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{4-2} = \frac{(2k+1)180^\circ}{2} = (2k+1)90^\circ$$

There are two asymptotes, therefore $k = 0, 1$

For $k = 0$

$$\theta_a = (0+1)90^\circ = 90^\circ$$

For $k = 1$

$$\theta_a = (2+1)90^\circ = 270^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{(s^2+5s+6)}{s^4+12s^3+41s^2+30s}$$

Let

$$\frac{(s^2+5s+6)}{s^4+12s^3+41s^2+30s} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^4+12s^3+41s^2+30s)(2s+5) - (s^2+5s+6)(4s^3+36s^2+82s+30)}{(s^4+12s^3+41s^2+30s)^2} = 0$$

$$(2s^5+5s^4+24s^4+60s^3+82s^3+205s^2+60s^2+150s) - (4s^5+36s^4+82s^3+30s^2+20s^4+180s^3+410s^2+150s+24s^3)$$

$$(2s^5+29s^4+142s^3+265s^2+150s) - (4s^5+56s^4+286s^3+656s^2+642s+180) = 0$$

$$2s^5+27s^4+144s^3+391s^2+492s+180=0$$

The roots are $s = -0.59$ or $-2.50 - j2.29$ or $-2.50 + j2.29$ or -5.46 or -2.45 . The real roots of this equation are $-0.59, -5.46$ which are the breakaway points and -2.45 which is the break-in point.

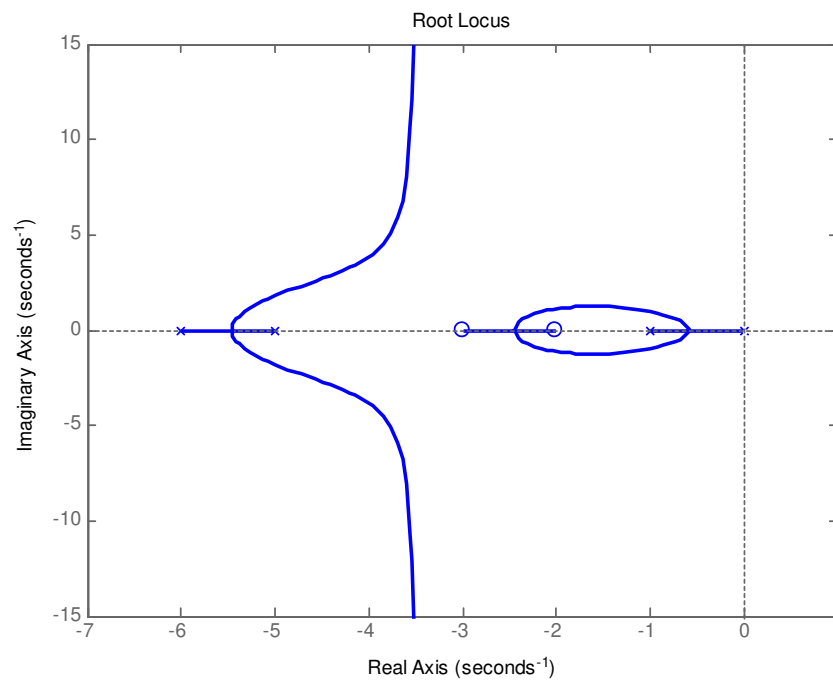
$j\omega$ - crossing

There is no $j\omega$ - crossing

Angles of Departure and Arrival

There are no Angles of Departure and Arrival

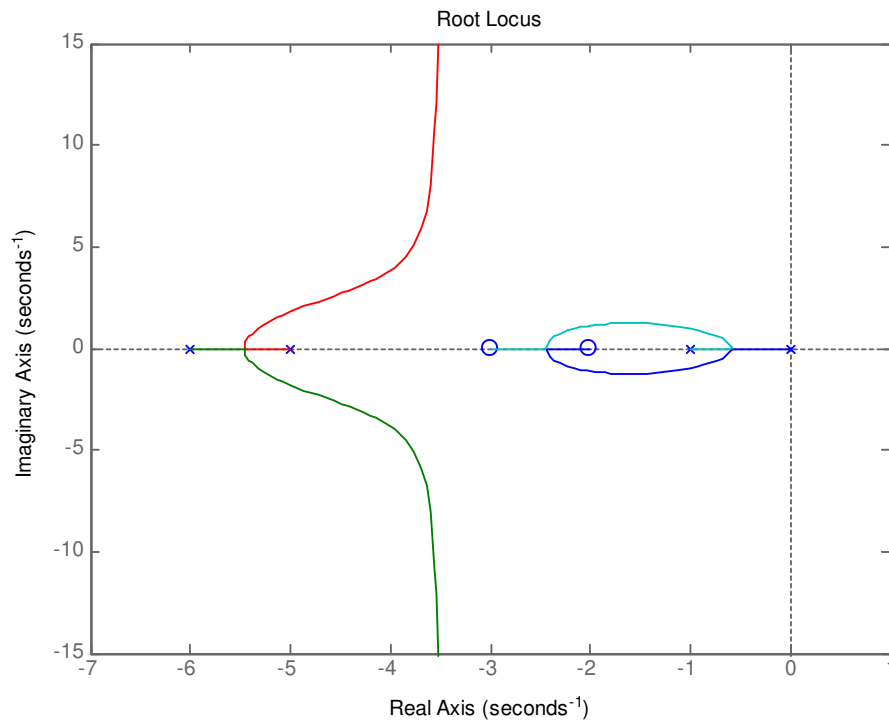
Using the above information the root locus is plotted as:



Matlab can be used to generate the root locus using the following code:

```
num=[1 5 6]  
den=[1 12 41 30 0]  
rlocus(num,den)  
grid
```

Running any of the above codes gives the following:



Determination of the existing of closed loop pole on the root locus

For a closed loop pole to exist on the root locus, the following condition must be satisfied:

$$\sum \text{angles of zeros of } G(s)H(s) - \sum \text{angles of poles of } G(s)H(s) = r(180)^\circ$$

Where $r = \pm 1, \pm 3, \pm 5, \dots$

$$K = \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}}$$

Example 1

Given a unity feedback that has a forward transfer function

$$G(s) = \frac{K(s+2)}{(s^2+4s+13)} = \frac{K(s+2)}{(s+2+j3)(s+2-j3)}$$

Determine if a point, $-3 + j0$ is on the root locus. If the point is on the root locus, find the gain, K that satisfies the requirement.

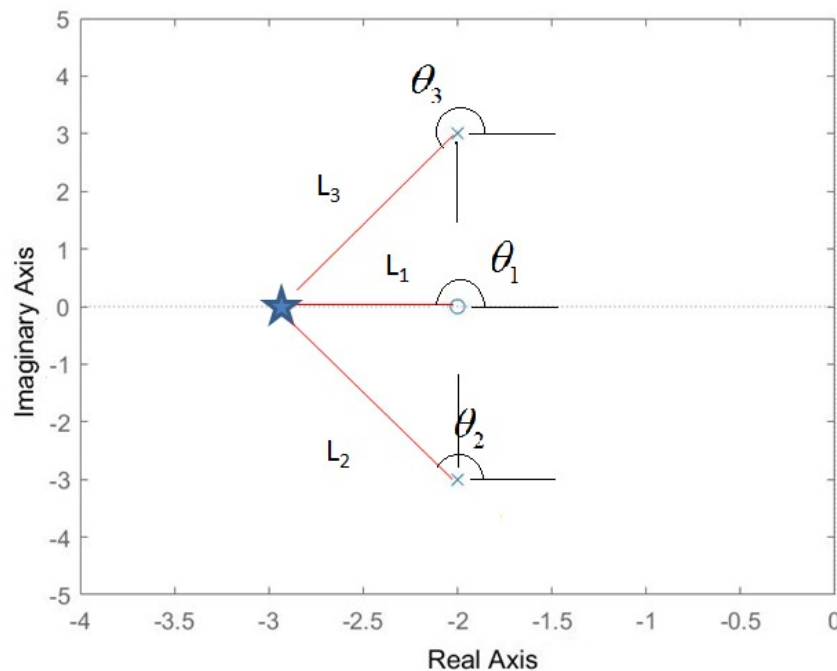
Solution

It is given that $G(s) = \frac{K(s+2)}{(s+2+j3)(s+2-j3)}$, $H(s) = 1$

$$G(s)H(s) = \frac{K(s+2)}{(s+2+j3)(s+2-j3)}$$

There is one zero at $s = -2$ and two poles at $s = -2 - j3$ and $s = -2 + j3$

The poles and zeros can be mapped on the s-plane as shown below:



For the point to be on the root locus the following condition must be satisfied:

$$\sum \text{angles of zeros of } G(s)H(s) - \sum \text{angles of poles of } G(s)H(s) = r(180)^\circ$$

$$\sum \text{angles of zeros of } G(s)H(s) - \sum \text{angles of poles of } G(s)H(s)$$

$$\angle \theta_1 - (\theta_2 + \theta_3) = 180^\circ - \left(\left(90^\circ + \tan^{-1} \frac{1}{3} \right) + \left(270^\circ - \tan^{-1} \frac{1}{3} \right) \right)$$

$$\angle 180^\circ - ((90^\circ + 18.43^\circ) + (270^\circ - 18.43^\circ))$$

$$\angle 180^\circ - (108.43^\circ + 251.57^\circ)$$

$$\angle 180^\circ - 360^\circ = -180^\circ$$

Hence the point is on the root locus since

$$\sum \text{angles of zeros of } G(s)H(s) - \sum \text{angles of poles of } G(s)H(s) = -180^\circ$$

The gain, K that satisfies the requirement can be obtained as follows:

$$K = \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = \frac{L_2 L_3}{L_1} = \frac{\sqrt{3^2+1^2} \times \sqrt{3^2+1^2}}{1} = 10$$

Hence $K = 10$

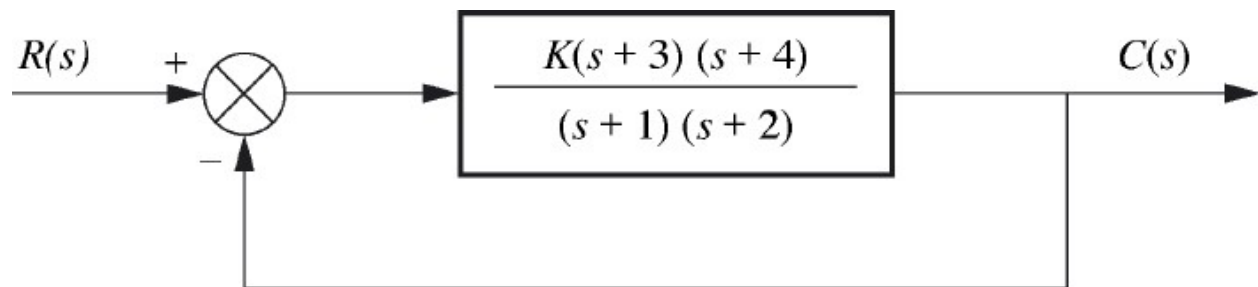
Solution of example 1 Matlab

Example 1 can be solve using the following Matlab code:

```
s=-3+0j
GH=(s+2)/(s^2+4*s+13)
Theta=(180/pi)*angle(GH)
M=abs(GH)
K=1/M
```

Example 2

Consider the closed loop system



Examine whether the point, $s = -2 + j3$ is a closed loop pole of the system.

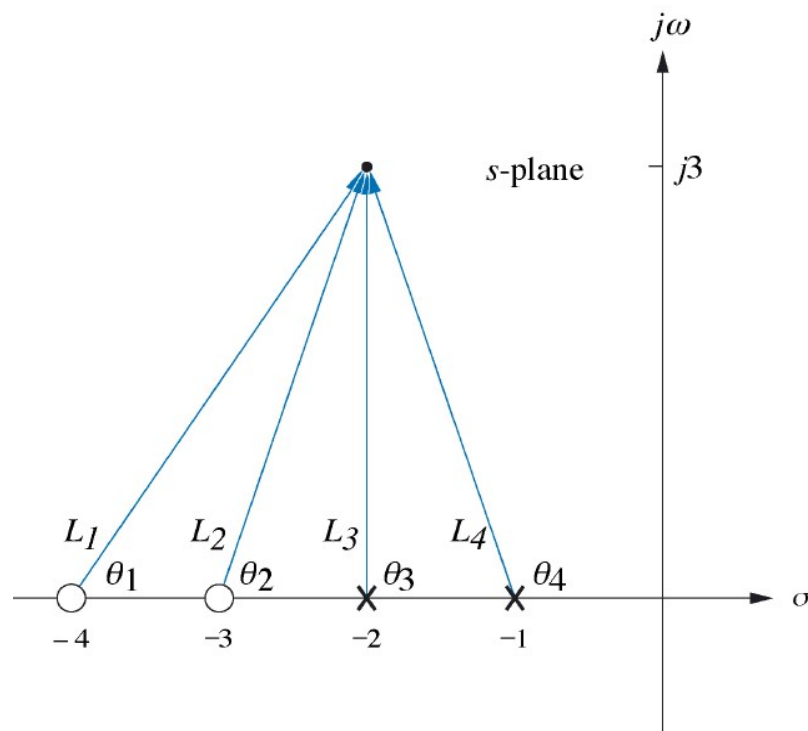
Solution

It is given that $G(s) = \frac{K(s+3)(s+4)}{(s+1)(s+2)}$, $H(s) = 1$

$$G(s)H(s) = \frac{K(s+3)(s+4)}{(s+1)(s+2)}$$

There are two zero at $s = -3$ and $s = -4$ and two poles at $s = -1$ and $s = -2$

The poles and zeros can be mapped on the s-plane as shown below:



For the point to be on the root locus the following condition must be satisfied:

$$\sum \text{angles of zeros of } G(s)H(s) - \sum \text{angles of poles of } G(s)H(s) = r(180)^\circ$$

$$\sum \text{angles of zeros of } G(s)H(s) - \sum \text{angles of poles of } G(s)H(s)$$

$$\angle(\theta_1 + \theta_2) - (\theta_3 + \theta_4) = \left(\tan^{-1} \frac{3}{2} + \tan^{-1} \frac{3}{1} \right) - \left(90^\circ + \left(180^\circ - \tan^{-1} \frac{3}{1} \right) \right)$$

$$\angle(56.31^\circ + 71.57^\circ) - (90^\circ + (180^\circ - 71.57^\circ))$$

$$\angle 127.88^\circ - (90^\circ + 108.43^\circ)$$

$$\angle -70.55^\circ$$

Hence the point is **not** on the root locus since

$$\sum \text{angles of zeros of } G(s)H(s) - \sum \text{angles of poles of } G(s)H(s) \neq r(180)^\circ$$

Solution of example 2 Matlab

Example 2 can be solve using the following Matlab code:

$$\begin{aligned}
 s &= -2 + 3j \\
 GH &= (s+3) * (s+4) / ((s+1) * (s+2)) \\
 \text{Theta} &= (180/\pi) * \text{angle}(GH) \\
 M &= \text{abs}(GH) \\
 K &= 1/M
 \end{aligned}$$

Example 3

Given a unity feedback that has a forward transfer function

$$G(s) = \frac{K(s+2)(s^2+2s+17)}{(s+10)(s^2+9)}$$

Determine the value of gain, K to be used if one of the CL poles must be at $s = -9$.

Solution

It is given that $G(s) = \frac{K(s+2)(s^2+2s+17)}{(s+10)(s^2+9)}$, $H(s) = 1$

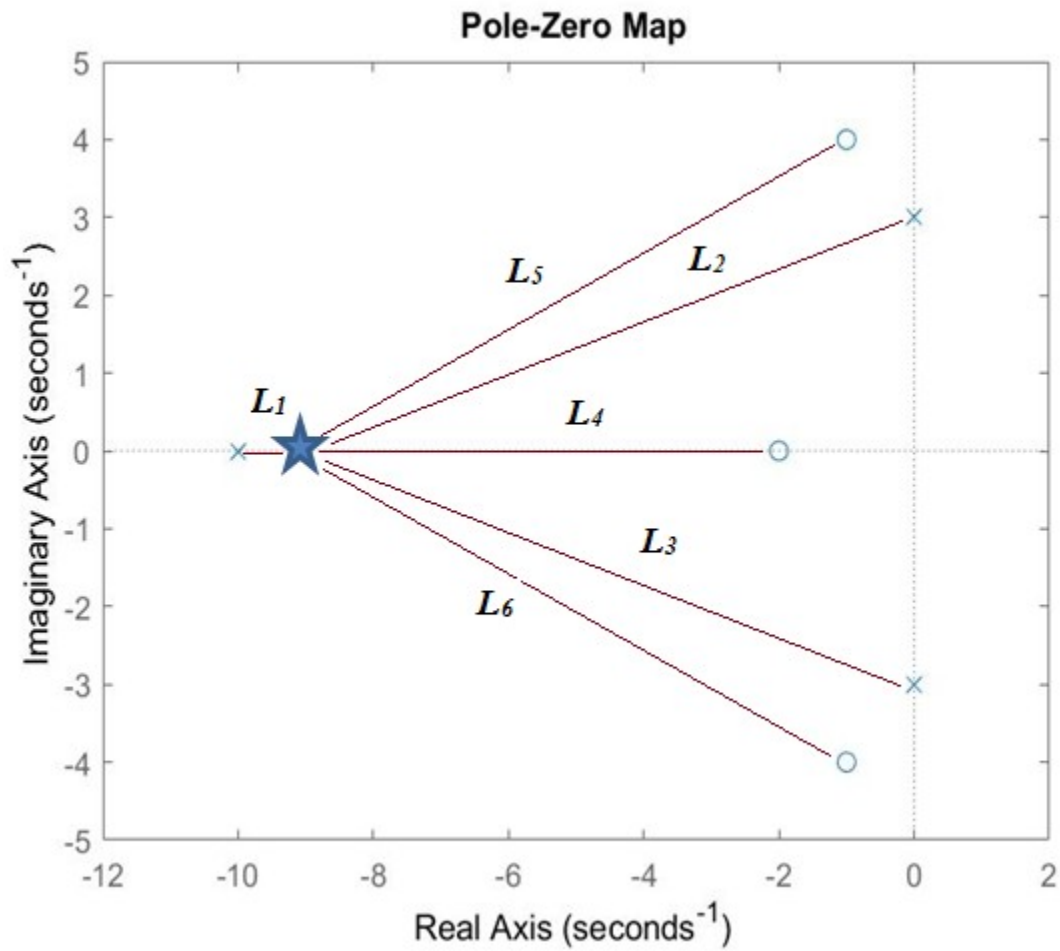
$$G(s)H(s) = \frac{K(s+2)(s^2+2s+17)}{(s+10)(s^2+9)}$$

There are three zero at $s = -2$ and

$$s_{1,2} = \frac{-2 \pm \sqrt{4 - 4 \times 17}}{2} = \frac{-2 \pm \sqrt{-64}}{2} = \frac{-2 \pm j8}{2} = -1 \pm j4$$

There are three poles at $s = -10$, $s = j3$ and $s = -j3$

The poles and zeros can be mapped on the s-plane as shown below:



$$K = \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = \frac{L_1 L_2 L_3}{L_4 L_5 L_6} = \frac{1 \times \sqrt{9^2 + 3^2} \times \sqrt{9^2 + 3^2}}{7 \times \sqrt{8^2 + 4^2} \times \sqrt{8^2 + 4^2}} = \frac{1 \times 9^2 + 3^2}{7 \times 8^2 + 4^2} = \frac{81 + 9}{7 \times (64 + 16)} =$$

Solution of example 3 Matlab

Example 3 can be solve using the following Matlab code:

```
s = -9
GH = ((s+2)*(s^2+2*s+17))/((s+10)*(s^2+9))
Theta = (180/pi)*angle(GH)
M = abs(GH)
K = 1/M
```

Example 4

For a unity feedback system that has the forward transfer function,

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

- Sketch the root locus.
- Find K so that the system operates at damping ratio of 0.5.

Solution

- There are three finite open loop poles located at 0, -1, and -4.

There is no finite open loop zero.

Asymptotes

$$\text{Number of asymptotes} = \# \text{ finite poles} - \# \text{ finite zeros} = 3 - 0 = 3$$

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{ finite poles} - \# \text{ finite zeros}}$$

$$\sigma_a = \frac{((0) + (-1) + (-4)) - (0)}{3 - 0} = \frac{-5}{3} = -1.67$$

Angle

$$\theta_a = \frac{(2k+1)180^\circ}{\# \text{ finite poles} - \# \text{ finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k+1)180^\circ}{3-0} = \frac{(2k+1)180^\circ}{3} = (2k+1)60^\circ$$

There are two asymptotes, therefore $k = 0, 1, 2$

For $k=0$

$$\theta_a = (0+1)60^\circ = 60^\circ$$

For $k=1$

$$\theta_a = (2+1)60^\circ = 180^\circ$$

For $k=2$

$$\theta_a = (4+1)60^\circ = 300^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s))=0$$

$$G(s)H(s) = \frac{K}{s(s+1)(s+4)} = \frac{K}{s^3+5s^2+4s}$$

Let

$$\frac{1}{s^3+5s^2+4s} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^3+5s^2+4s)(0) - (1)(3s^2+10s+4)}{(s^3+5s^2+4s)^2} = 0$$

$$3s^2+10s+4=0$$

The roots are $s = -2.87$ or -0.46 . $s = -0.46$ lies on the root-locus for $K > 0$, therefore this is the breakaway point.

$j\omega$ - crossing

$$1 + KG(s)H(s) = 0$$

$$1 + \frac{K}{s^3+5s^2+4s} = 0$$

$$\frac{s^3+5s^2+4s+K}{s^3+5s^2+4s} = 0$$

$$s^3+5s^2+4s+K=0$$

$$s^3+5s^2+4s+K=0$$

Routh table

s^3	1	4
s^2	5	K
s^1	$\frac{20-K}{5}$	0
s^0	K	

At the limit of stability $\frac{20-K}{5}=0$

$$20 - K = 0$$

Hence $K = 2.73$

The auxiliary equation is formed using the row above the zero row as:

$$5s^2 + K = 0$$

$$5s^2 + 20 = 0$$

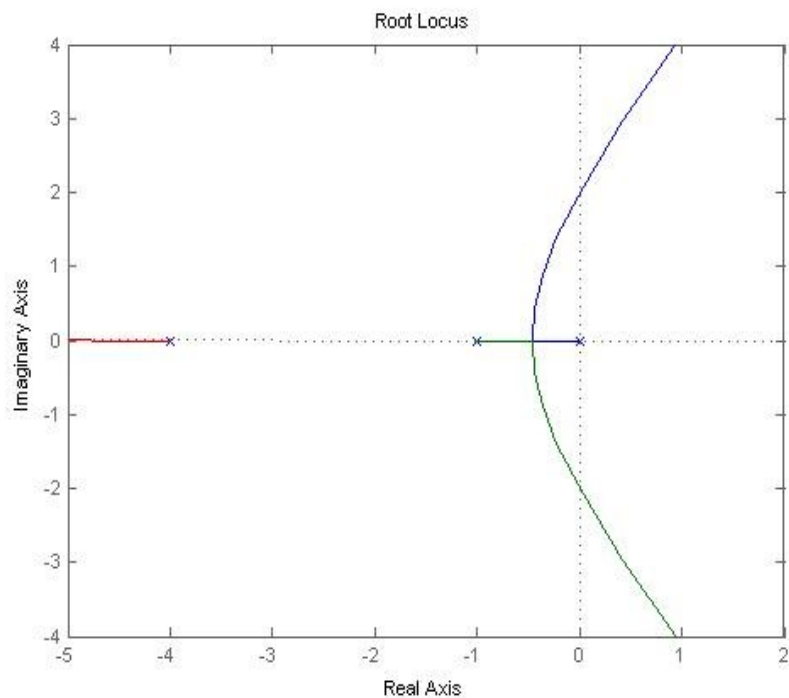
Solving for s in the auxiliary equation gives the $j\omega$ -crossing values as:

$$s = \pm \sqrt{\frac{-20}{5}} = \pm j2$$

Angles of Departure and Arrival

There are no Angles of Departure and Arrival

Using the above information the root locus is plotted as:



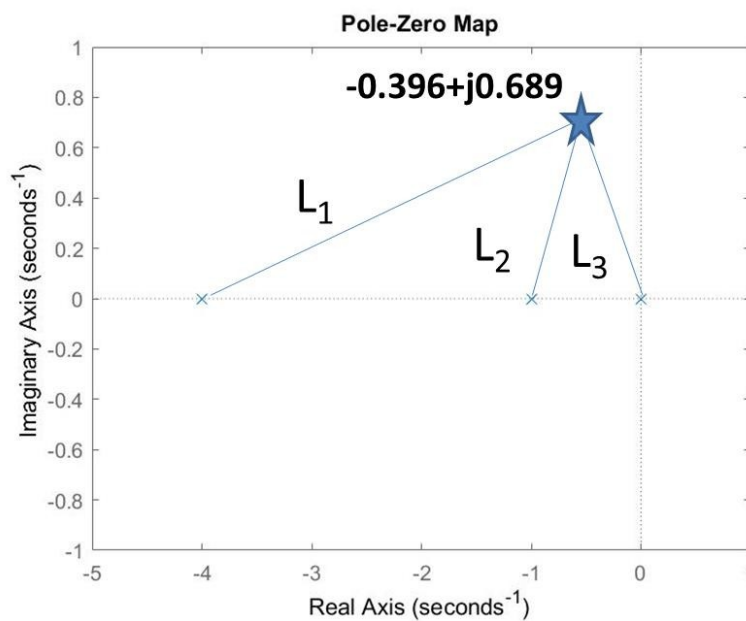
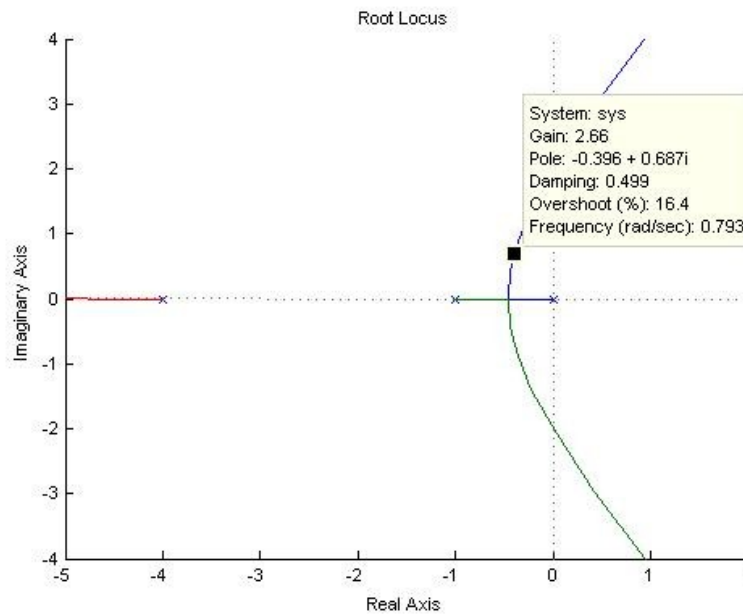
b. Given damping ratio, $\zeta = 0.5$

But, $\cos\theta = \zeta$

So, $\theta = \cos^{-1}(0.5) = 60^\circ$

Draw straight line from origin to cross the RL at an angle of 60°

The intersection between Straight line and RL yields the point $(-0.396 + j0.689)$



$$K = \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = L_1 L_2 L_3 = \sqrt{0.689^2 + (4 - 0.396)^2} \times \sqrt{0.689^2 + (1 - 0.396)^2} \times \sqrt{0.689^2 + 0.396^2} =$$

Solution of example 4 Matlab

Example 4 can be solve using the following Matlab code:

```
s=tf('s')
g=1/(s*(s+1)*(s+4))
rlocus(g)
z=0.5
sgrid(z,0)
axis([-5 1 -3 3])
```

Running the above code the root locus will be plotted. From the plot the pole location is obtained as $s = -0.396 + 0.689j$.

Using the code below the gain K will be obtained.

```
s = -0.396+0.689j
GH=1/(s*(s+1)*(s+4))
Theta=(180/pi)*angle(GH)
M=abs(GH)
K=1/M
```

Design of control systems using Root locus

From the previous section, it can be observed that, the root locus graphically displayed both transient response and stability information. The locus can be sketched quickly to get a general idea of the changes in transient response generated by changes in gain. Specific points on the locus also can be found accurately to give quantitative design information.

Design with Gain Adjustment

The root locus typically allows us to choose the proper loop gain to meet a transient response specification. As the gain is varied, we move through different regions of response. Setting the gain at a particular value yields the transient response dictated by the poles at that point on the root locus.

The formulae describing transient response specification (percent overshoot, settling time, and peak time) were derived only for a system with two complex poles and no zeros. That is they are based on the second order transfer function which is given by:

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

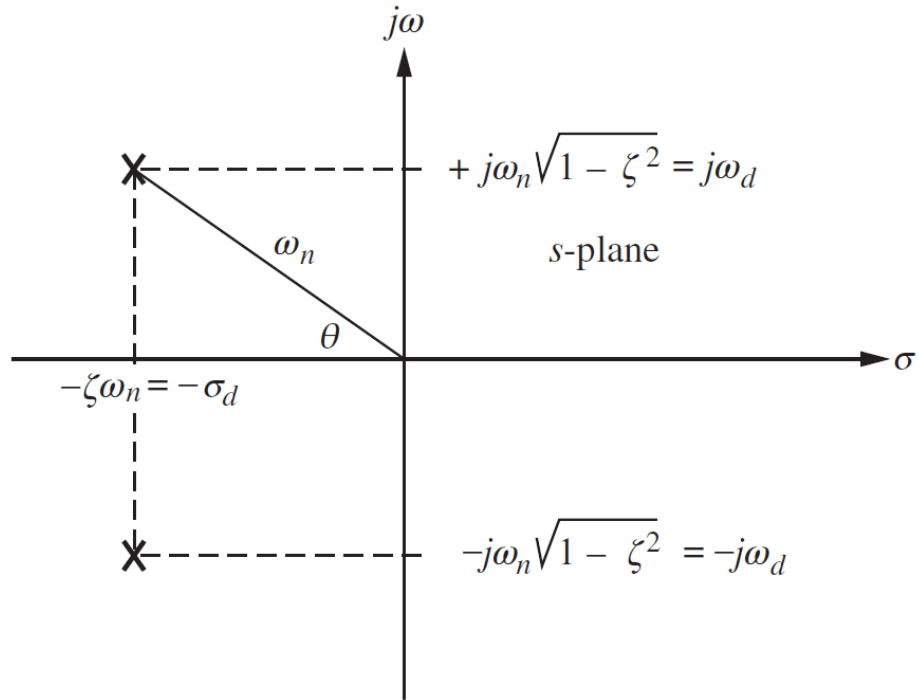
The poles of the system can be determined from the characteristics equation:

$$\begin{aligned} s^2 + 2\zeta\omega_n s + \omega_n^2 &= 0 \\ a &= 1, \quad b = 2\zeta\omega_n, \quad c = \omega_n^2 \\ s &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ s &= \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2} \\ s &= \frac{-2\zeta\omega_n \pm 2\omega_n\sqrt{\zeta^2 - 1}}{2} \\ s &= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \end{aligned}$$

When $0 < \zeta < 1$

$$\begin{aligned} s &= -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2} \\ s &= -\sigma_d \pm j\omega_d \\ &= -\text{Real} \pm j\text{Imaginary} \end{aligned}$$

The poles are mapped on the s-plane as follows:



Where $\omega_n = \text{natural frequency}$

$\zeta = \text{damping ratio}$

$\omega_d = \text{damped frequency}$

From the above information, the following can be defined:

1. Pole angle from origin, θ

$$\theta = \tan^{-1} \left(\frac{\text{Imaginary}}{\text{real}} \right) = \tan^{-1} \left(\frac{\omega_d}{\sigma_d} \right)$$

2. Damping ratio, ζ

$$\zeta = \cos \theta$$

3. Natural frequency, ω_n

$$\omega_n = \sqrt{\text{real}^2 + \text{Imaginary}^2} = \sqrt{\sigma_d^2 + \omega_d^2}$$

4. Settling time, T_s

$$T_s = \frac{4}{|\text{Real}|} = \frac{4}{\sigma_d} = \frac{4}{\zeta \omega_n}$$

5. Peak time, T_p

$$T_p = \frac{\pi}{|\text{Imaginary}|} = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

6. Percent overshoot, %OS

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

7. Rise time, T_r

The rise time can be obtained from the table below based on the corresponding damping ratio ζ and natural frequency ω_n

damping ratio, ζ	Normalized rise time, $\omega_n T_r$
0.1	1.104
0.2	1.203
0.3	1.321
0.4	1.463
0.5	1.638
0.6	1.854
0.7	2.126
0.8	2.467
0.9	2.883

For example a system with damping ratio of 0.5 and natural frequency of 4, the rise time can be obtained as follows:

From the table the normalized rise time that corresponds to damping ratio of 0.5 is 1.638.

Therefore

$$\omega_n T_r = 1.638$$

Hence, the rise time is

$$T_r = \frac{1.638}{\omega_n} = \frac{1.638}{4} = 0.4095s$$

Example 1

Consider a second order closed loop system having the its poles located at $-3 - j7$ and $-3 + j7$. Determine:

1. Pole angle from origin, θ
2. Damping ratio, ζ

3. Natural frequency, ω_n
4. Settling time, T_s
5. Peak time, T_p
6. Percent overshoot, %OS
7. Rise time, T_r

Solution

Given:

$$s = -3 \pm j7 = -\sigma_d \pm j\omega_d$$

1. Pole angle from origin, θ

$$\theta = \tan^{-1} \left(\frac{\text{Imaginary}}{\text{real}} \right) = \tan^{-1} \left(\frac{\omega_d}{\sigma_d} \right) = \tan^{-1} \left(\frac{7}{3} \right) = 66.8^\circ$$

2. Damping ratio, ζ

$$\zeta = \cos\theta = \cos 66.8 = 0.394$$

3. Natural frequency, ω_n

$$\omega_n = \sqrt{\text{real}^2 + \text{Imaginary}^2} = \sqrt{\sigma_d^2 + \omega_d^2} = \sqrt{3^2 + 7^2} = 7.616 \text{ rad/s}$$

4. Settling time, T_s

$$T_s = \frac{4}{|\text{Real}|} = \frac{4}{\sigma_d} = \frac{4}{3} = 1.333 \text{ s}$$

5. Peak time, T_p

$$T_p = \frac{\pi}{|\text{Imaginary}|} = \frac{\pi}{\omega_d} = \frac{\pi}{7} = 0.449 \text{ s}$$

6. Percent overshoot, %OS

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = e^{-(0.394\pi/\sqrt{1-(0.394)^2})} \times 100 = 26.01\%$$

7. Rise time, T_r

From the table the normalized rise time that corresponds to damping ratio of 0.394 is 1.463.

Therefore

$$\omega_n T_r = 1.463$$

Hence, the rise time is

$$T_r = \frac{1.463}{\omega_n} = \frac{1.638}{7.616} = 0.215s$$

Steady state error

Steady state error (e_{ss}) is the difference between the input and output for a prescribed test input as time approaches infinity. The steady state error depends on the input and system type. The table below gives the relationship between the steady state error, the input and system type:

TABLE 7.2 Relationships between input, system type, static error constants, and steady-state errors

Input	Steady-state error formula	Type 0		Type 1		Type 2	
		Static error constant	Error	Static error constant	Error	Static error constant	Error
Step, $u(t)$	$\frac{1}{1 + K_p}$	$K_p = \text{Constant}$	$\frac{1}{1 + K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	∞	$K_v = \text{Constant}$	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = \text{Constant}$	$\frac{1}{K_a}$

The static error constants can be obtained as follows:

Position constant, K_p is given as:

$$K_p = \lim_{s \rightarrow 0} G(s)$$

Velocity constant, K_v is given as:

$$K_v = \lim_{s \rightarrow 0} sG(s)$$

Acceleration constant, K_a is given as:

$$K_a = \lim_{s \rightarrow 0} s^2G(s)$$

Example 2

A unity feedback system has the following forward transfer function:

$$G(s) = \frac{1000(s + 8)}{(s + 7)(s + 9)}$$

- Evaluate the system type, K_p , K_v , and K_a .
- Use the answer in a to find the steady state errors for step, ramp and parabolic inputs.

Solution

- a. The system is type 0, since there is no open loop zero at the origin.

$$K_p = \lim_{s \rightarrow 0} G(s) = \frac{1000(8)}{(7)(9)} = 127$$

$$K_v = \lim_{s \rightarrow 0} sG(s) = (0) \left(\frac{1000(8)}{(7)(9)} \right) = 0$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = (0)^2 \left(\frac{1000(8)}{(7)(9)} \right) = 0$$

- b. For step input $e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+127} = 0.0078$

$$\text{For step input } e_{ss} = \frac{1}{K_v} = \frac{1}{0} = \infty$$

$$\text{For step input } e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \infty$$

Second-order approximation of higher order systems

If a system has more than two poles or has zeros, we cannot use the formulae to calculate the performance specifications that we derived. However, under certain conditions, a system with more than two poles or with zeros can be approximated as a second-order system that has just two complex dominant poles.

The conditions justifying a second-order approximation are:

- 1) Higher-order poles are much farther (at least **five** times) into the left half of the s-plane than the dominant second-order pair of poles.
- 2) Closed-loop zeros near the closed-loop second-order pole pair are nearly canceled by the close proximity of higher-order closed-loop poles.
- 3) Closed-loop zeros not canceled by the close proximity of higher-order closed-loop poles are far removed from the closed-loop second-order pole pair.

Once we justify this approximation, the formulas for percent overshoot, settling time, and peak time can be applied to these higher-order systems by using the location of the dominant poles.

The figure below illustrates condition 1) and 2).

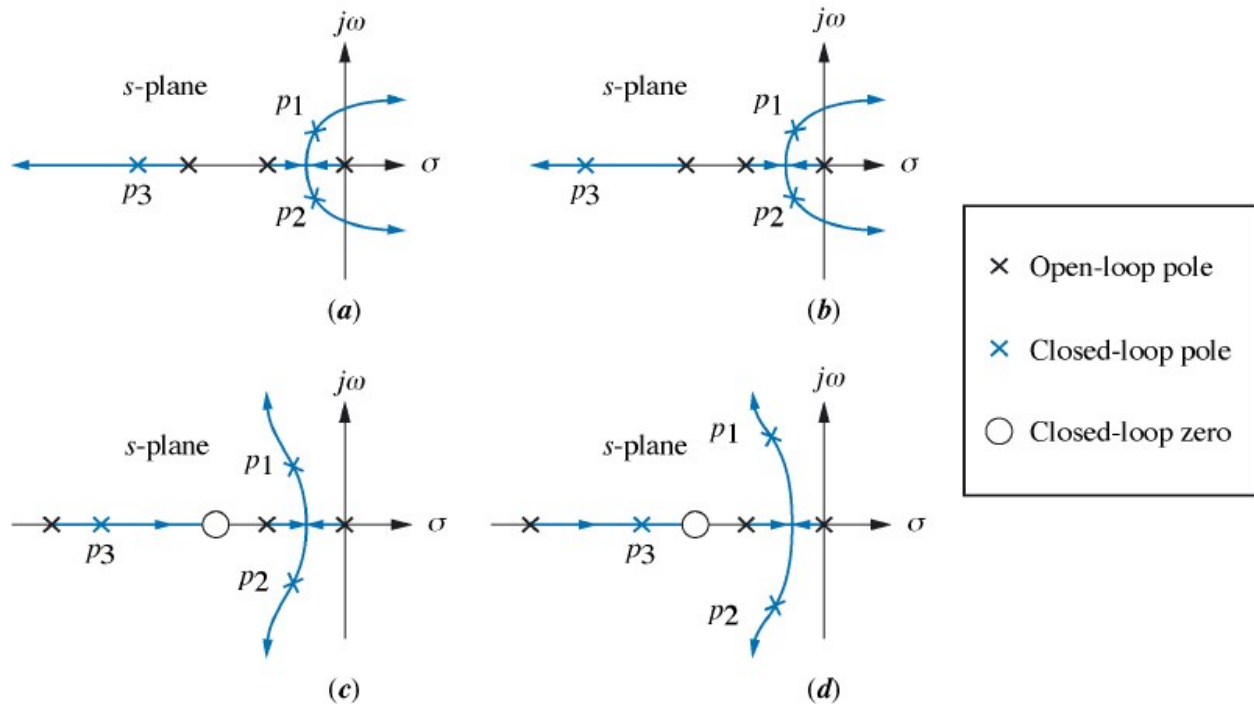


Figure (b) would yield a much better second-order approximation than Figure (a), since closed-loop pole p_3 is farther from the dominant, closed-loop second order pair, p_1 and p_2 .

Figure (d) would yield a much better second-order approximation than Figure (c), since closed-loop pole p_3 is closer to cancelling the closed-loop zero.

Design procedure for higher order systems

The design procedure for higher order systems can be summarized as follows:

1. Sketch the root locus.
2. Assume the system is a second order system without zero and find the gain to meet the specifications.
3. Justify the second-order approximation by finding higher order poles.

Example 1

Given a unity feedback system that has the forward-path transfer function:

$$G(s) = \frac{K}{(s+2)(s+4)(s+6)} = \frac{K}{s^3 + 12s^2 + 44s + 48}$$

- a. Sketch the root locus
- b. Design the value of K to yield 10% OS
- c. Estimate the settling time, peak time, rise time and steady-state error for the value of K in (b).

- d. Determine the validity of your second-order approximation.

Solution

- a. There are three finite open loop poles located at -2, -4, and -6.

There is no finite open loop zero.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 3 - 0 = 3

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((-2) + (-4) + (-6)) - (0)}{3 - 0} = \frac{-12}{3} = -4$$

Angle

$$\theta_a = \frac{(2k + 1)180^\circ}{\# \text{finite poles} - \# \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k + 1)180^\circ}{3 - 0} = \frac{(2k + 1)180^\circ}{3} = (2k + 1)60^\circ$$

There are three asymptotes, therefore $k = 0, 1, 2$

For $k = 0$

$$\theta_a = (0 + 1)60^\circ = 60^\circ$$

For $k = 1$

$$\theta_a = (2 + 1)60^\circ = 180^\circ$$

For $k = 2$

$$\theta_a = (4 + 1)60^\circ = 300^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{1}{s^3 + 12s^2 + 44s + 48}$$

Let

$$\frac{1}{s^3 + 12s^2 + 44s + 48} = \frac{u}{v}$$

$$\frac{vdu - udv}{v^2} = \frac{(s^3 + 12s^2 + 44s + 48)(0) - (1)(3s^2 + 24s + 44)}{(s^3 + 12s^2 + 44s + 48)^2} = 0$$

$$3s^2 + 24s + 44 = 0$$

The roots are $s = -2.85$ or -5.15 . $s = -2.85$ lies on the root-locus for $K > 0$, therefore this is the breakaway point.

j ω - crossing

$$1 + KG(s)H(s) = 0$$

$$1 + \frac{K}{s^3 + 12s^2 + 44s + 48} = 0$$

$$\frac{s^3 + 12s^2 + 44s + 48 + K}{s^3 + 12s^2 + 44s + 48} = 0$$

$$s^3 + 12s^2 + 44s + 48 + K = 0$$

$$s^3 + 12s^2 + 44s + 48 + K = 0$$

Routh table

s^3	1	44
s^2	12	$K + 48$
s^1	$480 - K$	0
s^0	12	
	$K + 48$	

At the limit of stability

$$\frac{480 - K}{12} = 0$$

$$480 - K = 0$$

Hence $K = 480$

The auxiliary equation is formed using the row above the zero row as:

$$12s^2 + K + 48 = 0$$

$$12s^2 + 528 = 0$$

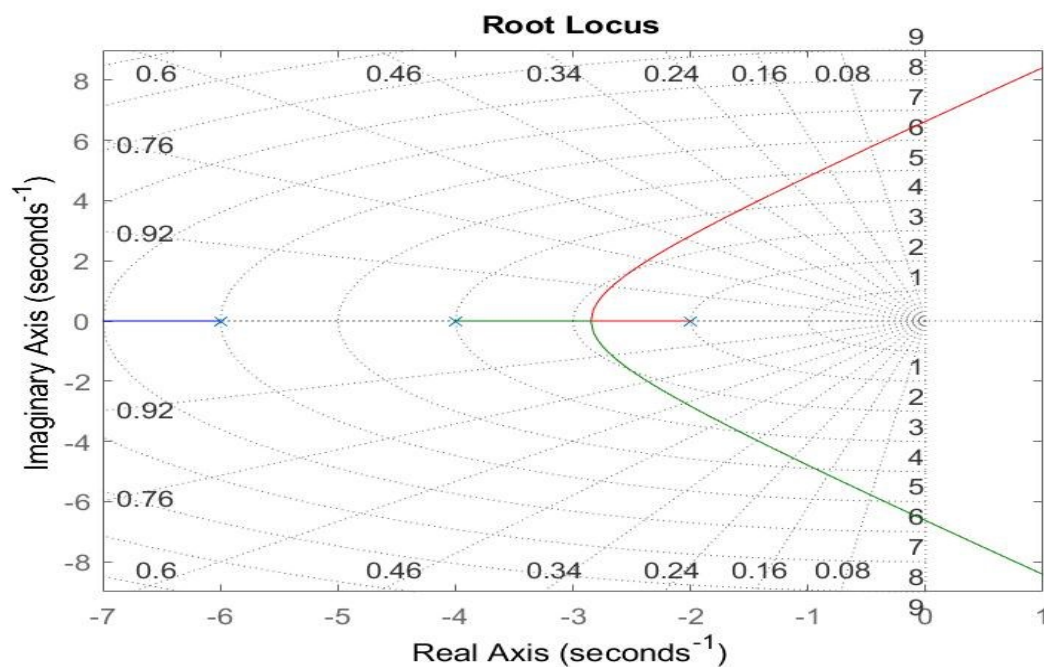
Solving for s in the auxiliary equation gives the $j\omega$ -crossing values as:

$$s = \pm \sqrt{-\frac{528}{12}} = \pm j6.63$$

Angles of Departure and Arrival

There are no Angles of Departure and Arrival

Using the above information the root locus is plotted as:



- b. Given Percent overshoot = 10%

$$\text{Percent overshoot} = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 10\%$$

Taking the natural log of both sides gives:

$$-(\zeta\pi/\sqrt{1-\zeta^2}) = \ln(0.1)$$

Squaring both sides gives:

$$\zeta^2\pi^2 = (\ln(0.1))^2(1-\zeta^2)$$

$$\begin{aligned}\zeta^2 \pi^2 + (\ln(0.1))^2 \zeta^2 &= \ln(0.1) \\ \zeta^2 (\pi^2 + (\ln(0.1))^2) &= (\ln(0.1))^2 \\ \zeta^2 &= \frac{(\ln(0.1))^2}{(\pi^2 + (\ln(0.1))^2)}\end{aligned}$$

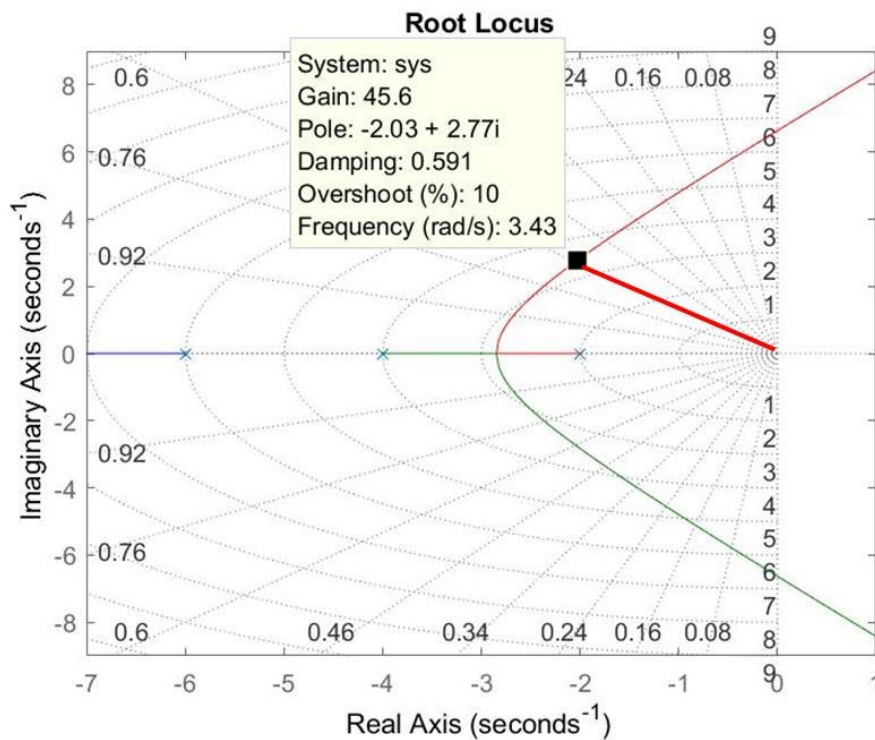
Therefore, the damping ratio ζ , can be obtained as:

$$\zeta = \sqrt{\frac{(\ln(0.1))^2}{(\pi^2 + (\ln(0.1))^2)}} = \sqrt{\frac{5.3019}{15.1715}} = 0.59$$

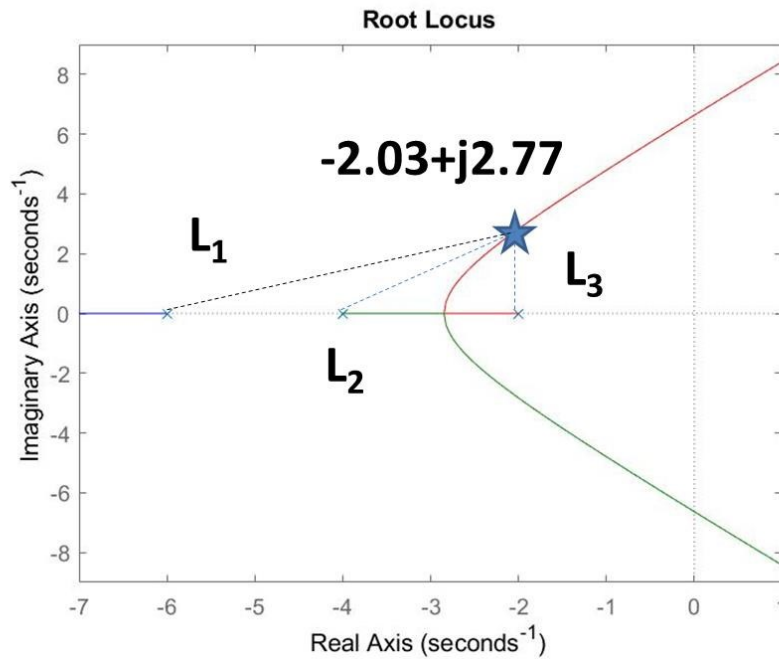
But, $\cos\theta = \zeta$

So, $\theta = \cos^{-1}(0.59) = 53.84^\circ$

Draw straight line from origin to cross the RL at an angle of 53.84°



The intersection between Straight line and RL yields the point $(-2.03 + j2.77)$



$$K = \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = L_1 L_2 L_3$$

$$= \sqrt{(6 - 2.03)^2 + 2.77^2} \times \sqrt{(4 - 2.03)^2 + 2.77^2} \times 2.77 = 4.84 \times 3.4 \times 2.77$$

$$= 45.58$$

c. $s = -2.03 + j2.77 = -\sigma_d \pm j\omega_d$

Settling time $T_s = \frac{4}{|Real|} = \frac{4}{\sigma_d} = \frac{4}{2.03} = 1.97s$

Peak time, $T_p = \frac{\pi}{|Imaginary|} = \frac{\pi}{\omega_d} = \frac{\pi}{2.77} = 1.13s$

Rise time

From the table the normalized rise time that corresponds to damping ratio of 0.59 is 1.834.

Therefore

$$\omega_n T_r = 1.834$$

$$\text{But } \zeta \omega_n = \sigma_d = 2.03 \Rightarrow \omega_n = \frac{\sigma_d}{\zeta} = \frac{2.03}{0.59} = 3.44$$

Hence, the rise time is

$$T_r = \frac{1.834}{\omega_n} = \frac{1.834}{3.44} = 0.53s$$

For step input the steady-state error is:

$$e_{ss} = \frac{1}{1+K_p}$$

$$\text{But } K_p = \lim_{s \rightarrow 0} G(s) = \frac{45.58}{(2)(4)(6)} = 0.95$$

$$\text{Hence } e_{ss} = \frac{1}{1+0.95} = 0.51$$

$$G(s) = \frac{K}{(s+2)(s+4)(s+6)} = \frac{K}{s^3 + 12s^2 + 44s + 48}$$

d. The closed loop transfer function can be obtained as:

$$\begin{aligned} \text{CLTF} &= \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{K}{s^3 + 12s^2 + 44s + 48}}{1 + \frac{K}{s^3 + 12s^2 + 44s + 48}} = \frac{\frac{K}{s^3 + 12s^2 + 44s + 48}}{\frac{s^3 + 12s^2 + 44s + 48 + K}{s^3 + 12s^2 + 44s + 48}} \\ &= \frac{K}{s^3 + 12s^2 + 44s + 48 + K} \end{aligned}$$

But $K=45.58$

Hence

$$\text{CLTF} = \frac{45.58}{s^3 + 12s^2 + 44s + 93.58}$$

The characteristic equation is

$$s^3 + 12s^2 + 44s + 93.58 = 0$$

Hence the closed loop poles are located at: $s = -2.03 - 2.77i$, $s = -2.03 + 2.77i$ and $s = -7.94$.

The ratio of dominant poles to the 3rd pole is $\frac{7.94}{2.03} = 3.91$. Hence the approximation is not valid, since the ratio is less than 5.

Solution of example 4 Matlab

Example 4 can be solved using the following Matlab code:

```
s=tf('s')
g=1/(s*(s+1)*(s+4))
rlocus(g)
z=0.5
```

```
sgrid(z,0)
axis([-5 1 -3 3])
```

Running the above code the root locus will be plotted. From the plot the pole location is obtained as $s = -0.396 + 0.689j$.

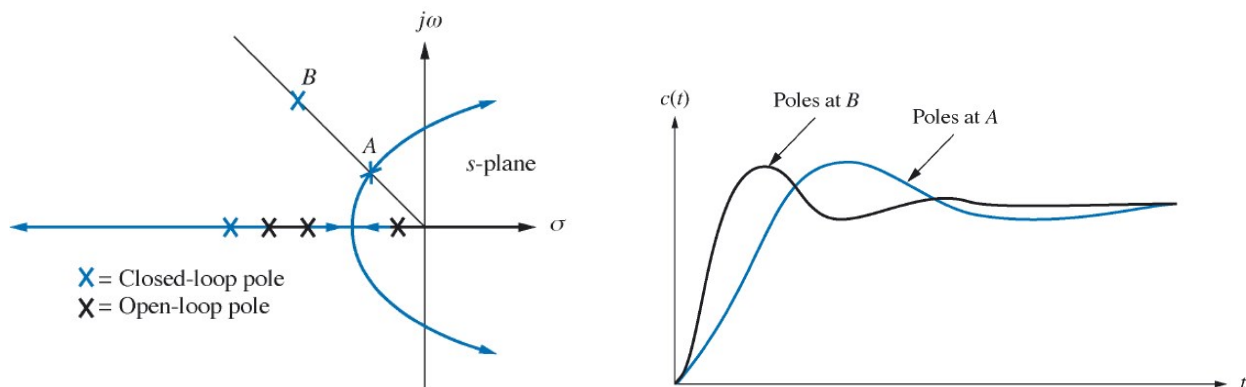
Using the code below the gain K will be obtained.

```
s = -0.396 + 0.689j
GH = 1/(s*(s+1)*(s+4))
Theta = (180/pi)*angle(GH)
M = abs(GH)
K = 1/M
```

Design with Compensators

With gain adjustment, we can only design for poles that are located on the root locus. Flexibility in the design can be increased if we can design for responses that are not on the root locus.

Consider the desired transient response defined by OS and settling time illustrated in the figure below:



We can obtain the settling time at A to satisfy the desired OS with gain adjustment. This is not possible for point B, since the point is not on the root locus.

However, this can be achieved by compensating the system with additional poles and zeros, so that the compensated system has a root locus that goes through the desired pole location for some value of gain.

Design with compensator is based on reshaping the root locus of the system by adding poles and zeros to the system's open-loop transfer function and forcing the root loci to pass through desired closed-loop poles in the s-plane.

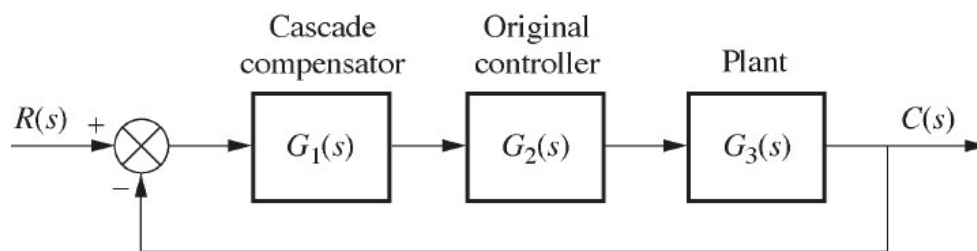
Compensators are not only used to improve the transient response of a system; they are also used independently to improve the steady-state error characteristics.

In the design with gain adjustment, when the system gain was adjusted to meet the transient response specification, steady-state error performance deteriorated, since both the transient response and the static error constant were related to the gain. The higher the gain, the smaller the steady-state error, but the larger the percent overshoot. On the other hand, reducing gain to reduce overshoot increased the steady-state error.

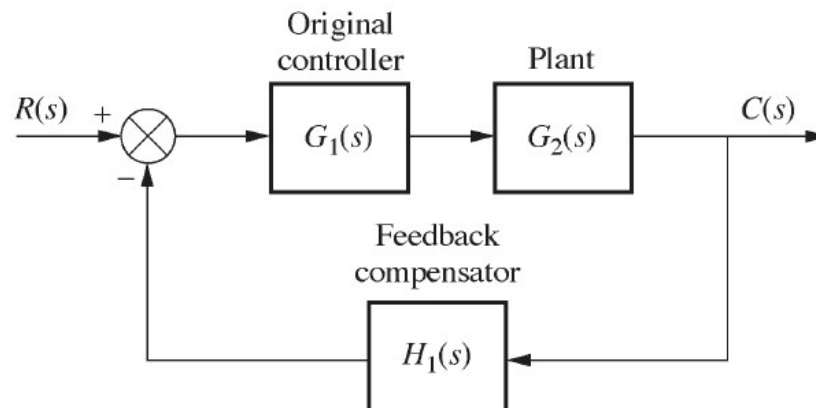
Design with compensators allows us to meet transient and steady-state error specifications simultaneously. We no longer need to compromise between transient response and steady-state error, as long as the system operates in its linear range.

There are two configurations of compensation:

Cascade compensation: The compensation network is placed in cascade with the plant as illustrated in the figure below:



Feedback compensation: The compensator is placed in the feedback path as illustrated in the figure below:



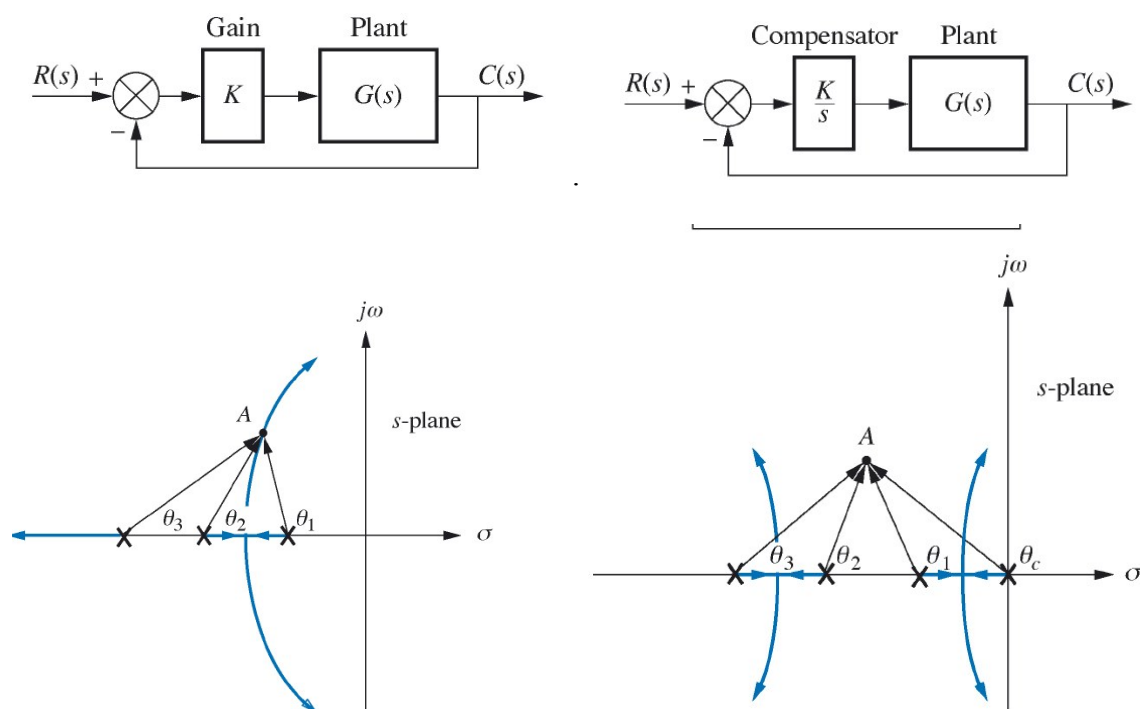
Both configurations change the OL poles and zeros, thus creating a new root locus that goes through the desired CL pole location.

A compensator can be PI controller/compensator, PD controller/compensator or PID controller/compensator.

PI controller/compensator

PI controller/compensator is used to improve the steady-state error. Steady state error (e_{ss}) can be improved by placing an open-loop pole at the origin as this increase the system type by one.

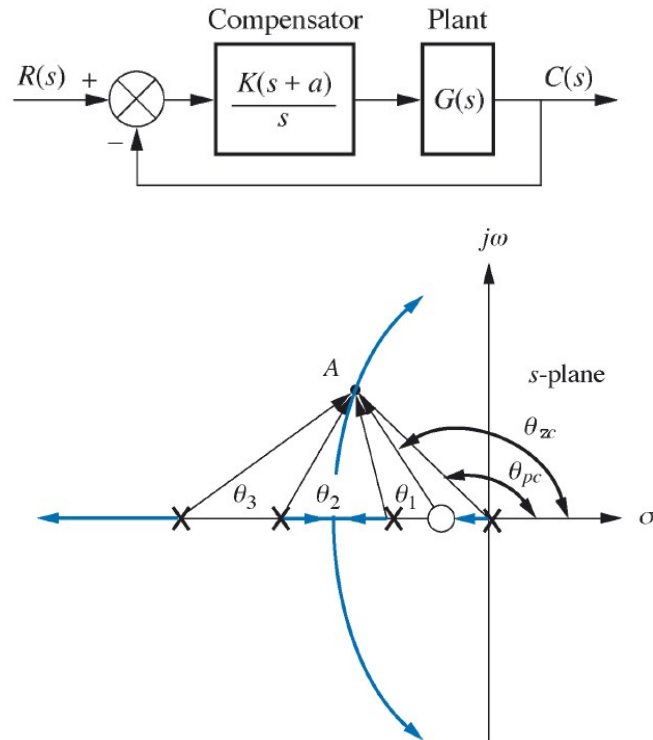
Consider a system operating at a desirable response with CL pole at A as illustrated in the figures below:



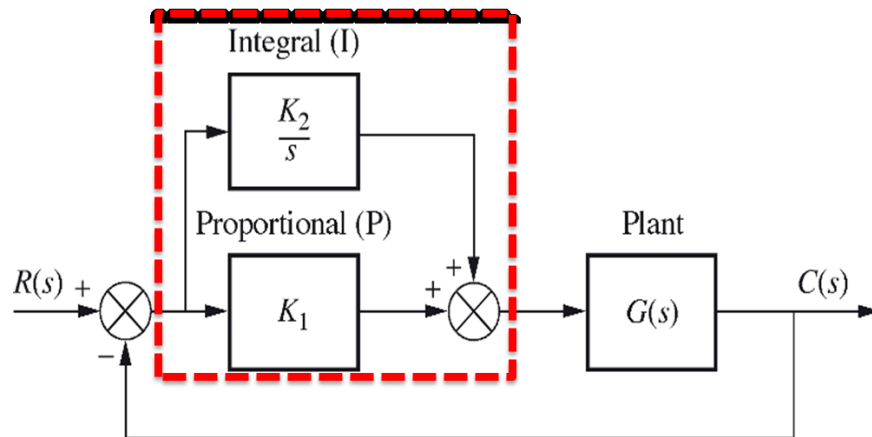
By adding a pole to increase the system type, A is no longer a CL pole on the root locus.

To solve this problem, a zero close to the pole at the origin has to be added. Thus A is now a CL pole on the root locus.

Thus, we have improved the e_{ss} without affecting the transient response. This is known as PI controller/compensator.



The figure below shows the block diagram of a PI compensated system:



From the above figure the transfer function of the PI controller can be obtained as:

$$G_{PI}(s) = K_1 + \frac{K_2}{s} = \frac{K_1 s + K_2}{s} = \frac{K_1 \left(s + \frac{K_2}{K_1} \right)}{s} = \frac{K_1 (s + z)}{s}$$

The PI controller is designed by adding a pole at the origin and a zero at $s = z$

The zero is chosen to satisfy transient response specifications.

The value of the zero z can be adjusted by varying $\frac{K_2}{K_1}$

If the same transient response as the uncompensated system is required, choose the zero close to the origin. For example: $s = -0.1$, $s = -0.01$.

Example 1

For the unity feedback system with open loop transfer function (OLTF):

$$G(s) = \frac{K}{(2s + 1)(0.5s + 1)} = \frac{K}{s^2 + 2.5s + 1}$$

Design a PI controller to achieve the following:

1. %OS = 10 %
2. Settling time, $T_s = 3.2$ s
3. Zero steady-state error, e_{ss} to unit step input.

Solution

Step 1: Sketch the root locus of the system without the PI Controller.

There are two finite open loop poles located at -0.5 and -2 .

There is no finite open loop zero.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = $2 - 0 = 2$

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((-0.5) + (-2)) - (0)}{2 - 0} = \frac{-2.5}{2} = -1.25$$

Angle

$$\theta_a = \frac{(2k + 1)180^\circ}{\# \text{finite poles} - \# \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k + 1)180^\circ}{2 - 0} = \frac{(2k + 1)180^\circ}{2} = (2k + 1)90^\circ$$

There are two asymptotes, therefore $k = 0, 1$

For $k = 0$

$$\theta_a = (0 + 1)90^\circ = 90^\circ$$

For $k = 1$

$$\theta_a = (2 + 1)90^\circ = 270^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{1}{s^2 + 2.5s + 1}$$

Let

$$\frac{1}{s^2 + 2.5s + 1} = \frac{u}{v}$$

$$\frac{vdu - u dv}{v^2} = \frac{(s^2 + 2.5s + 1)(0) - (1)(2s + 2.5)}{(s^2 + 2.5s + 1)^2} = 0$$

$$2s + 2.5 = 0$$

Therefore the breakaway point is $s = 1.25$

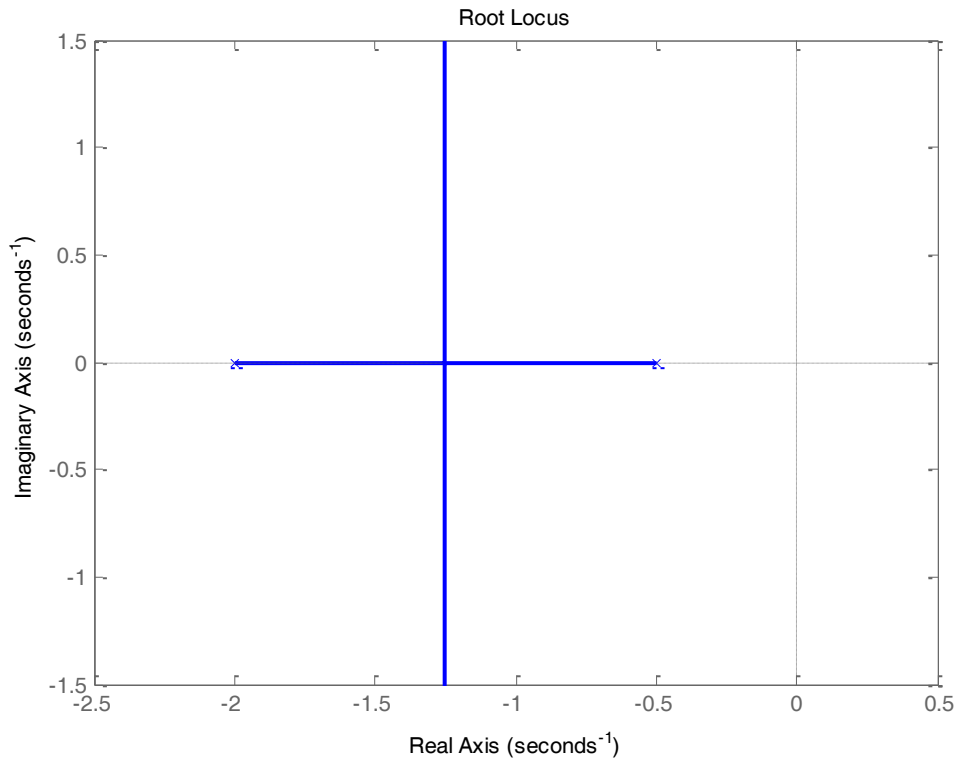
j ω - crossing

No jw crossing

Angles of Departure and Arrival

There are no Angles of Departure and Arrival

Using the above information the root locus is plotted as:



Step 2: Find the CL poles for the given %OS

Given Percent overshoot = 10%

$$\text{Percent overshoot} = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 10\%$$

Taking the natural log of both sides gives:

$$-(\zeta\pi/\sqrt{1-\zeta^2}) = \ln(0.1)$$

Squaring both sides gives:

$$\begin{aligned}\zeta^2\pi^2 &= (\ln(0.1))^2(1-\zeta^2) \\ \zeta^2\pi^2 + (\ln(0.1))^2\zeta^2 &= \ln(0.1) \\ \zeta^2(\pi^2 + (\ln(0.1))^2) &= (\ln(0.1))^2 \\ \zeta^2 &= \frac{(\ln(0.1))^2}{(\pi^2 + (\ln(0.1))^2)}\end{aligned}$$

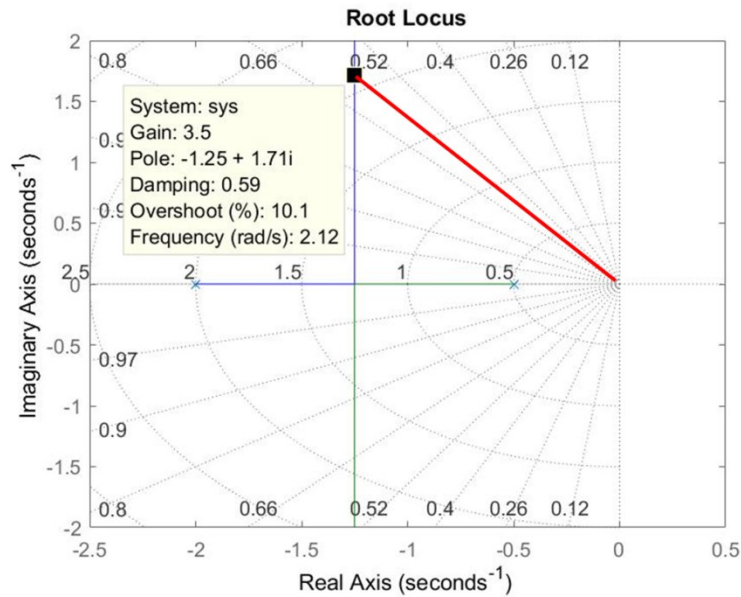
Therefore, the damping ratio ζ , can be obtained as:

$$\zeta = \sqrt{\frac{(\ln(0.1))^2}{(\pi^2 + (\ln(0.1))^2)}} = \sqrt{\frac{5.3019}{15.1715}} = 0.59$$

But, $\cos\theta = \zeta$

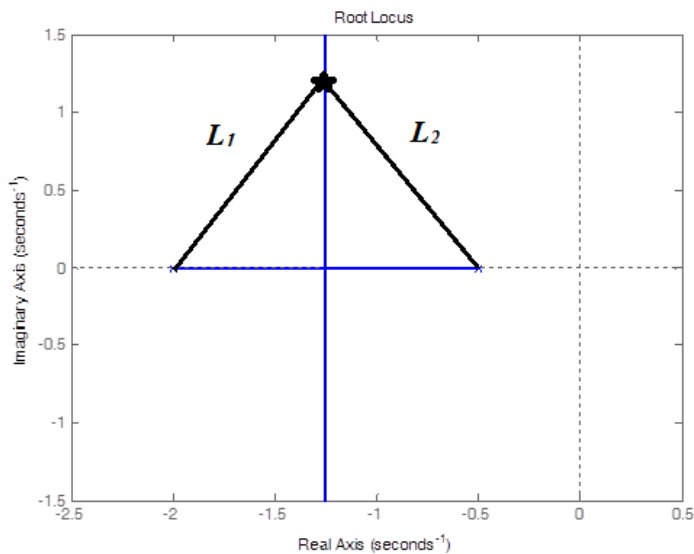
So, $\theta = \cos^{-1}(0.59) = 53.84^\circ$

Draw straight line from origin to cross the RL at an angle of 53.84°



The intersection between Straight line and RL yields the point $(-1.25 + j1.71)$. Therefore, the desired closed loop poles are $-1.25 \pm j1.71$

The uncompensated gain (K) can be obtained as follows:



$$K_1 = \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = L_1 L_2 = \sqrt{(0.75)^2 + 1.71^2} \times \sqrt{(0.75)^2 + 1.71^2} = 3.5$$

Step 3: Obtain the PI controller transfer function

The transfer function of the PI controller is given as:

$$G_{PI}(s) = \frac{K_1 \left(s + \frac{K_2}{K_1} \right)}{s} = \frac{K_1(s + z)}{s}$$

The location of the zero is chosen as $z = -0.01$

Hence,

$$G_{PI}(s) = \frac{K_1(s + 0.01)}{s}$$

Step 4: Compute the gains of the PI controller (K_1 and K_2)

The open loop transfer function of the system with the PI controller can be obtained as:

$$G_{PI}(s)G(s) = \frac{K_1(s + 0.01)}{s(2s + 1)(0.5s + 1)}$$

The gain of the compensated system (K_1) can be obtained as follows:

Note that: the pole at the origin and the zero located at 0.01 almost cancels each other.

Therefore, K_1 is approximately the same as the uncompensated gain K

Hence

$$K_1 = K = 3.5$$

$$z = \frac{K_2}{K_1} = 0.01$$

Hence

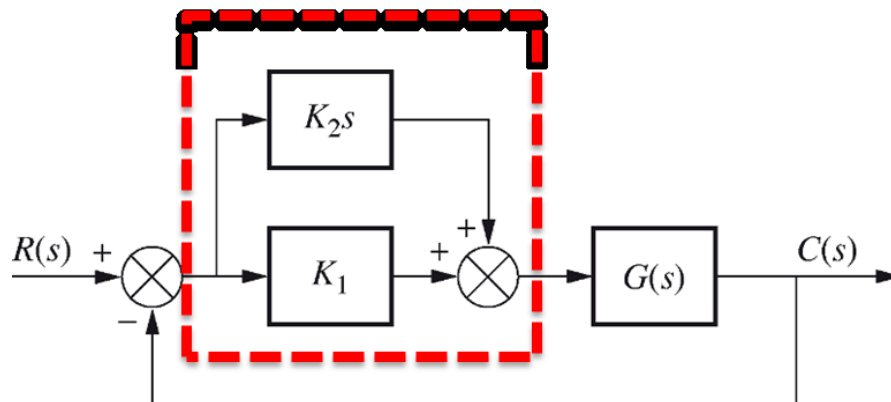
$$K_2 = 0.01K_1 = 0.01 \times 3.5 = 0.035$$

Therefore the PI controller gains are $K_1 = 3.5$ and $K_2 = 0.035$

PD controller/compensator

PD controllers are used to improve transient response and maintaining the steady-state error.

Transient response can be improved by adding a single zero to the forward path of the feedback control system as illustrated in the figure below:



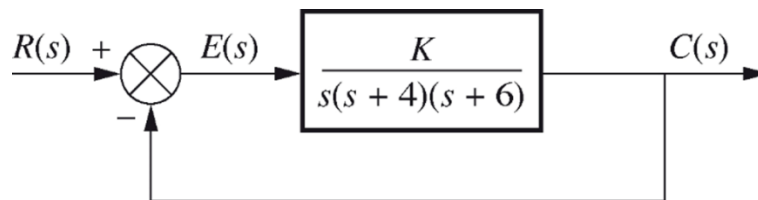
From the above figure the transfer function of the PD controller can be obtained as:

$$G_{PD}(s) = K_1 + K_2s = K_2 \left(s + \frac{K_1}{K_2} \right) = K_2(s + z)$$

A proper selection of the zero z can improve the system's transient response.

Example 1

Given the system, design a PD controller to yield 16 % overshoot, with a threefold reduction in the settling time.



Solution

Step 1: Sketch the root locus of the uncompensated system.

There are three finite open loop poles located at 0, -4 and -6.

There is no finite open loop zero.

Asymptotes

$$\text{Number of asymptotes} = \# \text{ finite poles} - \# \text{ finite zeros} = 3 - 0 = 3$$

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((0) + (-4) + (-6)) - (0)}{3 - 0} = \frac{-10}{3} = -3.33$$

Angle

$$\theta_a = \frac{(2k + 1)180^\circ}{\# \text{finite poles} - \# \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k + 1)180^\circ}{3 - 0} = \frac{(2k + 1)180^\circ}{3} = (2k + 1)60^\circ$$

There are two asymptotes, therefore $k = 0, 1$

For $k = 0$

$$\theta_a = (0 + 1)60^\circ = 60^\circ$$

For $k = 1$

$$\theta_a = (2 + 1)60^\circ = 180^\circ$$

For $k = 2$

$$\theta_a = (4 + 1)60^\circ = 300^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{1}{s^3 + 10s^2 + 24s}$$

Let

$$\frac{1}{s^3 + 10s^2 + 24s} = \frac{u}{v}$$

$$\frac{vdu - udv}{v^2} = \frac{(s^3 + 10s^2 + 24s)(0) - (1)(3s^2 + 20s + 24)}{(s^3 + 10s^2 + 24s)^2} = 0$$

$$3s^2 + 20s + 24 = 0$$

The roots are $s = -5.0972$ or -1.5695 . $s = -1.5695$ lies on the root-locus for $K > 0$, therefore this is the breakaway point.

$j\omega$ - crossing

$$1 + KG(s)H(s) = 0$$

$$1 + \frac{K}{s^3 + 20s^2 + 24s} = 0$$

$$\frac{s^3 + 20s^2 + 24s + K}{s^3 + 20s^2 + 24s} = 0$$

$$s^3 + 20s^2 + 24s + K = 0$$

$$s^3 + 20s^2 + 24s + K = 0$$

Routh table

s^3	1	24
s^2	20	K
s^1	$\frac{480 - K}{20}$	0
s^0	K	

At the limit of stability

$$\frac{480 - K}{20} = 0$$

$$480 - K = 0$$

Hence $K = 480$

The auxiliary equation is formed using the row above the zero row as:

$$20s^2 + K = 0$$

$$20s^2 + 480 = 0$$

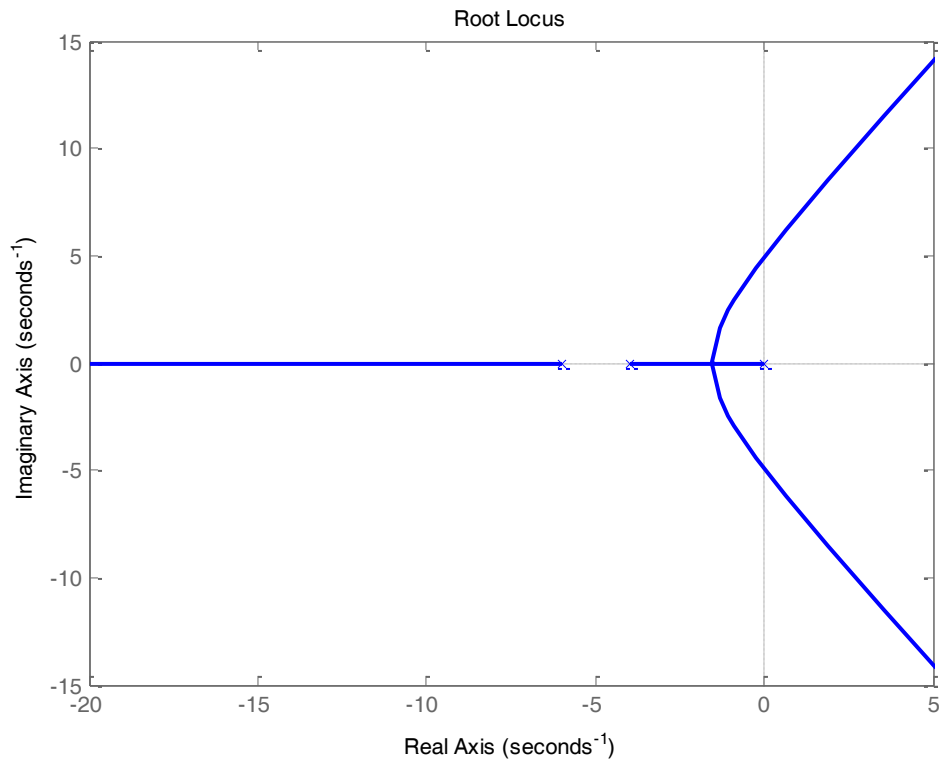
Solving for s in the auxiliary equation gives the $j\omega$ -crossing values as:

$$s = \pm \sqrt{-\frac{480}{20}} = \pm j4.9$$

Angles of Departure and Arrival

There are no Angles of Departure and Arrival

Using the above information the root locus is plotted as:



Step 2: Find the CL poles for the given %OS

Given Percent overshoot = 16%

$$\text{Percent overshoot} = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 16\%$$

Taking the natural log of both sides gives:

$$-(\zeta\pi/\sqrt{1-\zeta^2}) = \ln(0.16)$$

Squaring both sides gives:

$$\begin{aligned}\zeta^2\pi^2 &= (\ln(0.16))^2(1-\zeta^2) \\ \zeta^2\pi^2 + (\ln(0.16))^2\zeta^2 &= (\ln(0.16))^2 \\ \zeta^2(\pi^2 + (\ln(0.16))^2) &= (\ln(0.16))^2 \\ \zeta^2 &= \frac{(\ln(0.16))^2}{(\pi^2 + (\ln(0.16))^2)}\end{aligned}$$

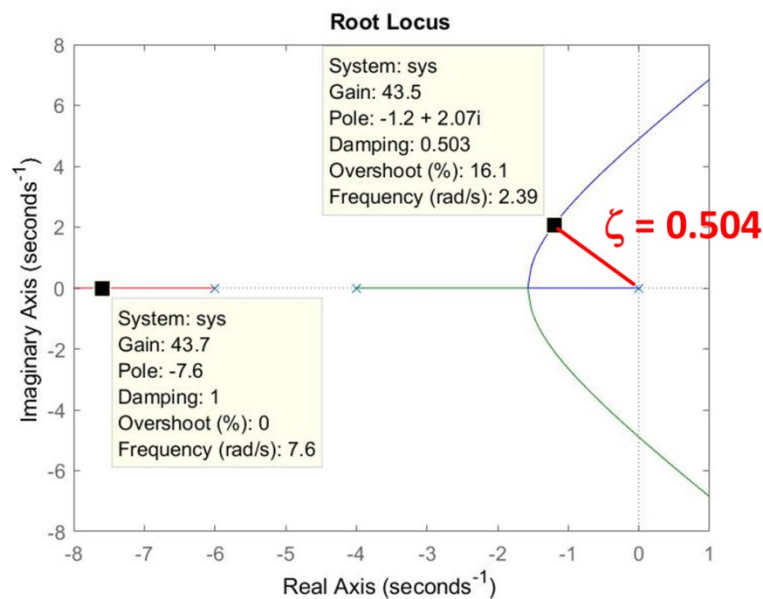
Therefore, the damping ratio ζ , can be obtained as:

$$\zeta = \sqrt{\frac{(\ln(0.16))^2}{(\pi^2 + (\ln(0.16))^2)}} = \sqrt{\frac{3.3584}{13.2280}} = 0.504$$

But, $\cos\theta = \zeta$

So, $\theta = \cos^{-1}(0.504) = 59.74^\circ$

Draw straight line from origin to cross the RL at an angle of 59.74°



The intersection between Straight line and RL yields the point $(-1.2 + j2.07)$. Therefore, the desired closed loop poles are $-1.2 \pm j2.07$

The settling time, T_s

$$T_s = \frac{4}{|Real|} = \frac{4}{1.2} = 3.33s$$

From the question it is required to make threefold reduction in the settling time.

Therefore, the new settling time will be one-third of the old settling time, that is:

$$T_{snew} = \frac{3.33}{3} = 1.11s$$

Step 3: Obtain the desired closed loop poles based on the new settling time

Form the new settling time:

$$1.11 = \frac{4}{\zeta \omega_n}$$

$$\text{Hence } \omega_n = \frac{4}{\zeta(1.11)} = \frac{4}{0.504 \times 1.11} = 7.15 \text{ rad/s}$$

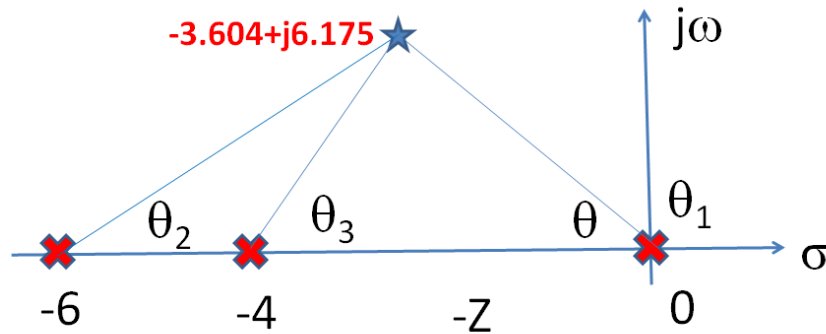
The desired dominant poles are given as:

$$\begin{aligned} s &= -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2} \\ &= -0.504 \times 7.15 \pm j 7.15 \times \sqrt{1 - 0.504^2} \\ &= -3.604 + j 6.175 \end{aligned}$$

Step 4: Obtain the location of the PD compensator zero z

The zero is calculated using angle property:

$$\text{total angles from poles to the root} - \text{total angles from zeros to the root} = 180^\circ$$



$$\theta = \tan^{-1} \left(\frac{6.175}{3.604} \right) = 59.73^\circ$$

$$\theta_1 = 180^\circ - 59.73^\circ = 120.27^\circ$$

$$\theta_2 = \tan^{-1} \left(\frac{6.175}{6 - 3.604} \right) = 68.79^\circ$$

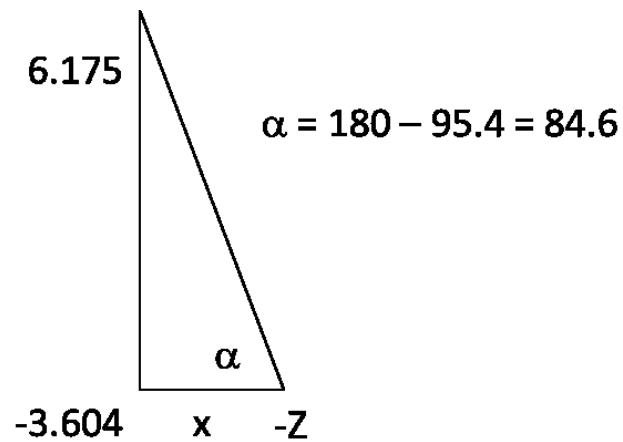
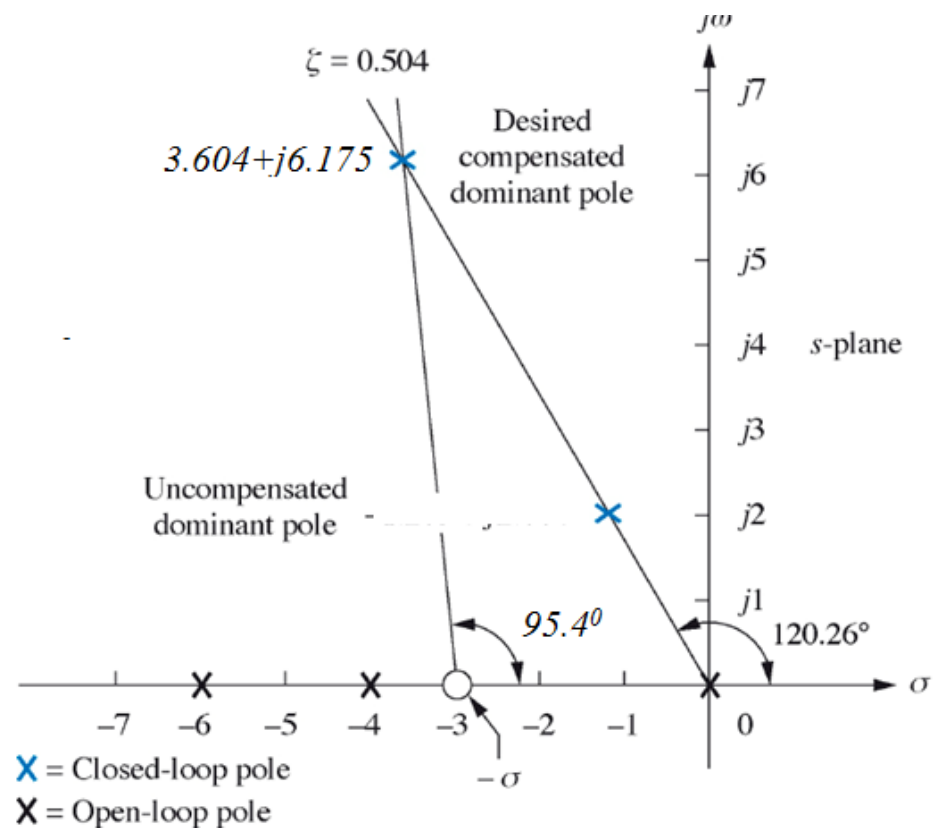
$$\theta_3 = \tan^{-1} \left(\frac{6.175}{4 - 3.604} \right) = 86.33^\circ$$

$$\theta_1 + \theta_2 + \theta_3 - \theta_{\text{zero}} = 180^\circ$$

Hence,

$$\theta_{\text{zero}} = 120.27^\circ + 68.79^\circ + 86.33^\circ - 180^\circ = 95.39^\circ$$

Then, the location of z can be calculated as follows:



$$\tan \alpha = \frac{6.175}{x}$$

$$x = \frac{6.175}{\tan \alpha} = \frac{6.175}{\tan(84.6)} = 0.584$$

$$z = 3.604 - 0.584 = 3.02$$

Hence, the transfer function of the PD controller can be obtained as:

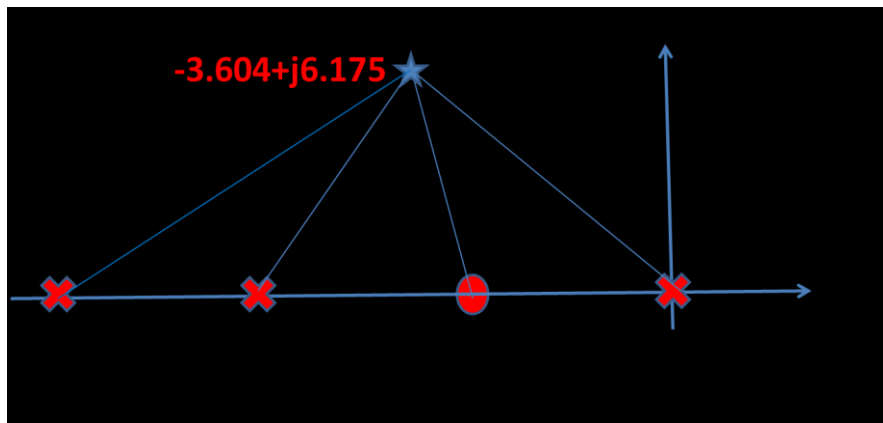
$$G_{PD}(s) = K_2 \left(s + \frac{K_1}{K_2} \right) = K_2(s + z) = K_2(s + 3.02)$$

The open loop transfer function of the system with the PD controller (compensated system) can be obtained as:

$$G_{PD}(s)G(s) = \frac{K_2(s + 3.02)}{s(s + 4)(s + 6)}$$

Step 5: Compute the gains of the PD controller (K_1 and K_2)

The gain of the compensated system (K_2) can be computed as:



$$\begin{aligned} K_2 &= \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = \frac{L_1 L_2 L_3}{L_4} \\ &= \frac{\sqrt{3.604^2 + 6.175^2} \times \sqrt{(6 - 3.604)^2 + 6.175^2} \times \sqrt{(4 - 3.604)^2 + 6.175^2}}{\sqrt{(3.604 - 3)^2 + 6.175^2}} \\ &= \frac{7.15 \times 6.62 \times 6.19}{6.2} = 47.26 \end{aligned}$$

Recall $z = \frac{K_1}{K_2}$

Then $3 = \frac{K_1}{47.26}$

Hence $K_1 = 3(47.26) = 141.78$

Therefore the PD controller gains are $K_1 = 141.78$ and $K_2 = 47.26$

Example 2

For a unity feedback system with open loop transfer function,

$$G(s) = \frac{50}{s(s+5)}$$

- Determine **percent OS** and **settling time** of the system
- Design a controller to have **50 % reduction in the OS** and **fourfold improvement in the settling time**.

Solution

- The closed loop transfer function can be obtained as:

$$\text{CLTF} = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{50}{s(s+5)}}{1 + \frac{50}{s(s+5)}} = \frac{\frac{50}{s(s+5)}}{\frac{s(s+5) + 50}{s(s+5)}} = \frac{50}{s^2 + 5s + 50}$$

Comparing with the general second order system

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Gives:

$$2\zeta\omega_n = 5 \Rightarrow \zeta\omega_n = 2.5$$

$$\omega_n^2 = 50 \Rightarrow \omega_n = 7.07$$

Hence,

$$\zeta = \frac{2.5}{7.07} = 0.354$$

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = e^{-(0.354\pi/\sqrt{1-(0.354)^2})} \times 100 = 30.45\%$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{2.5} = 1.6s$$

It is required that the %OS should be reduced by 50%.

Hence the desired %OS is 15.23%

Also, it is required that the settling time should be reduced to 1/4.

Hence the desired settling time is 0.4s

- Step 1:** Find the desired CL poles based on the new %OS and settling time.

New Percent overshoot = 15.23%

New settling time = 0.4s

$$\text{Percent overshoot} = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 15.23\%$$

Taking the natural log of both sides gives:

$$-(\zeta\pi/\sqrt{1-\zeta^2}) = \ln(0.1523)$$

Squaring both sides gives:

$$\begin{aligned}\zeta^2\pi^2 &= (\ln(0.1523))^2(1-\zeta^2) \\ \zeta^2\pi^2 + (\ln(0.1523))^2\zeta^2 &= (\ln(0.1523))^2 \\ \zeta^2(\pi^2 + (\ln(0.1523))^2) &= (\ln(0.1523))^2 \\ \zeta^2 &= \frac{(\ln(0.1523))^2}{(\pi^2 + (\ln(0.1523))^2)}\end{aligned}$$

Therefore, the damping ratio ζ , can be obtained as:

$$\zeta = \sqrt{\frac{(\ln(0.1523))^2}{(\pi^2 + (\ln(0.1523))^2)}} = \sqrt{\frac{3.5416}{13.4112}} = 0.514$$

$$T_{snew} = \frac{4}{\zeta\omega_n} = 0.4$$

Hence

$$\omega_n = \frac{4}{0.4\zeta} = \frac{4}{0.4(0.514)} = 19.46 \text{ rad/s}$$

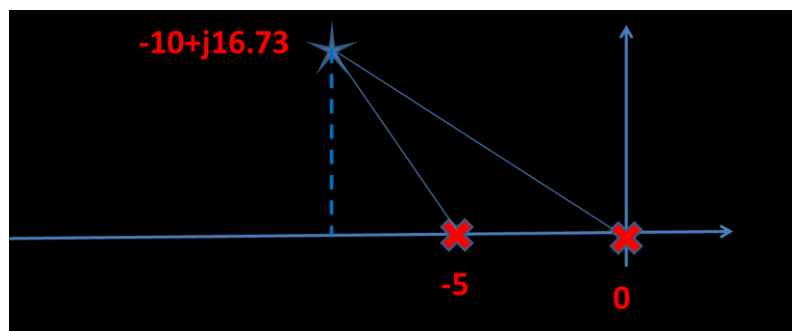
The closed loop poles can be obtained as:

$$\begin{aligned}s &= -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} \\ &= -0.514 \times 19.46 \pm j19.46 \times \sqrt{1-0.514^2} \\ &= -10 \pm j16.69\end{aligned}$$

Step 2: Obtain the location of the PD compensator zero z

The zero is calculated using angle property:

total angles from poles to the root – total angles from zeros to the root = 180°



$$\alpha_1 = \tan^{-1}\left(\frac{16.69}{10}\right) = 59.07^\circ$$

$$\theta_1 = 180^\circ - \alpha_1 = 120.93^\circ$$

$$\alpha_2 = \tan^{-1}\left(\frac{16.69}{5}\right) = 73.32^\circ$$

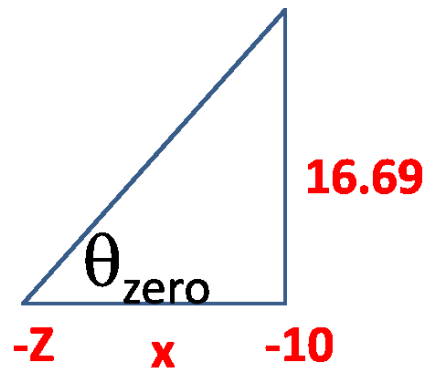
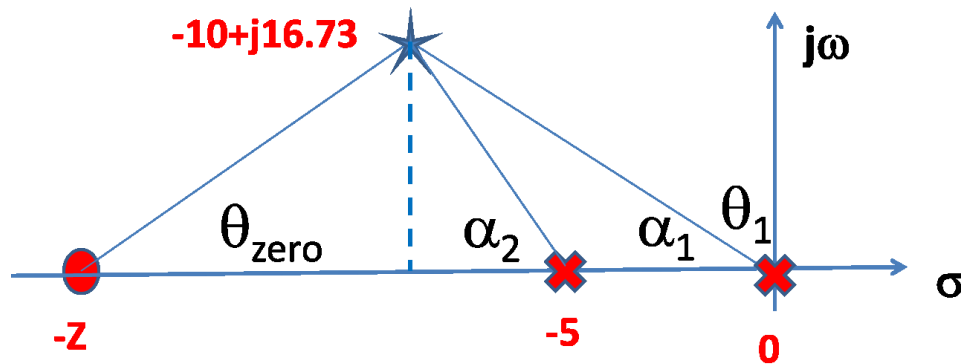
$$\theta_2 = 180^\circ - \alpha_2 = 106.68^\circ$$

$$\theta_1 + \theta_2 - \theta_{zero} = 180^\circ$$

Hence,

$$\theta_{zero} = 120.93^\circ + 106.68^\circ - 180^\circ = 47.61^\circ$$

Then, the location of z can be calculated as follows:



$$\tan\theta_{zero} = \frac{16.69}{x}$$

$$x = \frac{16.69}{\tan\theta_{zero}} = \frac{16.69}{\tan(47.61)} = 15.23$$

$$z = -10 - 15.23 = -25.23$$

Hence, the transfer function of the PD controller can be obtained as:

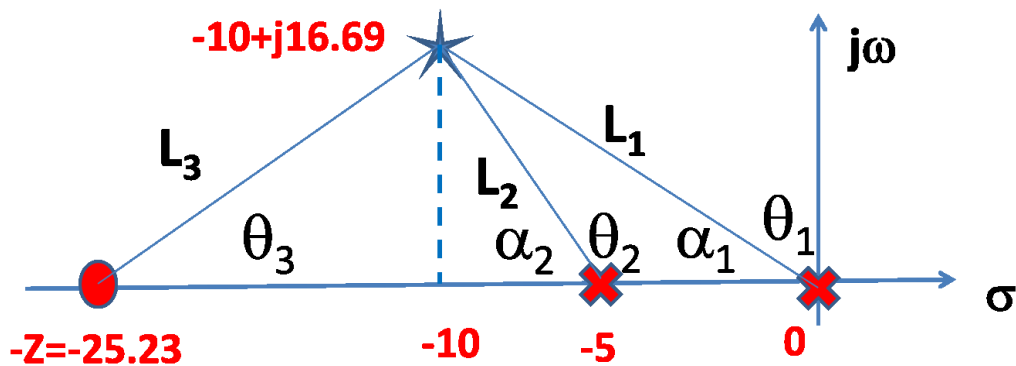
$$G_{PD}(s) = K_2 \left(s + \frac{K_1}{K_2} \right) = K_2(s + z) = K_2(s + 25.23)$$

The open loop transfer function of the system with the PD controller (compensated system) can be obtained as:

$$G_{PD}(s)G(s) = \frac{K_2(s + 25.23)}{s(s + 5)}$$

Step 3: Compute the gains of the PD controller (K_1 and K_2)

The gain of the compensated system (K_2) can be computed as:



$$K_2 = \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = \frac{L_1 L_2}{L_3} = \frac{\sqrt{10^2 + 16.69^2} \times \sqrt{5^2 + 16.69^2}}{\sqrt{15.23^2 + 16.69^2}} = \frac{19.46 \times 17.42}{22.59} = 15$$

Recall $z = \frac{K_1}{K_2}$

Then $25.23 = \frac{K_1}{15}$

Hence $K_1 = 25.23(15) = 378.45$

Therefore the PD controller gains are $K_1 = 378.45$ and $K_2 = 15$

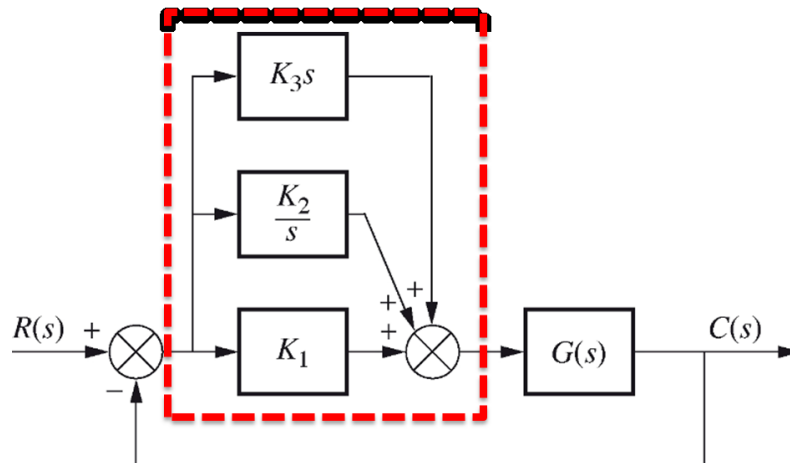
PID controller/Compensator

PID controllers are used to improve steady-state error and transient response independently.

A PID controller is a combination of PI and PD controllers.

We first design for transient response and then design for steady-state error.

The figure below illustrates the PID controller configuration:

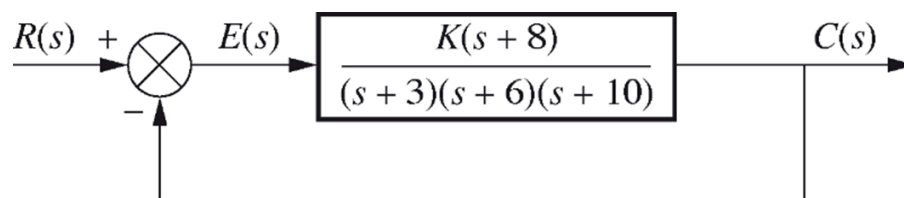


From the above figure the transfer function of the PID controller can be obtained as:

$$G_{PID}(s) = K_1 + \frac{K_2}{s} + K_3s = \frac{K_1s + K_2 + K_3s^2}{s} = \frac{K_3 \left(s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3} \right)}{s}$$

Example 1

Given the system, design a PID controller so that the system can operate with a peak time that is two-thirds of the uncompensated system at 20 % overshoot, and with zero steady-state error for a step input.



Solution

Step 1: Sketch the root locus of the uncompensated system.

There are three finite open loop poles located at -3, -6 and -10.

There is one finite open loop zero located at -8.

Asymptotes

Number of asymptotes = # finite poles - # finite zeros = 3-1=2

Real-axis intercept,

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{((-3) + (-6) + (-10)) - (-8)}{3 - 2} = \frac{-11}{2} = -5.5$$

Angle

$$\theta_a = \frac{(2k + 1)180^\circ}{\# \text{finite poles} - \# \text{finite zeros}}$$

where $k = 0, \pm 1, \pm 2, \dots$

$$\theta_a = \frac{(2k + 1)180^\circ}{3 - 0} = \frac{(2k + 1)180^\circ}{2} = (2k + 1)90^\circ$$

There are two asymptotes, therefore $k = 0, 1$

For $k = 0$

$$\theta_a = (0 + 1)90^\circ = 90^\circ$$

For $k = 1$

$$\theta_a = (2 + 1)90^\circ = 270^\circ$$

Real-axis Breakaway and Break-in Points

$$\frac{d}{ds}(G(s)H(s)) = 0$$

$$G(s)H(s) = \frac{(s + 8)}{(s^3 + 19s^2 + 108s + 180)}$$

Let

$$\frac{(s + 8)}{(s^3 + 19s^2 + 108s + 180)} = \frac{u}{v}$$

$$\frac{vdu - udv}{v^2} = \frac{(s^3 + 19s^2 + 108s + 180)(1) - (s + 8)(3s^2 + 38s + 108)}{(s^3 + 19s^2 + 108s + 180)^2} = 0$$

$$(s^3 + 19s^2 + 108s + 180) - (3s^3 + 38s^2 + 108s + 24s^2 + 304s + 864) = 0$$

$$2s^3 + 43s^2 + 304s + 684 = 0$$

The roots are $s = -8.4385 + j1.6640$ or $-8.4385 - j1.6640$ or $s = -4.6231$. $s = -4.6231$ lies on the root-locus for $K > 0$, therefore this is the breakaway point.

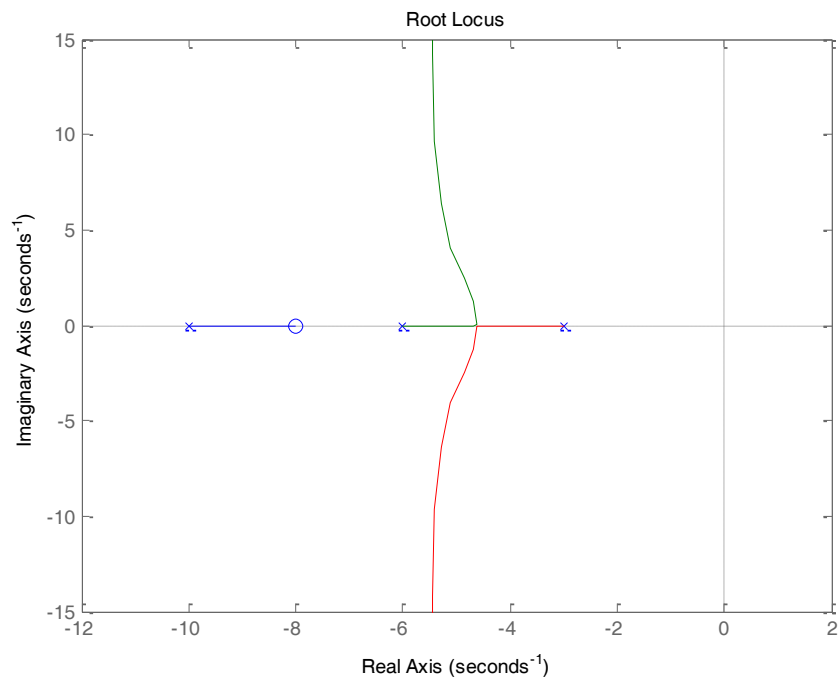
j ω - crossing

There is no **j ω - crossing**

Angles of Departure and Arrival

There are no Angles of Departure and Arrival

Using the above information the root locus is plotted as:



Step 2: Find the CL poles for the given %OS

Given Percent overshoot = 20%

$$\text{Percent overshoot} = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 20\%$$

Taking the natural log of both sides gives:

$$-(\zeta\pi/\sqrt{1-\zeta^2}) = \ln(0.2)$$

Squaring both sides gives:

$$\begin{aligned}\zeta^2 \pi^2 &= (\ln(0.2))^2 (1 - \zeta^2) \\ \zeta^2 \pi^2 + (\ln(0.2))^2 \zeta^2 &= \ln(0.2) \\ \zeta^2 (\pi^2 + (\ln(0.2))^2) &= (\ln(0.2))^2 \\ \zeta^2 &= \frac{(\ln(0.2))^2}{(\pi^2 + (\ln(0.2))^2)}\end{aligned}$$

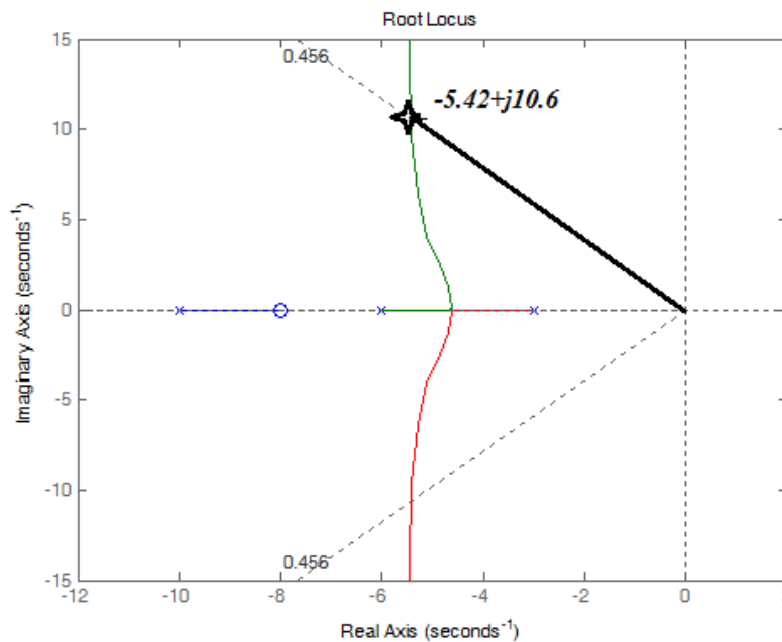
Therefore, the damping ratio ζ , can be obtained as:

$$\zeta = \sqrt{\frac{(\ln(0.2))^2}{(\pi^2 + (\ln(0.2))^2)}} = \sqrt{\frac{2.59}{12.46}} = 0.456$$

But, $\cos\theta = \zeta$

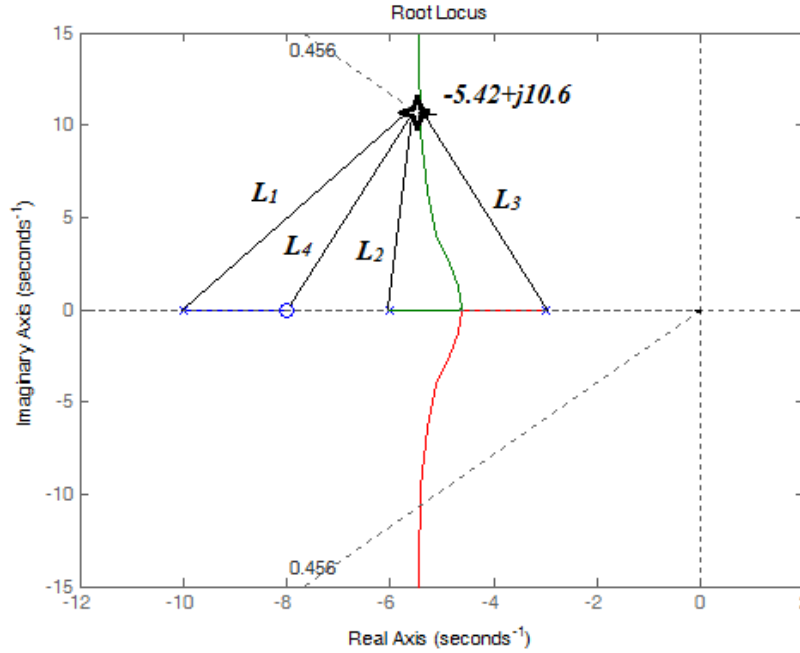
So, $\theta = \cos^{-1}(0.456) = 62.87^\circ$

Draw straight line from origin to cross the RL at an angle of 62.87°



The intersection between Straight line and RL yields the point $(-5.42 + j10.6)$. Therefore, the desired closed loop poles are $-5.42 \pm j10.6$

The uncompensated gain K can be obtained as follows:



$$\begin{aligned}
 K &= \frac{1}{|G(s)H(s)|} = \frac{\prod \text{poles lengths}}{\prod \text{zeros lengths}} = \frac{L_1 L_2 L_3}{L_4} \\
 &= \frac{\sqrt{(10 - 5.42)^2 + 10.6^2} \times \sqrt{(6 - 5.42)^2 + 10.6^2} \times \sqrt{(5.42 - 3)^2 + 10.6^2}}{\sqrt{(8 - 5.42)^2 + 10.6^2}} \\
 &= \frac{11.56 \times 10.61 \times 10.87}{10.91} = 122
 \end{aligned}$$

The peak time, T_p

$$T_p = \frac{\pi}{|\text{Imaginary}|} = \frac{\pi}{10.6} = 0.296s$$

From the question it is required to operate the system with a peak time of two-third of the original peak time.

Therefore, the new peak time will be two-third of the old settling time, that is:

$$T_{pnew} = \frac{2 \times 0.296}{3} = 0.196s$$

Step 3: Obtain the desired closed loop poles based on the new peak time

Form the new peak time:

$$T_{pnew} = \frac{\pi}{|\text{Imaginary}|} = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = 0.196$$

Hence

$$\omega_d = \frac{\pi}{0.196} = 16.03$$

$$\omega_n = \frac{\pi}{0.196 \times \sqrt{1 - \zeta^2}} = \frac{\pi}{0.196 \times \sqrt{1 - 0.456^2}} = 18$$

$$\text{Hence } \omega_n = \frac{4}{\zeta(1.11)} = \frac{4}{0.504 \times 1.11} = 7.15 \text{ rad/s}$$

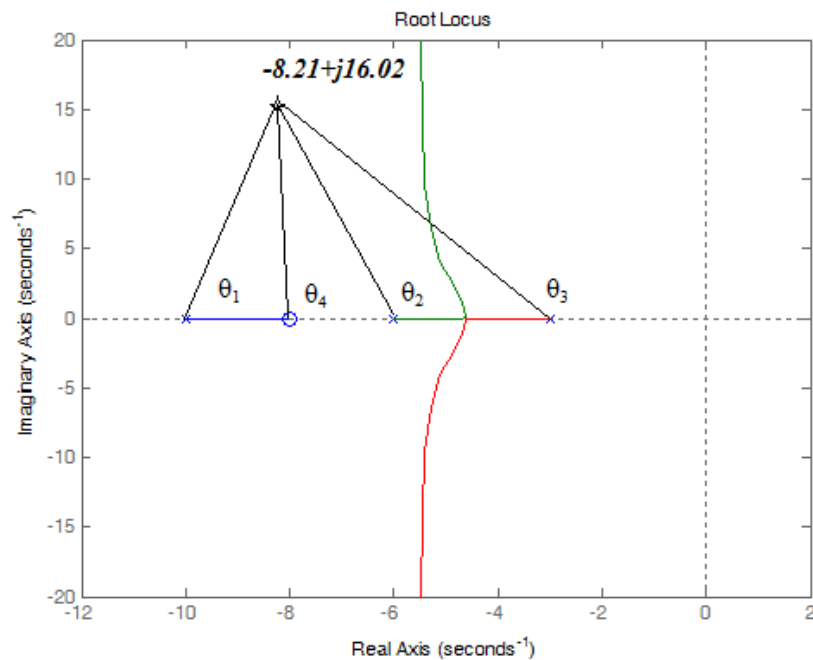
The desired dominant poles are given as:

$$\begin{aligned} s &= -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2} \\ &= -0.456 \times 18 \pm j18 \times \sqrt{1 - 0.456^2} \\ &= -8.21 \pm j16.02 \end{aligned}$$

Step 4: Obtain the location of the PD compensator zero z

The zero is calculated using angle property:

total angles from poles to the root – total angles from zeros to the root = 180°



$$\theta_1 = \tan^{-1}\left(\frac{16.02}{10 - 8.21}\right) = 83.62^\circ$$

$$\theta_2 = 180^\circ - \tan^{-1}\left(\frac{16.02}{8.21 - 6}\right) = 97.85^\circ$$

$$\theta_3 = 180^\circ - \tan^{-1}\left(\frac{16.02}{8.21 - 3}\right) = 108.02^\circ$$

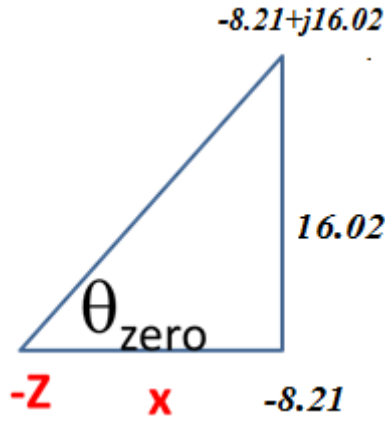
$$\theta_4 = 180^\circ - \tan^{-1}\left(\frac{16.02}{8.21 - 8}\right) = 90.75^\circ$$

$$\theta_1 + \theta_2 + \theta_3 - (\theta_4 + \theta_{zero}) = 180^\circ$$

Hence

$$\theta_{zero} = \theta_1 + \theta_2 + \theta_3 - \theta_4 - 180^\circ = 83.62^\circ + 97.85^\circ + 108.02^\circ - 90.75^\circ - 180^\circ = 18.74^\circ$$

Then, the location of z can be calculated as follows:



$$\tan\theta_{zero} = \frac{16.02}{x}$$

$$x = \frac{16.02}{\tan\theta_{zero}} = \frac{16.02}{\tan(18.74)} = 47.22$$

$$z = -8.21 - 47.22 = -55.43$$

Hence, the transfer function of the PD controller can be obtained as:

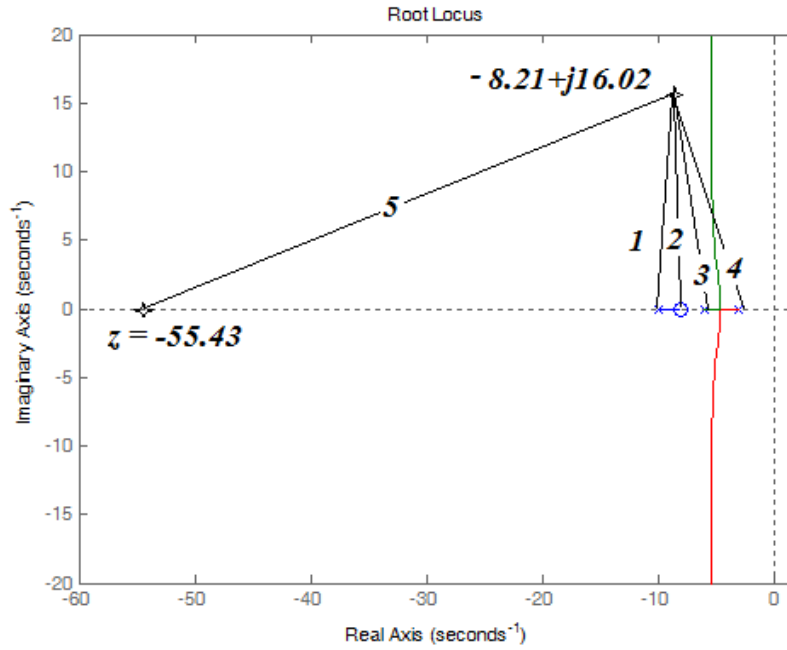
$$G_{PD}(s) = K_2 \left(s + \frac{K_1}{K_2} \right) = K_2(s + z) = K_2(s + 55.43)$$

The open loop transfer function of the system with the PD controller (compensated system) can be obtained as:

$$G_{PD}(s)G(s) = \frac{K_2(s + 8)(s + 55.43)}{(s + 3)(s + 6)(s + 10)}$$

Step 5: Compute the compensated gain (K_2)

The gain of the compensated system (K_2) can be computed as follows:



$$K_2 = \frac{L_1 L_3 L_4}{L_2 L_5} = \frac{\sqrt{(10 - 8.21)^2 + 16.02^2} \times \sqrt{(8.21 - 6)^2 + 16.02^2} \times \sqrt{(8.21 - 3)^2 + 16.02^2}}{\sqrt{(8.21 - 8)^2 + 16.02^2} \times \sqrt{(55.43 - 8.21)^2 + 16.02^2}}$$

$$= \frac{16.12 \times 16.17 \times 16.85}{16.02 \times 49.86} = 5.5$$

Hence

$$G_{PD}(s)G(s) = \frac{5.5(s + 8)(s + 55.43)}{(s + 3)(s + 6)(s + 10)}$$

Step 6: Design PI controller to reduce the steady state error to zero.

The transfer function of the PI controller is given as:

$$G_{PI}(s) = \frac{K_1(s + z)}{s}$$

The location of the zero is chosen as $z = 0.01$ and let $K_1 = 1$

Hence,

$$G_{PI}(s) = \frac{(s + 0.01)}{s}$$

The open loop transfer function can be written as:

$$G_{PD}(s)G_{PI}(s)G(s) = \frac{5.5(s+8)(s+55.43)(s+0.01)}{s(s+3)(s+6)(s+10)}$$

Step 6: Obtain the gains of the PID controller Design PI controller to reduce the error to zero

$$G_{PID}(s) = \frac{K_3 \left(s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3} \right)}{s} = \frac{5.5(s+55.43)(s+0.01)}{s} = \frac{5.5(s^2 + 55.44s + 0.5543)}{s}$$

Hence

$$K_3 = 5.5$$

$$\frac{K_1}{K_3} = 55.44 \Rightarrow K_1 = 5.5 \times 55.44 = 304.92$$

$$\frac{K_2}{K_3} = 0.5543 \Rightarrow K_2 = 5.5 \times 0.5543 = 3.05$$

Therefore the PID controller gains are: $K_1 = 304.92$, $K_2 = 3.05$, $K_3 = 5.5$

MCS4206: Control Engineering II

Chapter 4

Frequency Response and Specifications

Introduction

The performance of a control system is normally measured in terms of its time-domain characteristics.

It may equally be measured in frequency-domain, since there are correlations between the frequency-domain and time-domain performances in a linear system, so that the time-domain properties of the system can be predicted based on the frequency-domain analysis.

Besides, the frequency domain approach provides an alternative perspective to the complex analysis and design problems of linear control systems in addition to the time-domain studies.

Frequency Response

The **frequency response** of a system is defined as the steady-state response (output) of the system to a sinusoidal input signal.

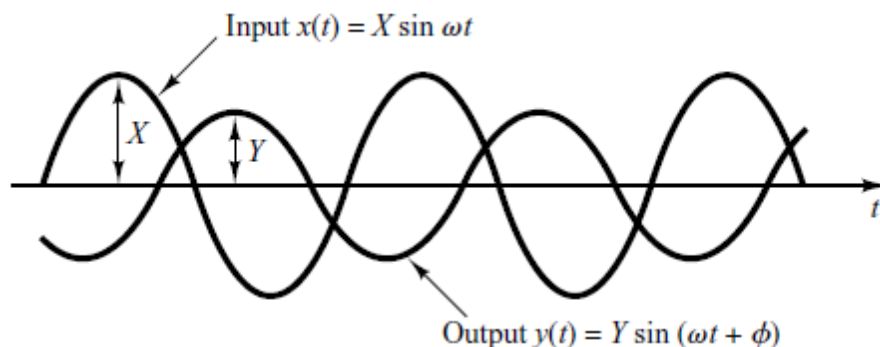
Assume that a linear time-invariant system with transfer function $G(s)$ is subjected to sinusoidal input with amplitude X and frequency ω given by

$$x(t) = X \sin \omega t$$

The steady-state output of the system $y(t)$ will also be a sinusoid with the same frequency, but possibly with different amplitude and phase; i.e.

$$y(t) = Y \sin(\omega t + \phi)$$

where Y is the amplitude of the output sine wave and ϕ is the phase shift in degrees or radians introduced by the system.



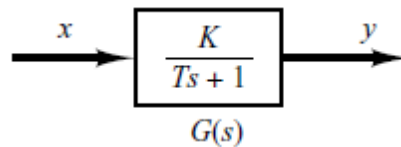
The frequency response of a system with a transfer function $G(s)$ can be obtained by replacing s with $j\omega$ in the transfer function. i.e

$$G(j\omega) = Me^{j\phi} = M \angle \phi$$

Where M is the magnitude (amplitude) and ϕ is the phase angle

Example

Consider the system shown below



Obtain the frequency response (the steady-state output $y_{ss}(t)$) for a sinusoidal input $x(t) = X \sin \omega t$

Solution

The transfer function $G(s)$ is

$$G(s) = \frac{K}{Ts + 1}$$

The frequency response can be obtained by replacing s with $j\omega$ in the transfer function. i.e

$$G(j\omega) = \frac{K}{jT\omega + 1}$$

The magnitude can be obtained as

$$|G(j\omega)| = \frac{K}{\sqrt{1 + T^2\omega^2}}$$

The phase angle can be obtained as

$$\phi = \tan^{-1}\left(\frac{0}{K}\right) - \tan^{-1}\left(\frac{T\omega}{1}\right) = -\tan^{-1} T\omega$$

Hence the frequency response is obtained as:

$$y_{ss}(t) = \frac{XK}{\sqrt{1 + T^2\omega^2}} \sin(\omega t - \tan^{-1} T\omega)$$

Example 2

Consider a system with the following Transfer Function:

$$G(s) = \frac{s + 0.1}{s + 5}$$

- a) Obtain the frequency response and hence the amplitude and the phase.
b) Obtain the output response $y(t)$ for the following inputs:

- i. $x(t) = \cos 2t$
ii. $x(t) = \cos(10t - 50^\circ)$

Solution

a)

The frequency response can be obtained by replacing s with $j\omega$ in the transfer function i.e:

$$G(j\omega) = \frac{j\omega + 0.1}{j\omega + 5}$$

The amplitude (magnitude) can be obtained as:

$$|G(j\omega)| = \frac{\sqrt{\omega^2 + 0.01}}{\sqrt{\omega^2 + 25}}$$

The phase angle can be obtained as:

$$\Phi(j\omega) = \tan^{-1}\left(\frac{\omega}{0.1}\right) - \tan^{-1}\left(\frac{\omega}{5}\right)$$

b)

i.

The input signal has a frequency of $\omega = 2 \text{ rad/s}$, therefore

$$|G(j\omega)| = \frac{\sqrt{2^2 + 0.01}}{\sqrt{2^2 + 25}} = 0.372$$

$$\Phi(j\omega) = \tan^{-1}\left(\frac{2}{0.1}\right) - \tan^{-1}\left(\frac{2}{5}\right) = 87.14 - 21.08 = 65.34^\circ$$

Hence the output response

$$y(t) = \cos(2t + 65.34^0)$$

ii.

The input signal has a frequency of $\omega = 10\text{rad/s}$, therefore

$$|G(j\omega)| = \frac{\sqrt{10^2 + 0.01}}{\sqrt{10^2 + 25}} = 0.894$$

$$\Phi(j\omega) = \tan^{-1}\left(\frac{10}{0.1}\right) - \tan^{-1}\left(\frac{10}{5}\right) = 89.42 - 63.43 = 25.99^0$$

Hence the output response

$$y(t) = \cos(10t - 50^0 + 25.99^0) = \cos(10t - 24.01^0)$$

Frequency response plots

As explained above, the frequency response is characterized by three parameters; the magnitude, the phase angle, and the frequency. Using these parameters, the frequency response can be represented graphically

There are three commonly used graphical representations of frequency response:

1. Bode diagram or logarithmic plot
2. Nyquist plot or polar plot
3. Log-magnitude-versus-phase plot (Nichols plots)

According to the course content Nyquist plot and Bode plot will be discussed in this course

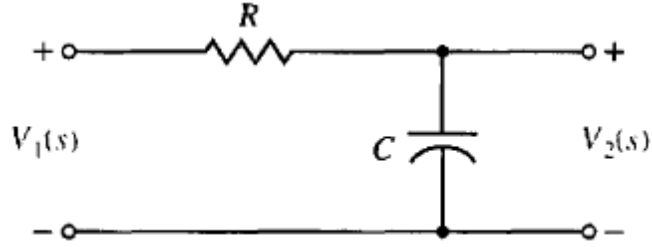
Nyquist plot

The Nyquist plot of $G(j\omega)$ is a plot of the magnitude of $G(j\omega)$ versus the phase angle of $G(j\omega)$ on polar coordinates as ω is varied from zero to infinity.

Note that in Nyquist plots a positive phase angle is measured counterclockwise from the positive real axis, while the negative phase angle is measured clockwise from the positive real axis.

Example 1

Obtain the Nyquist plot for the circuit below



Solution

$$G(s) = \frac{V_2(s)}{V_1(s)} = \frac{1}{RCs + 1},$$

$$G(j\omega) = \frac{1}{j\omega(RC) + 1} = \frac{1}{j(\omega/\omega_1) + 1},$$

Where

$$\omega_1 = \frac{1}{RC}.$$

The magnitude can be obtained as

$$|G(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_1)^2}}$$

The phase angle can be obtained as

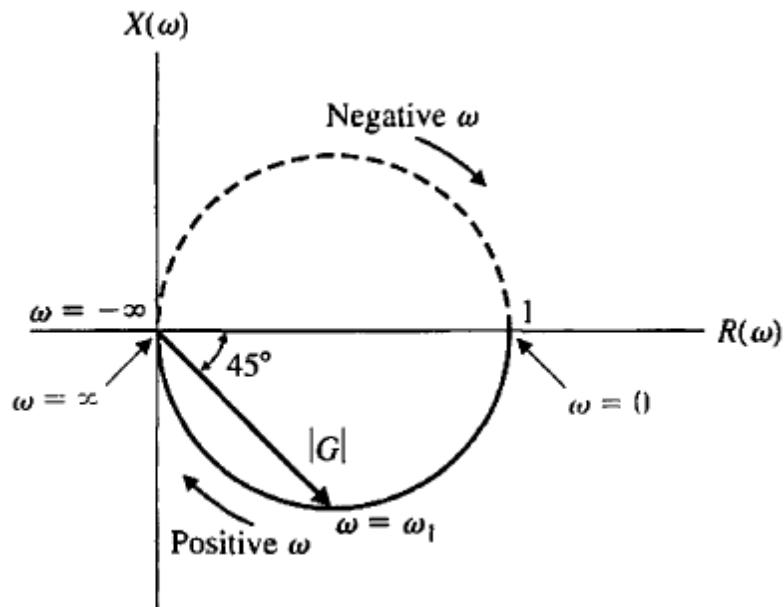
$$\phi = \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{\omega/\omega_1}{1}\right) = -\tan^{-1} \omega/\omega_1$$

At $\omega = 0$, we have $|G(j\omega)| = 1$ and $\phi = 0$

At $\omega = \infty$, we have $|G(j\omega)| = 0$ and $\phi = -90^\circ$

At $\omega = \omega_1$, we have $|G(j\omega)| = \frac{1}{\sqrt{2}} = 0.7071$ and $\phi = -45^\circ$

These three points were used to plot the Nyquist plot as shown in the figure below.



Example 2

Consider the transfer function below:

$$G(s) = \frac{K}{s(s\tau + 1)}$$

Plot the Nyquist plot to the system

Solution:

$$G(j\omega) = \frac{K}{j\omega(j\omega\tau + 1)} = \frac{K}{j\omega - \omega^2\tau}$$

The magnitude can be obtained as

$$|G(j\omega)| = \frac{K}{\sqrt{\omega^2 + \omega^4\tau^2}}$$

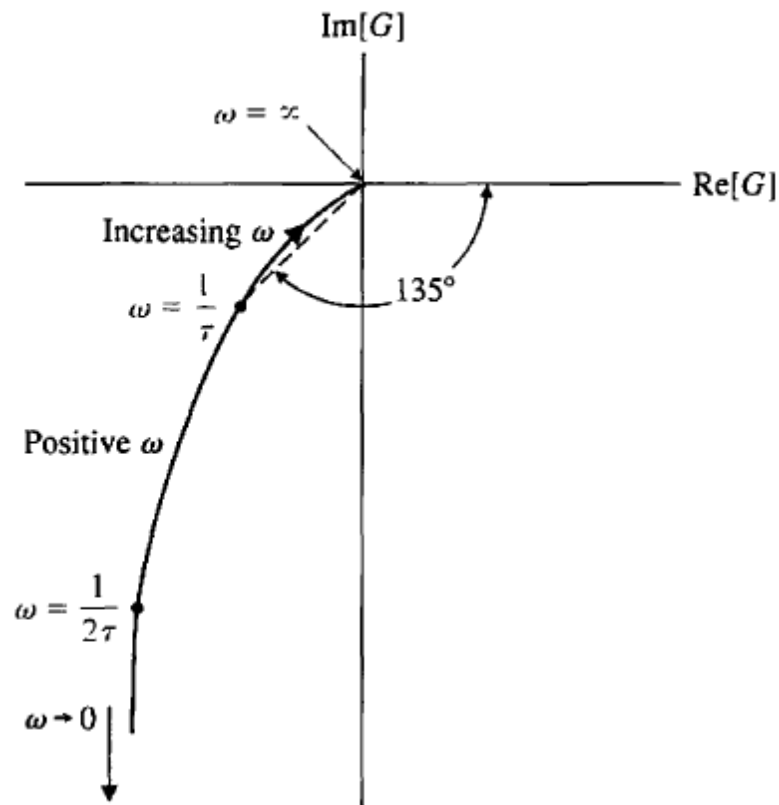
The phase angle can be obtained as

$$\phi = \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{\omega}{-\omega^2\tau}\right) = -\tan^{-1}\left(\frac{1}{-\omega\tau}\right)$$

Using the values of ω from 0 to ∞ the following table is constructed.

ω	0	$1/2\tau$	$1/\tau$	∞
$ G(j\omega) $	∞	$4K\tau/\sqrt{5}$	$K\tau/\sqrt{2}$	0
$\phi(\omega)$	-90°	-117°	-135°	-180°

The nyquist plot is drawn using the data in the above table as shown below:



Bode plot

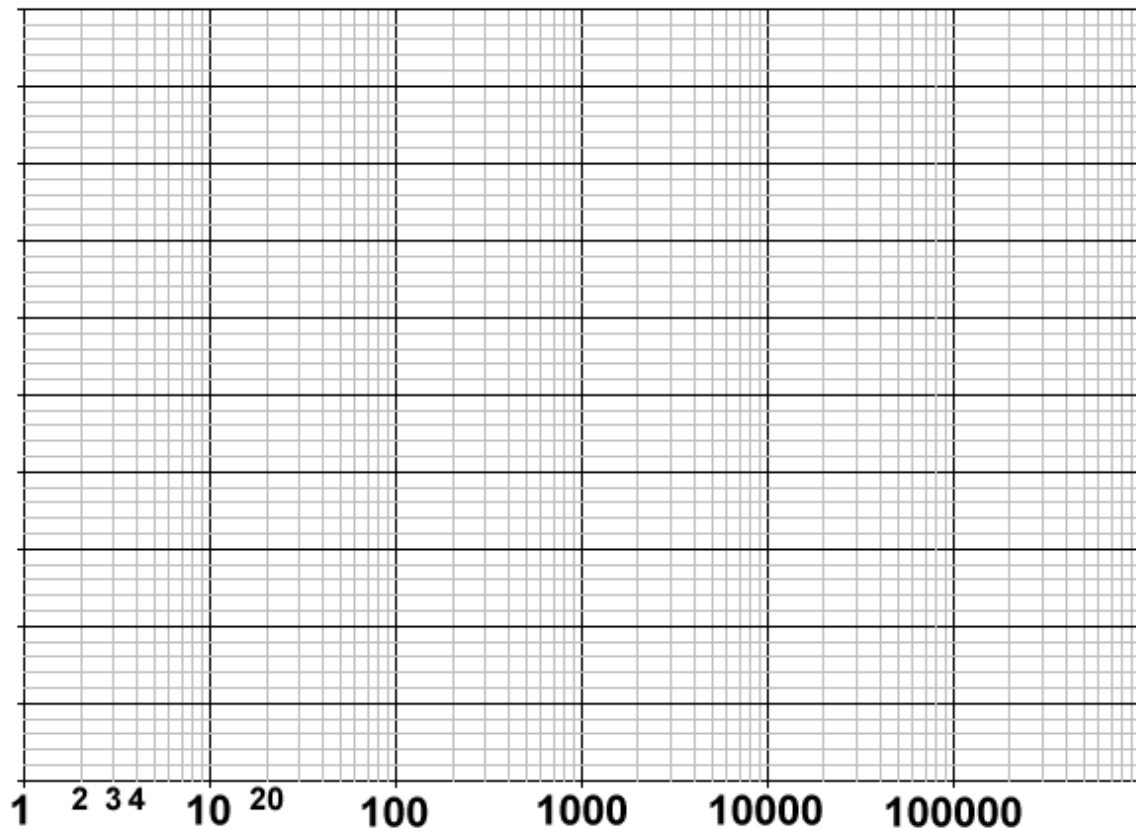
The bode plot represent the frequency response data using two graphs, namely:

- Amplitude or gain versus frequency and
- Phase angle versus frequency

Unlike in the Nyquist plot, in the bode the magnitude is expressed in decibel (dB).

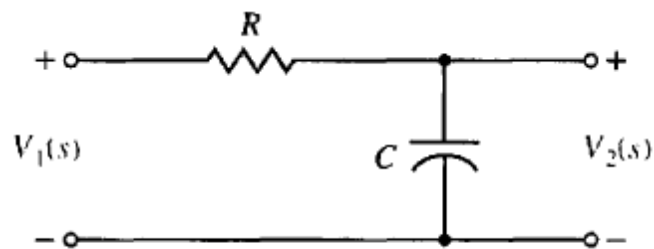
$$M = 20\log|G(j\omega)| \text{ dB}$$

The paper used for Bode plot is called semi-log graph paper. The figure below shows a semi-log paper



Example 1

On a semi-log graph paper draw the bode plot for the RC circuit below



Solution

$$G(s) = \frac{V_2(s)}{V_1(s)} = \frac{1}{RCs + 1},$$

$$G(j\omega) = \frac{1}{j\omega\tau + 1}$$

Where

$$\tau = RC$$

The magnitude can be obtained as

$$|G(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}}$$

The magnitude in dB can be obtained as

$$M = 20\log|G(j\omega)| = 20\log\left(\frac{1}{1 + (\omega\tau)^2}\right)^{1/2} = 10\log\left(\frac{1}{1 + (\omega\tau)^2}\right)$$

The phase angle can be obtained as

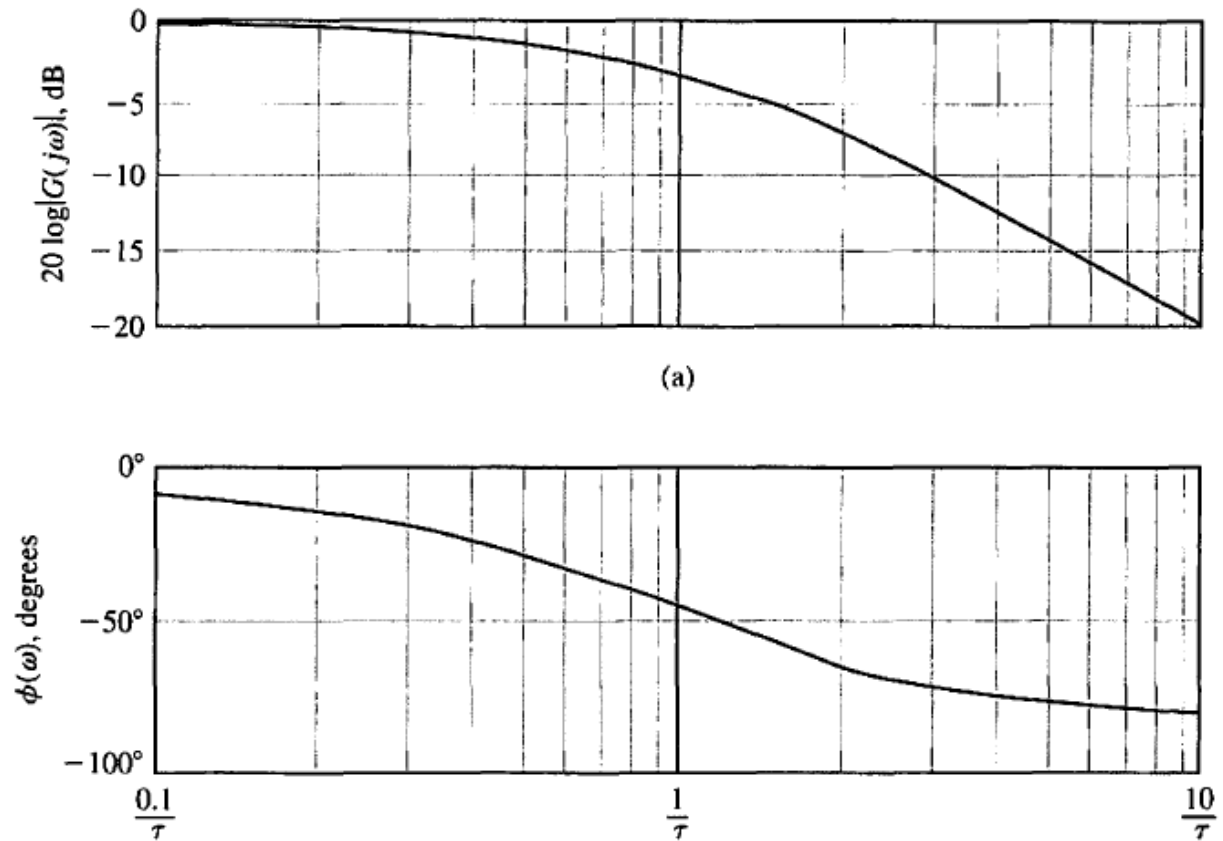
$$\phi = \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{\omega\tau}{1}\right) = -\tan^{-1}\omega\tau$$

At $\omega = \frac{0.1}{\tau}$, we have $M = -0.04dB$ and $\phi = -5.71^\circ$

At $\omega = \frac{1}{\tau}$, we have $M = -3.01dB$ and $\phi = -45^\circ$

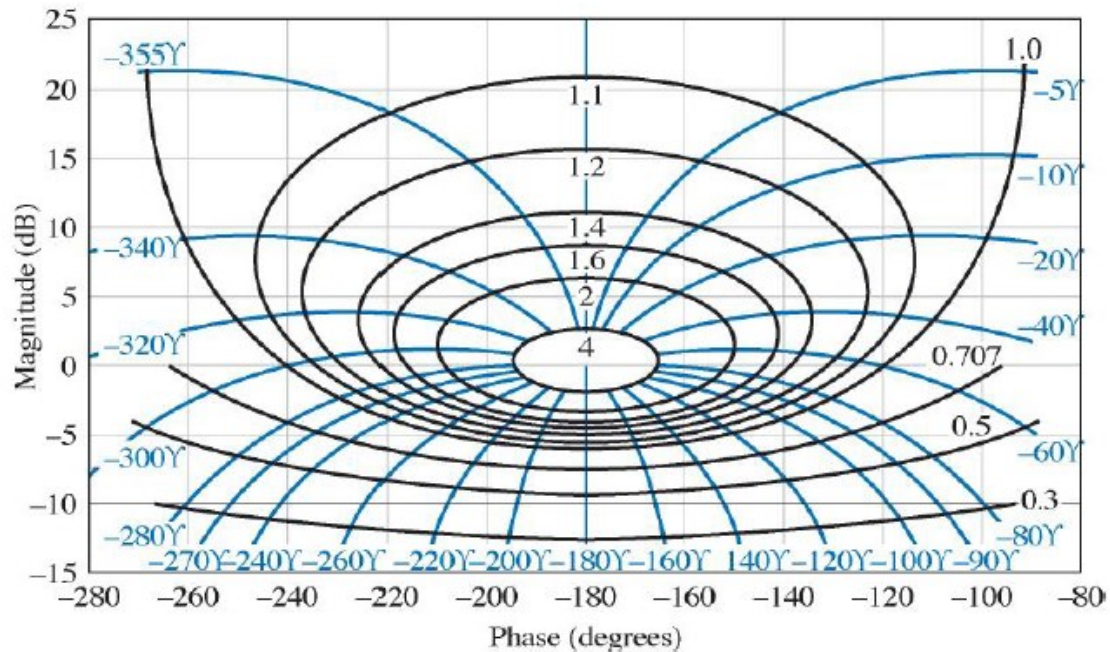
At $\omega = \frac{10}{2\tau}$, we have $M = -20.04dB$ and $\phi = -84.29^\circ$

These three points were used to plot the Bode plot as shown in the figure below.



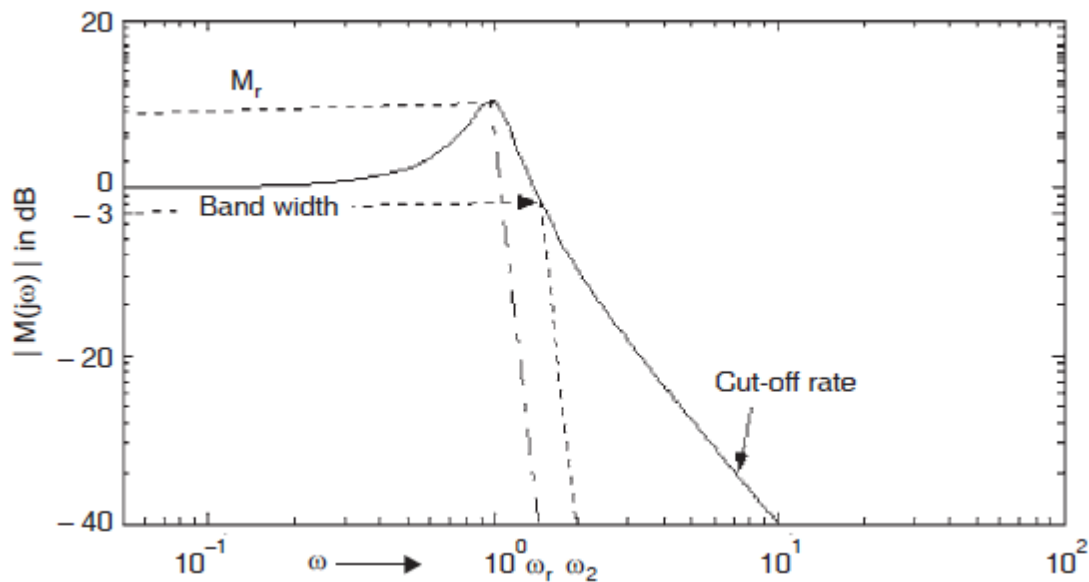
Nichols plots

The Nichols plot of $G(j\omega)$ is a plot of the magnitude of $G(j\omega)$ in dB versus the phase angle of $G(j\omega)$ on polar coordinates as ω is varied from zero to infinity. The figure below illustrates the Nichols plot:



Frequency-Domain Specifications

In the design of linear control systems using the frequency-domain methods, it is necessary to define a set of specifications so that the performance of the system can be properly assessed in terms of these specifications. The following frequency-domain specifications are often used in practice.



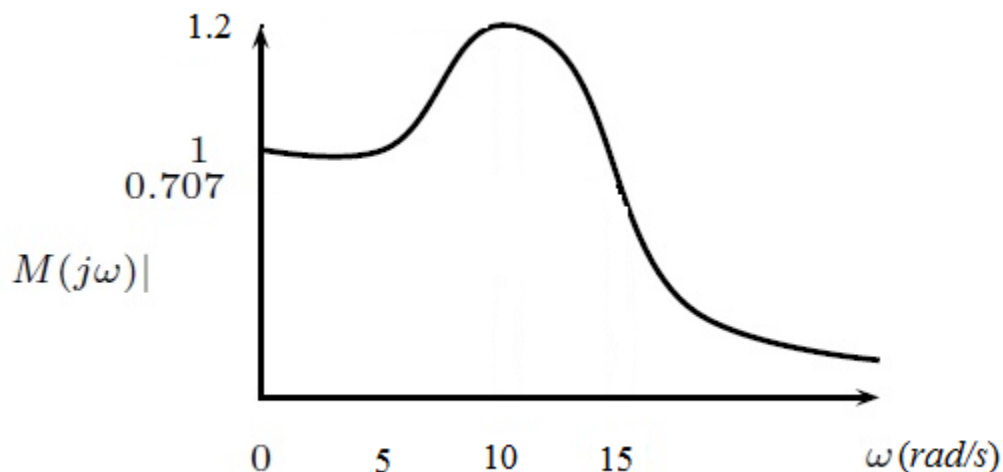
1. **Resonance Peak M_r :** This is the maximum value of the amplitude M and it indicates a measure of relative stability. The large value of M_r is associated with large overshoot in

transient input response. Recommended value of M_r is in the range of 1.1 to 1.5 for a closed loop system.

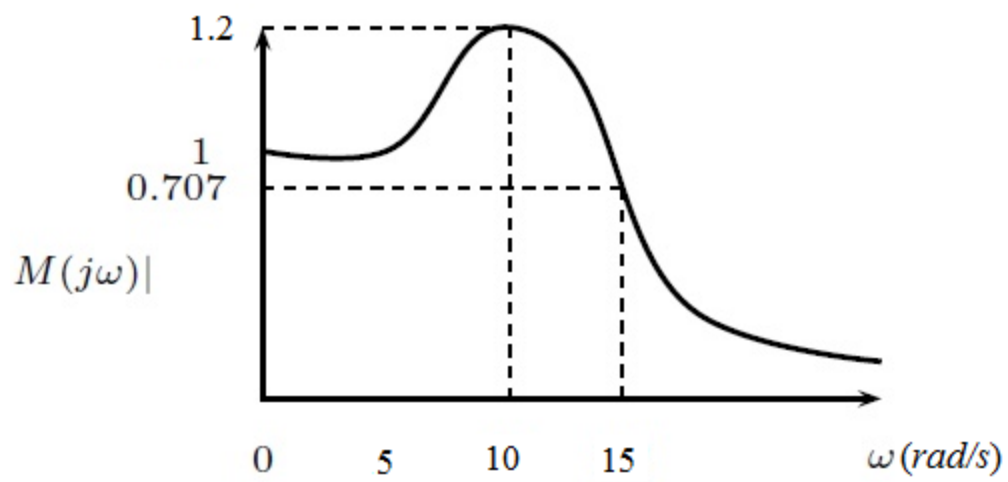
2. **Resonant Frequency (ω_r):** The resonant frequency ω_r is defined as the frequency at which the peak amplitude M_r occurs.
3. **Bandwidth: The bandwidth (BW)** is defined as the frequency at which the magnitude of $G(j\omega)$ drops to 70.7 percent of its zero-frequency value, or 3 dB down from the zero frequency value. In general, the bandwidth of a control system gives a measure of the transient response properties, in that a large bandwidth corresponds to a faster rise time, since higher frequency signals are more easily passed on to the outputs. Conversely, if the bandwidth is small, signals of relatively low frequencies only can pass to the output which makes the time response sluggish. Bandwidth also indicates the noise-filtering characteristics and the robustness of the system.
4. **Cutoff Rate:** Often, bandwidth alone is inadequate in the indication of the characteristics of the system in distinguishing signals from noise. Sometimes it may be necessary to specify the cutoff rate of the frequency response, which is the slope of magnitude of $G(j\omega)$ at the high frequency end of its spectrum. Apparently, two systems can have the same bandwidth, but the cutoff rates of the frequency responses may be different.

Example

Find the Resonance peak (M_r), Resonance Frequency (ω_r) and Bandwidth (BW) from the frequency response plot of figure below:



Solution



$$M_r = 1.2$$

$$\omega_r = 10 \text{ rad/s}$$

$$BW = 15 \text{ rad/s}$$